Swirl-String-Theory Canon-v0.8.19

Canonical Reference and Research Framework

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Abstract

This document presents Swirl-String Theory (SST) as a structured canonical synthesis rather than as a claim that every phenomenological sector has already been derived to theorem level. The text consolidates the primitive constant chain linking the effective fluid density, characteristic swirl speed, Compton anchoring, and horn/circulation scale; records delay-induced mode selection as the preferred non-operator route to spectral discreteness; and organizes the atomic bridge, spectroscopic filter, relational-time sector, and topology-based mass program within one explicitly stratified framework. Throughout, orthodox inputs, SST-internal derivations, bridge-level constructions, and conjectural extensions are kept distinct so that the canon remains internally coherent, externally comparable, and open to falsification.

Keywords: Swirl String Theory, SST, Canon, Topological Hydrodynamics, Knot Theory, Relational Time, Delay Dynamics, Epistemic Classification, orthodox ropelength foundation, screening functional scalarization, thin-filament energy scale, geometric mass audit, geometric impedance bridge, reintegration rules

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Canon reading guide.

[ORTHODOX]	established physics or mathematics.
[DERIVED]	consequences obtained from definitions and assumptions of this SST layer.
[CALIBRATED]	fixed by, or algebraically equivalent to, measured constants in the primitive calibration chain; exact as accounting, not independent prediction.
[CONDITIONAL]	dependent on an explicitly stated premise, closure, or benchmark assumption.
[SPECULATIVE]	conjectural extensions or identifications not yet closed as theorems.

Optional *role* labels (*e.g.*, *Definition*, *Benchmark*, *Constraint*) qualify how to read a statement but do not replace the primary tag. Orthodox constraints—including quantum predictions, atomic spectroscopy, and relativistic consistency—are not overridden: bridge constructions, benchmark sections, and research-track material remain subordinate to the consistency stratification introduced later in the canon.

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1 Philosophical Prelude

1.1 Why a philosophical prelude is necessary

The SST canon is not only a collection of equations but also a statement about what counts as fundamental. Standard quantum theory and general relativity are highly successful formalisms, yet they begin from different ontological primitives: operator-valued states in one case and spacetime geometry in the other. SST instead adopts a single material substrate as the underlying level of description and treats particles, radiation, and clock structure as organized states of that substrate.

1.2 The ontological shift

The canon therefore makes the following conceptual replacement:

$$\text{point particles} \longrightarrow \text{stable swirl-string configurations,} \tag{1}$$
$$\text{vacuum as background} \longrightarrow \text{vacuum as structured medium,} \tag{2}$$
$$\text{quantization as postulate} \longrightarrow \text{discreteness from dynamics and topology.} \tag{3}$$

This is not a denial of effective particle language. Rather, it is a claim that particle language is secondary and emerges from a more primitive hydrodynamic and topological layer.

1.3 Relation to orthodox physics

SST is not introduced as a replacement for the empirical content of orthodox physics. Its canonical role is instead threefold:

1. to reproduce known low-energy structures in the appropriate limits,
2. to identify which apparently fundamental constants or rules may be structurally redundant,
3. to supply a dynamical mechanism for phenomena that are otherwise postulated axiomatically.

In this sense the canon follows a *Rosetta principle*: whenever a standard description already works, SST must map to it before claiming any deeper reinterpretation.

1.4 Why delay and topology matter

Two themes recur across the development of the canon. First, finite circulation delay on closed carriers provides a natural source of temporal nonlocality. Second, topological invariants constrain admissible configurations and stabilize structures that would otherwise collapse or disperse. Together they motivate the canonical SST slogan

delay selects modes, topology protects them.

This is the philosophical reason why the canon treats quantization neither as a primitive operator rule nor as a purely statistical statement.

1.5 Scope of the present canon

The present version of the canon should be read as a structured reference document rather than a claim that every sector has already been derived to theorem level. Some components are mature and reusable, such as the primitive constant chain, circulation quantization, and the Swirl-Clock. Other components are deliberately retained as research-track layers, such as full gauge emergence, topology-to-charge assignments, and complete many-body atomic closure. The point of the canon is therefore not maximal rhetorical closure, but explicit logical organization.

1.6 Interpretive summary

The philosophical stance of SST can be summarized in one sentence:

Physics is interpreted as the dynamics of a continuous medium whose stable, delayed, and topologically organized motions appear effectively as particles, clocks, fields, and spectra.

2 Formal Foundations

2.1 Primitive Structure of the Theory

We define the Swirl-String Theory (SST) as a continuum-based framework built on a minimal set of primitive quantities.

Primitive calibration set:

$$\mathcal{P}_{\text{cal}} = \{\rho_f, v_{\circlearrowleft}^{*}, \omega_c\} \tag{4}$$

where:

- ρ_f is the effective background fluid density,
- $v_{\circlearrowleft}^{*} = \|\mathbf{v}_{\circlearrowleft}\|$ is the canonical characteristic swirl-speed magnitude,
- ω_c is the Compton angular frequency.

[**CANON / NOTATION**]: $v_{\circlearrowleft}^{*}$ is a calibrated scalar speed. The local swirl-velocity field is denoted

$$\mathbf{u}_{\circlearrowleft}(\mathbf{x}, t), \qquad \boldsymbol{\omega}_{\circlearrowleft}(\mathbf{x}, t) = \nabla \times \mathbf{u}_{\circlearrowleft}(\mathbf{x}, t). \tag{5}$$

The symbol $\mathbf{v}_{\circlearrowleft}$ without explicit arguments is reserved for the canonical characteristic carrier velocity; it must not be used as the unrestricted local field in bridge equations.

[**ORTHODOX**]: $\omega_c = \frac{m_e c^2}{\hbar}$ is a standard physical constant.

[**SPECULATIVE**]: The identification of ω_c as the primitive Compton phase frequency of the SST carrier is an SST postulate; it is not a local vorticity variable.

2.2 Derived Horn/Circulation Radius

The quantity previously denoted as the characteristic "core radius" is fixed more precisely as a Compton-locked horn/circulation radius:

$$r_c \equiv R_{\text{horn}} = \frac{v_{\circlearrowleft}^{*}}{\omega_c} = \frac{\hbar v_{\circlearrowleft}^{*}}{m_e c^2} = \frac{\alpha \hbar}{2 m_e c} = \frac{r_e}{2}. \tag{6}$$

Here

$$\omega_c \equiv \frac{m_e c^2}{\hbar} \tag{7}$$

is the electron Compton angular frequency. Thus r_c is the radius at which the canonical tangential swirl speed $v_{\circlearrowleft}^{*}$ is obtained at the Compton angular frequency.

[**CANON / NOTATION**]: ω_c denotes the Compton angular frequency. It must not be confused with the local vorticity scale. If $\omega_{\text{vort},c}$ denotes the local vorticity of a rigid local rotation, then

$$\omega_{\text{vort},c} = 2\omega_c, \qquad R_{\text{horn}} = \frac{2 v_{\circlearrowleft}^{*}}{\omega_{\text{vort},c}}. \tag{8}$$

[**CANON / SEMANTIC RULE**]: r_c is not the resolved physical vortex-tube radius. The local tube/core thickness is denoted by a_{core} and remains a separate geometric or variational quantity:

$$a_{\text{core}} \neq r_c. \tag{9}$$

[**DERIVED**] **Consequence**: Eq. (6) removes r_c as an independent parameter while avoiding the earlier ambiguity between horn-envelope scale, tube radius, and vorticity.

[**CALIBRATED CHAIN GUARD**]: Eq. (6) is a Compton–electron-radius closure inside the primitive calibration chain. It must not be cited as an independent hydrodynamic derivation of either α, $v_{\circlearrowleft}^{*}$, or r_c. A future fluid-dynamical derivation would have to obtain the same radius from a resolved vortex-tube variational problem using a_{core} or an explicit profile, without inserting the electron-radius identity.

2.3 Circulation Quantization

The circulation is defined as:

$$\Gamma = \oint \mathbf{v} \cdot d\ell \tag{10}$$

We introduce the fundamental circulation quantum:

$$\Gamma_0 = 2\pi r_c \, v_{\circlearrowleft}^* \tag{11}$$

and impose:

$$\Gamma = n\Gamma_0, \quad n \in \mathbb{Z} \tag{12}$$

[**ORTHODOX**]: Circulation quantization appears in superfluid systems.
[**DERIVED**]: In SST, Γ_0 is no longer an input but follows from Eq. 6.

2.4 Energy and Mass Relation

Mass is interpreted as stored rotational energy:

$$E = mc^2 \tag{13}$$

[**ORTHODOX**]: Standard relativistic energy relation.
[**SPECULATIVE**]: SST interprets this energy as localized vortex kinetic energy.

2.5 Swirl Velocity Constraint

We define the dimensionless ratio:

$$\eta_0 = \frac{v_{\circlearrowleft}^*}{c} \tag{14}$$

Within the present calibrated constant chain:

$$\eta_0 = \frac{\alpha}{2} \tag{15}$$

[**CALIBRATED ALGEBRAIC IDENTITY**]: Once the Compton anchoring relation $r_c = v_{\circlearrowleft}^*/\omega_c$ is combined with the electron-radius closure $r_c = \alpha\hbar/(2m_e c)$, the identity $v_{\circlearrowleft}^* = \alpha c/2$ follows by direct substitution.

[**CRITICAL NOTE**] This is not an independent fluid-dynamical derivation of α or of the characteristic speed. The fine-structure constant enters through the adopted electron-radius / horn-radius calibration, so the result is a calibrated algebraic identity rather than an emergent theorem.

2.6 Swirl-Clock Definition

We introduce the Swirl-Clock as a local scalar field encoding relativistic time dilation:

$$S_{(t)}^{\circlearrowleft}(x,t) = \sqrt{1 - \frac{\|\mathbf{u}_{\circlearrowleft}(x,t)\|^2}{c^2}} \tag{16}$$

[**ORTHODOX**] **Functional form**: This matches the standard Lorentz time-dilation factor.

[**CANON / SYMBOL DISCIPLINE**] The argument-free constant $v_{\circlearrowleft}^*$ denotes the calibrated horn/circulation speed. It must not be used as the unrestricted local velocity field in clock bridge equations. The canonical benchmark value is recovered from Eq. (16) only at points or coarse-grained states where $\|\mathbf{u}_{\circlearrowleft}(x,t)\| = v_{\circlearrowleft}^*$.

[**SPECULATIVE**] **Identification**: In SST, the relevant velocity is interpreted as a local medium-kinematic state rather than only a standard worldline speed.

[**DERIVED**] **Consequence**: Once this identification is adopted, the Swirl-Clock provides a local scalar measure of kinematic time modulation within the medium description.

2.7 Two-Speed Discipline for Clock Emergence

The canon distinguishes the canonical horn/circulation speed $v_{\circlearrowleft}^{*}$ from the transverse causal speed c:

$$v_{\circlearrowleft}^{*} = 1.09384563 \times 10^{6}\ \mathrm{m\,s^{-1}}, \qquad c = 2.99792458 \times 10^{8}\ \mathrm{m\,s^{-1}}, \qquad \frac{v_{\circlearrowleft}^{*}}{c} = 3.64867628 \times 10^{-3}. \tag{17}$$

Thus $v_{\circlearrowleft}^{*}$ is a calibrated circulation/horn speed, not the universal causal speed.

[CANON / SEMANTIC RULE] The calibrated identity $v_{\circlearrowleft}^{*} = \alpha c/2$ must not be inverted into a claim that the internal core sound speed equals c. In a Gross–Pitaevskii / NLSE core model with circulation quantum and healing length

$$\Gamma_q = \frac{h_*}{m_*}, \qquad \xi = \frac{\hbar_*}{\sqrt{2}\, m_* c_s}, \tag{18}$$

imposing $\Gamma_q = \Gamma_0$ and $\xi = r_c$ gives

$$c_s = \frac{\Gamma_0}{2\sqrt{2}\pi r_c} = \frac{v_{\circlearrowleft}^{*}}{\sqrt{2}} = 7.73465663 \times 10^{5}\ \mathrm{m\,s^{-1}}. \tag{19}$$

Therefore a calculation that closes an inertia ratio as u^2/c_s^2 has produced an internal core-acoustic closure, not yet the Lorentz-compatible ratio u^2/c^2.

[RESEARCH-TRACK GATE] A Lorentz-compatible derivation of the Swirl-Clock requires a separate transverse torsion/shear sector with propagation speed $c_T = c$, plus a core–torsion impedance identity of the form

$$\mathsf{M}_{\mathrm{torsion}}[K] \stackrel{?}{=} \frac{2E_0[K]}{c_T^2}\,\mathsf{I}. \tag{20}$$

Until Eq. (20) is derived from a controlled field calculation, the Swirl-Clock remains an orthodox functional form with an SST interpretation, or a derived-conditional bridge result, rather than a hard theorem from the NLSE core alone.

2.8 Density Hierarchy

We distinguish the following densities:

$$\rho_f \quad \text{(effective background fluid density)}, \tag{21}$$
$$\rho_E = \tfrac{1}{2}\rho_f\, v_{\circlearrowleft}^{*\,2} \quad \text{(swirl energy density)}, \tag{22}$$
$$\rho_m = \rho_E/c^2 \quad \text{(mass-equivalent density)}, \tag{23}$$
$$\rho_{\mathrm{core}} \quad \text{(canonical saturated envelope-equivalent density)}. \tag{24}$$

[ORTHODOX] Definition: $\rho_m = \rho_E/c^2$ is the mass-equivalent density associated with the swirl kinetic accounting above; in orthodox limits, total energy E in a region still yields mass density E/c^2.

[SPECULATIVE] Interpretation: The distinction between background, energetic, and saturated envelope-equivalent sectors is elevated in SST to a structural ontology of the medium.

2.9 Horn-Envelope Density Closure

The density formerly denoted as a "core density" is now interpreted as an effective horn-envelope normalization density, not as a measured local material density of the resolved vortex tube:

$$\rho_{\mathrm{horn}}^{\mathrm{eff}} \equiv \frac{m_e c^2}{2\pi v_{\circlearrowleft}^{*\,2} R_{\mathrm{horn}}^3} = \frac{m_e c^2}{2\pi v_{\circlearrowleft}^{*\,2} {r_c}^3}. \tag{25}$$

[CANON / SEMANTIC RULE]: Equation (25) defines the canonical horn-envelope density closure used by the present SST calibration program. Legacy appearances of ρ_{core} in expressions proportional to $\rho_{\mathrm{core}} {r_c}^3$ should be read as $\rho_{\mathrm{horn}}^{\mathrm{eff}}$ unless a separate resolved-tube model is explicitly introduced.

[DERIVED] Consequence: Within that closure, $\rho_{\mathrm{horn}}^{\mathrm{eff}}$ is no longer an independent parameter. If a local tube mass density is required, it must be introduced through the separate tube radius a_{core} and a corresponding local tube-volume or profile model.

2.10 Chronos–Kelvin Invariant and Clock–Radius Transport

For a thin material loop of radius $R(t)$ carried by an incompressible, inviscid flow with no reconnection events, Kelvin's theorem implies conservation of circulation. In the near-solid-body core approximation this gives

$$\frac{D}{Dt}(R^2\omega) = 0, \tag{26}$$

where ω is the characteristic local vorticity on the loop. For rigid local rotation, $v_t = (\omega/2)R$; evaluated at the canonical horn scale this gives $v_t = (\omega/2)r_c$. Using this relation together with Eq. (16), one may rewrite

$$\omega = \frac{2c}{r_c}\sqrt{1 - S_{(t)}^2}, \tag{27}$$

so that Eq. (26) takes the equivalent form

$$\frac{D}{Dt}\left(\frac{2c}{r_c}R^2\sqrt{1 - S_{(t)}^2}\right) = 0. \tag{28}$$

[ORTHODOX]: Eq. (26) is the thin-loop form of Kelvin circulation conservation in inviscid barotropic flow [6, 23].

[CONDITIONAL DERIVED]: Eq. (28) is the SST rewriting of the same invariant in terms of the Swirl-Clock, conditional on evaluating the local tangential speed at the fixed horn/circulation scale r_c. It is not a generic radius-transport theorem for an arbitrary resolved tube radius $a_{\text{core}}(t)$.

Differentiating Eq. (28) gives a useful transport identity:

$$\frac{dS_{(t)}}{dt} = \frac{2\left(1 - S_{(t)}^2\right)}{S_{(t)}}\frac{1}{R}\frac{dR}{dt}. \tag{29}$$

[CONDITIONAL DERIVED]: within the horn-anchored approximation, contraction of a material loop drives stronger clock suppression, while expansion relaxes it. The identity deliberately mixes a variable loop radius $R(t)$ with the fixed horn-evaluation radius r_c; it should not be read as an unconstrained resolved-tube transport law.

2.11 Reparametrized Zero-Parameter Corollary

Older SST canon layers often used the triplet (Γ_0, ρ_f, r_c) as the primitive parameterization. In the present v0.8.x architecture, Eq. (6) and Eq. (11) show that this is a reparameterization of the present primitive calibration set $(\rho_f, v_{\circlearrowleft}^{*}, \omega_c)$ rather than a distinct axiom.

Explicitly,

$$\Gamma_0 = 2\pi r_c\, v_{\circlearrowleft}^{*} = 2\pi\frac{v_{\circlearrowleft}^{*\,2}}{\omega_c}. \tag{30}$$

Hence any dimensional SST quantity written in terms of (Γ_0, ρ_f, r_c) can be rewritten in terms of $(\rho_f, v_{\circlearrowleft}^{*}, \omega_c)$ with no new calibration input.

A coarse-grained density law retained from earlier canon layers is

$$\rho_f = K\,\Omega, \qquad K = \frac{\rho_{\text{core}}\, r_c}{v_{\circlearrowleft}^{*}}, \tag{31}$$

where Ω is a coarse-grained local rotation rate and K has dimensions $\mathrm{kg\,m^{-3}\,s}$. This relation records the canonical bridge between a sparse effective bulk density and a high-density horn-envelope normalization; it is not a resolved local-tube density law unless a_{core} is supplied.

[**CRITICAL NOTE**] Eq. (31) is not an additional zero-parameter theorem unless the averaging prescription that defines Ω is fixed independently of the data being fitted. If Ω is chosen phenomenologically, the relation is a bridge ansatz and K is a calibration transfer coefficient, not a new prediction.

[**DERIVED COROLLARY**]: the older "zero-parameter" language is retained only as a statement of reparameterization discipline: once one primitive triplet is fixed, no additional dimensional constants should be introduced without explicit justification.

2.12 Canonical Master Equation

The preceding subsections define the four ingredients that later SST papers often treat separately: a local mass-equivalent density, a local Swirl-Clock, a geometric gate, and a topological kernel. We therefore introduce a single canonical equation that organizes these sectors in one place.

First define the local swirl energy density and its mass-equivalent form:

$$\rho_E(x,t) = \frac{1}{2}\rho_f(x,t)\,\|\mathbf{u}_{\circlearrowleft}(x,t)\|^2 \tag{32}$$

$$\rho_m(x,t) = \frac{\rho_E(x,t)}{c^2} \tag{33}$$

Using Eq. 16, the local Swirl-Clock factor is

$$S_{(t)}^{\circlearrowleft}(x,t) = \sqrt{1 - \frac{\|\mathbf{u}_{\circlearrowleft}(x,t)\|^2}{c^2}} \tag{34}$$

We then define a binary geometric exposure gate

$$G(K) \in \{0,1\} \tag{35}$$

together with the canonical geometric factor

$$\Pi_K = \left(\frac{\lambda_c}{\pi r_c}\right)^{G(K)} = \left(\frac{4}{\alpha}\right)^{G(K)}. \tag{36}$$

[**CANON / ALPHA-GATE GUARD**] The equality $\lambda_c/(\pi r_c) = 4/\alpha$ is a calibrated geometric rewriting inside the present constant chain. It must not be cited as an independent derivation of the fine-structure constant. The finite-cell trefoil result is retained separately as an α-free sub-per-mille coincidence plus obstruction result in Subsection D.10.

Numerically,

$$\frac{\lambda_c}{\pi r_c} = \frac{4}{\alpha} \simeq 5.48144 \times 10^2. \tag{37}$$

This number is therefore an exposure or shielding gate inside the calibrated mass functional, not a new measurement or theorem for α.

The canonical topological kernel is taken to be

$$\Xi_K = [\alpha_C\, C(K) + \beta_L\, L(K) + \gamma_H\, \mathcal{H}(K)]\, \varphi^{-2k(K)}. \tag{38}$$

[**CANON / GOLDEN-LAYER GUARD**]**.** The golden-layer factor has an exact hyperbolic reparametrization,

$$\eta_\varphi \equiv \operatorname{asinh}\left(\frac{1}{2}\right) = \ln\varphi, \tag{39}$$

$$\varphi^{-2k(K)} = \exp[-2k(K)\eta_\varphi]\,. \tag{40}$$

This is a parametrization guard for the existing layer factor, not a derivation of the golden factor from Euler dynamics, ropelength criticality, or knot stability. It also does not assert that the electron trefoil is a hyperbolic knot; the trefoil remains a torus knot in the topology dictionary. Any future promotion of this notation must identify $k(K)$ with a mechanically defined discrete transfer, impedance, or relaxation step.

The canonical SST master equation is therefore

$$M_K(x,t) = \rho_m(x,t)\, {r_c}^3\, \Pi_K\, \Xi_K\, S^{\circlearrowleft}_{(t)}(x,t)^{-2} \tag{41}$$

$$= \left(\frac{\rho_f(x,t)\, \|\mathbf{u}_{\circlearrowleft}(x,t)\|^2}{2c^2}\right) {r_c}^3 \left(\frac{\lambda_c}{\pi r_c}\right)^{G(K)} \left[\alpha_C\, C(K) + \beta_L\, L(K) + \gamma_H\, \mathcal{H}(K)\right] \varphi^{-2k(K)} S^{\circlearrowleft}_{(t)}(x,t)^{-2} \tag{42}$$

[SPECULATIVE] Organizing principle: This equation factorizes SST into four sectors:

$$\text{medium scale} \times \text{geometric gate} \times \text{topological kernel} \times \text{clock impedance}. \tag{43}$$

[DERIVED] Dimensional check:

$$\left[\frac{\rho_f\, \|\mathbf{u}_{\circlearrowleft}\|^2}{c^2} {r_c}^3\right] = \text{kg} \tag{44}$$

and all remaining factors are dimensionless, so Eq. 42 has the correct mass dimension.

[CANON / SEMANTIC RULE]: the factor ${r_c}^3$ in Eq. (42) is a canonical horn/envelope normalization volume scale. It is not a resolved physical tube volume unless a separate model sets $a_{\text{core}} = r_c$. Resolved tube energies, cutoffs, and local vorticity estimates must use a_{core} or an explicit profile.

[SPECULATIVE] Interpretation: The same local kinematic state that slows the Swirl-Clock also enters the effective mass scale; mass and local proper-time modulation are therefore treated as two aspects of one medium configuration.

[CRITICAL NOTE] Equation (42) is presently a canonical synthesis formula, not yet a theorem derived from first-principles filament dynamics alone.

[NORMALIZATION TRANSPARENCY] With the canonical background-energy reading, $\rho_m = \rho_E/c^2 = 4.65949 \times 10^{-12}\,\text{kg}\,\text{m}^{-3}$ and $\rho_m {r_c}^3 = 1.30330 \times 10^{-56}\,\text{kg} = 1.43072 \times 10^{-26} m_e$. Therefore an exposed electron-scale benchmark with $\Pi_K = 4/\alpha$ requires a dimensionless kernel of order

$$\Xi^{(e)}_{3_1} \sim \frac{m_e}{\rho_m {r_c}^3 (4/\alpha)} = 1.28 \times 10^{23}. \tag{45}$$

Unless a branch explicitly re-anchors the medium scale to $\rho^{\text{eff}}_{\text{horn}}$, the coefficients inside Ξ_K are therefore calibrated matching coefficients carrying the medium-to-particle hierarchy, not order-unity topological invariants. This note is a scale-accounting guard; it does not alter the dimensional validity of Eq. (42).

In the stationary weak-swirl limit $S^{\circlearrowleft}_{(t)} \to 1$, it reduces to the static mass-functional form; in the local-kinematic limit at fixed K, it reduces to a pure clock-impedance correction. The hierarchy is therefore carried predominantly by the geometric and topological factors, while the Swirl-Clock supplies a coherent local correction rather than the primary source of mass separation.

2.13 Geometric Representation

A swirl-string is represented as a closed curve:

$$X(t,\sigma) = (x(t,\sigma), y(t,\sigma), z(t,\sigma)) \tag{46}$$

with $\sigma \in [0, 2\pi)$.

2.14 Summary of Dependency Structure

The full dependency chain is:

$$(\rho_f, v^*_{\circlearrowleft}, \omega_c) \to r_c \to \Gamma_0 \to (\rho_E, \rho_m, \rho_{\text{core}}) \to S^{\circlearrowleft}_{(t)} \to M_K. \tag{47}$$

[DERIVED] Consistency statement: No quantity is defined circularly within the stated calibration chain. The α-dependent identities remain calibrated algebraic identities rather than independent derivations, and the canonical closure is summarized by Eq. 42.

2.15 Consistency Statement

- All derived quantities depend on the stated calibration set, explicit orthodox constants, and explicitly labeled bridge assumptions.
- No *unlabeled* external parameters are introduced.
- Quantities fixed by α, m_e, $\hbar$, c, R_∞, or t_p are labeled **[CALIBRATED]** rather than independent predictions.

2.16 Interpretation

The Formal Foundations establish SST as:

- a continuum theory with minimal primitives
- a geometrically grounded framework
- a non-circular parameter system

2.17 Maximal Swirl Tension

2.17.1 Definition and canonical form

Within the primitive calibration set $\mathcal{P}_{\text{cal}} = \{\rho_f, v_{\circlearrowleft}^{*}, \omega_c\}$, a characteristic force scale emerges naturally from the Compton energy stored over the canonical circulation cross-section. We define the *maximal swirl tension* as

$$F_{\text{swirl}}^{\max} = \frac{v_{\circlearrowleft}^{*}\,\hbar}{2{r_c}^2} = \frac{\hbar\omega_c}{2r_c}. \tag{48}$$

[CALIBRATED ALGEBRAIC IDENTITY] Canonical form: substituting $v_{\circlearrowleft}^{*} = \omega_c r_c$ into Eq. (48) gives $F_{\text{swirl}}^{\max} = \hbar\omega_c/2r_c$, the Compton energy scale divided by twice the canonical circulation radius. Numerically, $F_{\text{swirl}}^{\max} = 29.053507$ N.

[CRITICAL NOTE] $F_{\text{swirl}}^{\max}$ is a mechanical reformulation of the Compton energy scale over the canonical circulation cross-section. It is *not* independent of $\hbar$ and m_e; its role in the canon is interpretive and organisational rather than that of an independent primitive.

2.17.2 Internal consistency identities

The following expressions are algebraically equivalent within the canonical constant chain and serve as coherence verifications rather than independent derivations:

$$F_{\text{swirl}}^{\max} = \frac{e^2}{16\pi\varepsilon_0 {r_c}^2}, \tag{49}$$

$$= \frac{h\alpha c}{8\pi {r_c}^2}, \tag{50}$$

$$= \pi {r_c}^2\, \rho_{\text{horn}}^{\text{eff}}\, {v_{\circlearrowleft}^{*}}^2, \tag{51}$$

$$= \frac{c^4}{4G}\,\alpha\left(\frac{r_c}{\ell_P}\right)^{-2}. \tag{52}$$

where Eq. (51) uses the same horn-envelope normalization density $\rho_{\text{horn}}^{\text{eff}}$ fixed by Eq. (25). No additional symbol ρ_{calc} is introduced; substituting $v_{\circlearrowleft}^{*} = \alpha c/2$ and Eq. (25) reduces the density form identically to Eq. (48).

[CALIBRATED ALGEBRAIC IDENTITY] Consistency check: Eqs. (49)–(52) are equivalent given $v_{\circlearrowleft}^{*} = \alpha c/2$, the standard identity $e^2/(4\pi\varepsilon_0) = \alpha\hbar c$, $r_c = \alpha\hbar/(2m_e c)$, and $\ell_P^2 = \hbar G/c^3$. The Planck-force form in Eq. (52) is a hierarchy reparameterization of the same calibrated tension scale, not an independent derivation from Planck physics. Their mutual agreement is an internal algebraic coherence test of the calibrated chain, not an independent dynamical prediction.

2.17.3 Rydberg consistency identity

[**CALIBRATED IDENTITY**] The Rydberg constant is recoverable from the three SST calibration variables without e, ε_0, or m_e appearing *explicitly*:

$$R_\infty = \frac{v_{\circlearrowleft}^{*\,3}}{\pi\, r_c\, c^3}. \tag{53}$$

[**DIMENSIONAL GUARD**] The direction of Eq. (53) is fixed: the numerator is $v_{\circlearrowleft}^{*\,3}$ and the denominator is $\pi r_c c^3$. Its units are m^{-1}. The reciprocal expression $\pi r_c c^3 / v_{\circlearrowleft}^{*\,3}$ has units of length and is not a Rydberg constant.

This is a benchmark for algebraic consistency of the calibration chain, not for independent non-redundancy of the primitive set. [**CRITICAL NOTE**] Substituting the canonical definitions $v_{\circlearrowleft}^{*} = \alpha c/2$ and $r_c = v_{\circlearrowleft}^{*}/\omega_c = \alpha\hbar/(2m_e c)$ reduces Eq. (53) *exactly* to the standard $R_\infty = \alpha^2 m_e c/(4\pi\hbar)$ (verified to machine precision): m_e is hidden inside r_c and α inside both $v_{\circlearrowleft}^{*}$ and r_c. The identity is therefore a consistency check, not an independent non-circular derivation of R_∞. Combining Eq. (53) with Eq. (48) yields the compact relation

$$\boxed{F_{\mathrm{swirl}}^{\max} = \frac{16\pi^2\hbar R_\infty^2 c}{\alpha^5},} \tag{54}$$

which expresses $F_{\mathrm{swirl}}^{\max}$ in terms of spectroscopic observables alone, with no reference to m_e, r_c, or e as separate inputs. A further identity connecting atomic and force scales is

$$R_\infty = F_{\mathrm{swirl}}^{\max} \cdot \frac{\alpha^3}{4\pi m_e c^2}, \tag{55}$$

which states that the Rydberg constant is the maximal swirl tension scaled by the Compton energy and a purely geometric factor $\alpha^3/4\pi$.

2.17.4 Threefold physical interpretation

The quantity $F_{\mathrm{swirl}}^{\max}$ admits three distinct physical interpretations, each arising from a different sector of physics. [**CRITICAL NOTE**] These three values are algebraically equivalent expressions built from the same primitive set $\{\alpha, m_e, \hbar, c\}$; their numerical agreement is an *identity*, not an independent coincidence, and is not on its own confirmation of the theory.

(i) Gravitational analogue: the Planck force ceiling. In general relativity, the combination $c^4/4G$ sets the maximum force that any physical process can transmit without forming a causal horizon [30]. We denote this scale the maximal gravitational tension:

$$F_{\mathrm{gr}}^{\max} = \frac{c^4}{4G}. \tag{56}$$

[**ORTHODOX**] $F_{\mathrm{gr}}^{\max}$ is the Planck force, an invariant upper bound in classical general relativity [30].

[**CALIBRATED ALGEBRAIC IDENTITY**] The ratio of the two tension scales reduces exactly to

$$\boxed{\frac{F_{\mathrm{swirl}}^{\max}}{F_{\mathrm{gr}}^{\max}} = \frac{4\alpha_g}{\alpha},} \tag{57}$$

where $\alpha_g = G m_e^2/\hbar c$ is the gravitational fine-structure constant of the electron and α is the electromagnetic fine-structure constant. Numerically, $4\alpha_g/\alpha \approx 9.6 \times 10^{-43}$, in agreement with the directly computed ratio $29.053507\ \mathrm{N} / 3.026 \times 10^{43}\ \mathrm{N}$.

[**SPECULATIVE**] **Interpretation**: the hierarchy between the electromagnetic and gravitational force scales may be organized by the ratio of the two electron coupling constants. This is an interpretive compression of calibrated constants; by itself it does not prove a new medium-dynamical stiffness law.

(ii) Electrostatic interpretation: the Coulomb ceiling at the canonical circulation scale. When two protons are compressed to a centre-to-centre separation equal to the canonical circulation radius r_c, the Coulomb repulsion is

$$F_{pp}(r_c) \;=\; \frac{e^2}{4\pi\varepsilon_0 {r_c}^2} \;=\; 4\,F_{\rm swirl}^{\rm max}. \tag{58}$$

[**CALIBRATED**] The maximal Coulomb repulsion at the canonical circulation scale is exactly four times $F_{\rm swirl}^{\rm max}$. [**CRITICAL NOTE**] The factor of four is an arithmetic identity once $v_{\circlearrowleft}^{*} = \alpha c/2$ is adopted (the $4/\alpha$ gate); it is a self-consistency of the calibrated chain, not an independent prediction.

[**SPECULATIVE**] **Interpretation**: $F_{\rm swirl}^{\rm max}$ is the quarter-Coulomb force at the canonical circulation scale. The medium sustains one quarter of the maximum proton–proton Coulomb repulsion before the neutral circulation envelope reaches its structural limit. This provides a physical ceiling on the compression of charged vortex configurations.

(iii) Mechanical interpretation: the spring constant of photon–electron interaction. The electron's internal vortex is modelled as a linear spring with stiffness

$$K_e \;=\; \frac{F_{\rm swirl}^{\rm max}}{n\,r_c}, \qquad n = 2, \tag{59}$$

where n counts the number of half-wavelength segments on the core and is fixed to $n = 2$ by the photon–electron swirl matching condition. The natural oscillation frequency of the electron core under this spring is

$$\omega_{\rm spring} \;=\; \sqrt{\frac{K_e}{m_e}} \;=\; \sqrt{\frac{F_{\rm swirl}^{\rm max}}{2\,r_c\,m_e}} \;=\; \frac{\omega_c}{\alpha} \;=\; \frac{m_e c^2}{\alpha\hbar}, \tag{60}$$

a frequency intermediate between the atomic Rydberg scale ($\sim \alpha^2\omega_c$) and the bare Compton scale (ω_c).

[**DERIVED**] The energy stored in this spring at maximum compression $\delta = r_c$ is

$$E_{\rm spring} \;=\; \tfrac{1}{2}\,K_e\,{r_c}^2 \;=\; \frac{F_{\rm swirl}^{\rm max}\,r_c}{4} \;=\; \frac{\hbar\omega_c}{8} \;=\; \frac{m_e c^2}{8}. \tag{61}$$

One eighth of the electron rest energy is stored as internal vortex-spring potential at maximum compression.

[**SPECULATIVE**] **Physical picture**: during photon absorption, the electron vortex core is compressed for a duration of order one Planck time t_p into a transient mass-like configuration. The restoring force scale of this compression is $F_{\rm swirl}^{\rm max}$, and the energy scale of the transient is $m_e c^2/8$. The spring releases at the Compton period, generating the recoil that initiates the Coulomb interaction with the proton.

[**CRITICAL NOTE**] Equation (60) assumes $n = 2$ from empirical matching. A first-principles derivation of the photon wrap number from the delay dynamics of Section 7 is an open derivational step.

2.17.5 Unified summary

The three interpretations are collected in Table 1. Together they organize $F_{\rm swirl}^{\rm max}$ as a *calibrated structural tension scale*. The table is useful for cross-sector bookkeeping, but the equalities are algebraic consequences of the same calibrated electron-scale constants unless a separate medium-dynamical derivation is supplied.

[**SPECULATIVE**] **Canonical interpretation**: $F_{\rm swirl}^{\rm max}$ is retained as a candidate tension scale for topological integrity of the structured medium. Its comparison with $F_{\rm gr}^{\rm max} = c^4/4G$ is a calibrated electron-normalized analogy, not a theorem that gravitational and electromagnetic stiffness have already been derived from the substrate.

Table 1: Threefold physical interpretation of $F_{\rm swirl}^{\max}$ within the SST canonical framework.

Context	**Expression**	**Physical meaning**	**Status**
Gravitational ceiling	$F_{\rm gr}^{\max}/F_{\rm swirl}^{\max} = \alpha/(4\alpha_g)$	Calibrated electron-coupling ratio $\alpha/4\alpha_g$	[Calibrated identity]
Coulomb ceiling	$F_{pp}(r_c) = 4\,F_{\rm swirl}^{\max}$	Quarter of max. Coulomb force at circulation scale	[Calibrated identity]
Vortex spring	$K_e = F_{\rm swirl}^{\max}/2r_c$	Spring constant of photon–electron interaction	[Speculative]

3 Axiomatic Framework

3.1 Preliminaries

We formalize Swirl-String Theory (SST) as an axiomatic continuum theory defined over a three-dimensional spatial domain $\mathcal{M} \subset \mathbb{R}^3$ equipped with a preferred time parameter $t \in \mathbb{R}$.

The theory is constructed from a minimal primitive set $\mathcal{P}$ (Section 2) and a set of axioms $\mathcal{A}$ governing admissible configurations and dynamics.

3.2 Axiom 1: Continuum Substrate

Statement. There exists a continuous, incompressible, inviscid medium filling $\mathcal{M}$, characterized by a constant background density ρ_f.

$$\nabla \cdot \mathbf{v} = 0 \tag{62}$$

[**ORTHODOX**]: This corresponds to the standard incompressible Euler framework.

Interpretation. All physical structures arise as configurations within this medium; no discrete particles are assumed at the fundamental level.

3.3 Axiom 2: Existence of Stable Vortex Filaments

Statement. The medium admits stable, localized, topologically nontrivial vortex filament solutions.

[**ORTHODOX**] Physics input: vortex filaments are known idealized structures in fluid dynamics.

[**SPECULATIVE**] Extension: such filaments are assumed here to be dynamically stable and persist as fundamental carriers.

Mathematical form. A filament is represented as a closed embedding:

$$X : S^1 \to \mathcal{M}, \quad \sigma \mapsto X(t,\sigma) \tag{63}$$

3.4 Axiom 3: Circulation Quantization

Statement. Circulation around any stable filament is quantized:

$$\Gamma = \oint \mathbf{v} \cdot d\ell = n\Gamma_0, \quad n \in \mathbb{Z} \tag{64}$$

[**ORTHODOX**]: Quantized circulation appears in superfluid systems.

[**SPECULATIVE**]: Elevated here to a universal invariant.

3.5 Axiom 4: Compton Anchoring

Statement. The Compton phase frequency of the canonical horn/circulation carrier satisfies:

$$\frac{v_{\circlearrowleft}^{*}}{r_c} = \omega_c \tag{65}$$

[**ORTHODOX**]: ω_c is the Compton frequency.

[**SPECULATIVE**]: Identified as the intrinsic Compton phase frequency of the SST carrier, not as the local vorticity of the resolved tube.

Consequence. The horn/circulation radius is not independent but derived (Eq. 6); the local tube radius remains $a_{\rm core}$.

3.6 Axiom 5: Delay as a Constitutive Principle

Statement. Finite propagation time along closed vortex structures is a fundamental dynamical ingredient.

$$\tau_{\mathrm{circ}}(K) = \frac{L_K}{v_{\circlearrowleft}^{*}} \tag{66}$$

[**DERIVED**] Kinematic consequence: the delay follows from finite signal speed along a closed carrier.

Axiomatic role. The delay is not treated as negligible, but as structurally relevant to the dynamics.

Interpretation. Temporal nonlocality is intrinsic to the dynamics.

3.7 Axiom 6: Energy Localization

Statement. Energy is localized in vortex structures and determines mass via:

$$E = mc^2 \tag{67}$$

[**ORTHODOX**]: Relativistic energy relation.

[**SPECULATIVE**]: Interpreted as rotational energy of the vortex configuration.

3.8 Axiom 7: Swirl-Clock Relational Time

Statement. Local time is encoded in a scalar field derived from fluid kinematics:

$$S_{(t)}^{\circlearrowleft}(x,t) = \sqrt{1 - \frac{\|\mathbf{u}_{\circlearrowleft}(x,t)\|^2}{c^2}} \tag{68}$$

[**ORTHODOX**] Functional form: this matches Lorentz time dilation.

[**CANON / SYMBOL DISCIPLINE**] The calibrated value $v_{\circlearrowleft}^{*}$ is obtained only when the local field is evaluated at $\|\mathbf{u}_{\circlearrowleft}(x,t)\| = v_{\circlearrowleft}^{*}$; it is not a spatially constant clock field.

[**SPECULATIVE**] Identification: the relevant velocity is taken to be the local medium kinematic state.

[**DERIVED**] Consequence: within SST this yields a relational clock field tied to local motion.

Interpretation. Time is not fundamental but relational to the state of motion in the medium.

3.9 Axiom 8: Density Saturation and Horn-Envelope Structure

Statement. The canonical horn-envelope exhibits an effective saturated density $\rho_{\mathrm{horn}}^{\mathrm{eff}}$ determined by the mechanical normalization constraints of the medium. A resolved local tube density is not specified until a_{core} and a tube profile are supplied.

[**SPECULATIVE**] Closure hypothesis: the effective saturated envelope density is fixed by the closure relation of Eq. (25).

[**DERIVED**] Consequence: once that closure is adopted, matter corresponds to configurations whose neutral circulation envelope reaches a structural normalization limit.

3.10 Logical Structure of the Axioms

The axioms form a dependency hierarchy:

$$\text{Continuum} \to \text{Vortex Structures} \to \text{Quantization} \to \text{Delay Dynamics} \to \text{Mass/Energy Emergence} \tag{69}$$

3.11 Minimality Principle

Statement. The axioms are minimal in the sense that removing any one of them destroys at least one of the following:

- existence of stable structures
- emergence of discrete spectra
- compatibility with known physics

3.12 Philosophical Interpretation

The axiomatic structure implies:

- Particles are not fundamental objects but stable dynamical configurations
- Discreteness is emergent, not postulated
- Time is relational, not absolute

3.13 Consistency Condition

A valid SST realization must satisfy:

- Compatibility with orthodox physics in appropriate limits
- Internal logical closure
- Explicit traceability from axioms to predictions

3.14 Canonical Statement

Swirl-String Theory is defined as the minimal continuum theory in which stable vortex structures, governed by quantized circulation and delay dynamics, give rise to discrete physical phenomena consistent with known physics.

4 Consistency and Classification Framework

4.1 Classification Scheme

All statements are labeled as:

- [**ORTHODOX**] — established physics
- [**DERIVED**] — logically deduced within SST
- [**SPECULATIVE**] — novel or unverified

4.2 Extended Labeling Convention

We distinguish between *epistemic status* and *editorial role.*

Epistemic status. The primary status labels used in the present Canon are:

- [**ORTHODOX**] — established in standard physics or mathematics
- [**DERIVED**] — obtained internally from the present SST assumptions, definitions, or closures, *without re-using the target quantity as an input*
- [**CALIBRATED**] — fixed by, or algebraically equivalent to, a measured constant (α, m_e, $\hbar$, R_∞, $\ldots$) that enters through the primitive set; numerically exact but not an independent prediction
- [**CONDITIONAL**] — follows only if a stated, not-yet-proven premise holds (e.g. an open lemma or selection rule)
- [**SPECULATIVE**] — conjectural SST extension, interpretation, or unverified identification

A [**CRITICAL NOTE**] editorial overlay flags a circularity risk, dimensional issue, or reviewer vulnerability on any of the above.

Editorial role. Additional descriptors such as *Definition*, *Constraint*, *Interpretation*, *Benchmark*, *Roadmap*, and *Dimensional check* are not independent epistemic classes. They indicate only the role played by the statement.

Usage rule. When both are needed, the Canon uses the format

$$\texttt{[STATUS] Role: statement.} \tag{70}$$

For example, one may write **[ORTHODOX] Constraint**, **[DERIVED] Consequence**, or **[SPECULATIVE] Interpretation**.

4.3 Orthodox Constraints

The following must remain intact:

- Quantum mechanics predictions
- Atomic spectroscopy
- Relativistic consistency

4.4 Internal Consistency Rules

- No circular definitions
- All primitives explicitly defined
- Derived quantities must follow from axioms

4.5 Conflict Resolution Strategy

Priority order:

1. Logical consistency
2. Experimental agreement
3. Canonical simplicity

4.6 Epistemic Status Tracking

Each major result must explicitly state:

- Assumptions
- Derivation status
- Empirical testability

In particular, a definitional choice must not be presented as a theorem, and an interpretive bridge must not be presented as an orthodox result unless the underlying mathematical structure is itself standard.

4.7 Canon Integrity Principle

A valid SST canon must be internally consistent, externally compatible, and explicitly stratified by epistemic status.

5 Geometric and Topological Sector

5.1 Role of geometry and topology in the canon

This sector collects those parts of SST in which the identity of a state depends not only on local field amplitudes but on the global organization of the carrier. Geometry enters through lengths, radii, and closure constraints; topology enters through linking, crossing, chirality, and other invariant features that survive smooth deformations. At canon level, the main role of this sector is selective rather than exhaustive: it provides the organizing variables that feed the topological kernel Ξ_K and the geometric gate Π_K in Eq. (42).

The canon therefore treats geometry and topology as follows:

1. geometry sets admissible scales and shielding ratios,
2. topology labels persistent identity classes,
3. only those quantities that survive smooth deformation are allowed to enter the long-lived state classification.

This is also why layer or dressing indices are not treated as primary identity labels: they may modify a state, but they do not replace the topology-first classification of the carrier itself.

5.2 Orthodox ropelength and thickness convention

[ORTHODOX / CANON CONVENTION]. For any tame closed centerline representative γ_K of a swirl-string topology class K, the radius-convention thickness is the reach of the embedded curve,

$$\mathrm{Thi}_{\mathrm{rad}}(\gamma_K) = \mathrm{reach}(\gamma_K) = \mathrm{NIR}(\gamma_K), \tag{71}$$

where NIR denotes the normal injectivity radius. The radius-convention ropelength is

$$\mathrm{Rop}_{\mathrm{rad}}(\gamma_K) = \frac{\mathrm{Len}(\gamma_K)}{\mathrm{Thi}_{\mathrm{rad}}(\gamma_K)}. \tag{72}$$

Some knot-theory literature uses a diameter convention. SST fixes the radius convention in Eq. (72); if a diameter convention is imported, it must be converted by

$$\mathrm{Rop}_{\mathrm{diam}} = \frac{1}{2}\,\mathrm{Rop}_{\mathrm{rad}}\,. \tag{73}$$

[ORTHODOX]. Ropelength minimizers exist for tame knot and link types. Such tight representatives are $C^{1,1}$ and have bounded curvature, but they need not be globally C^2 and need not be unique [45]. Polygonal constrained tightening represents the same finite-thickness problem by active struts, kinks, and a projected constrained gradient [46].

[CANON RULE]. The bare topological class K does not by itself determine a unique physical particle representative. Any mass, screening, or stability calculation that uses ropelength must state which representative γ_K, which radius convention, and which relaxation protocol are being used. The physical centerline length is written

$$L_{\mathrm{phys}}(K) = a_{\mathrm{core}}(K)\;\mathrm{Rop}_{\mathrm{rad}}(\gamma_K), \tag{74}$$

with $a_{\mathrm{core}}(K) = \mathrm{Thi}_{\mathrm{rad}}(\gamma_K)$ unless a separately derived resolved-tube radius is specified.

5.3 Derived thin-filament energy scale

[DERIVED / LEADING-ORDER FILAMENT LIMIT]. The canon may use a hydrodynamic thin-filament energy scale before any empirical topological screening functional is introduced. For a slender closed swirl tube with circulation Γ, physical centerline length $L_{\mathrm{phys}}(K)$, core cutoff r_c, and outer matching scale ℓ_{out}, the leading envelope contribution is

$$E^{(0)}_{\mathrm{fil}}[K] \simeq \frac{\rho_f \Gamma^2}{4\pi}\,L_{\mathrm{phys}}(K)\,\ln\!\left(\frac{\ell_{\mathrm{out}}}{r_c}\right) + T_{\mathrm{core}}\,L_{\mathrm{phys}}(K) + \cdots\,. \tag{75}$$

The first term is the usual slender-vortex logarithmic energy scale, while $T_{\rm core}$ records the resolved line-tension contribution of the core model.

[CANON / FREE-SYMBOL GUARD]. The line tension $T_{\rm core}$ has units $[T_{\rm core}] = \mathrm{J\,m^{-1}} = \mathrm{N}$ and is not fixed by Eq. (75). Any use of this symbol must identify its source: a resolved core variational model, a numerical relaxation scale, or an explicitly stated calibration. In the absence of such a source, $T_{\rm core}$ remains a symbolic line-tension placeholder and must not be fitted independently from the screening coefficients of Eq. (232).

[CANON GUARD]. Equation (75) is the derived length/logarithm anchor for later knot screening. The surrogate terms in Eq. (232) may be calibrated against this energy scale, but they are not themselves derived by merely being listed as topological descriptors.

5.4 Euler topological preservation and closure discipline

[ORTHODOX]. In a smooth incompressible inviscid Euler flow, with no boundaries, forcing, viscosity, singular core collapse, or phase-slip defects, vortex lines are materially transported by the flow. Consequently, linking and knot type of already closed vortex carriers are preserved. A closed unknot cannot become a trefoil by ordinary ideal-Euler evolution:

$$0_1 \not\longrightarrow 3_1 \qquad \text{under smooth ideal Euler flow.} \tag{76}$$

[DERIVED SST CONSTRAINT]. Trefoil creation must therefore not be described as reconnection of an already closed ideal-Euler vortex tube. It is admissible only as a boundary-supported closure, phase-slip, nucleation, or other non-Euler event occurring before the final carrier has acquired a protected closed knot type. After closure, smooth evolution may deform the embedding but must preserve the knot class.

[CANON RULE]. Topology protects existing closed carriers; it does not by itself provide an ideal-Euler mechanism for creating them.

5.5 Framed self-linking of the swirl string and the spinorial selection of the lepton ladder

This subsection records the framed-tube self-linking of a closed torus-knot swirl string and the boundary class that selects its physical winding. It is the first genuinely non-circular topological result in the canon, in deliberate contrast to the calibrated α-identities of the Formal Foundations (which re-express α rather than predict it).

[DERIVED] Torus-surface self-linking. Let $K = T(p,q)$ be a closed swirl string on the standard torus and let $U_{\rm tor}$ be the torus surface-normal framing. The framed self-linking is

$$SL_{\rm tor}(T(p,q)) \;=\; pq, \tag{77}$$

verified numerically to machine precision for $p \in \{2,3\}$ and $q = 3,\dots,11$ with *no* target injected. For the lepton family $T(2,q)$, $SL_{\rm tor} = 2q$.

[DEFINITIONAL] Physical-framing winding. For any single-valued physical framing $U_{\rm phys}$ of K (the direction of the local core swirl-velocity field $\mathbf{u}_\circlearrowleft$), the relative winding

$$\chi \;:=\; SL(K,U_{\rm phys}) - SL(K,U_{\rm tor}) \;=\; \frac{1}{2\pi}\oint_K \mathrm{d}[\angle_{U_{\rm tor}}(U_{\rm phys})] \;\in\; \mathbb{Z} \tag{78}$$

is an integer relative winding number, so $SL_{\rm phys}(T(p,q)) = pq + \chi$.

[CRITICAL NOTE] Any test that fixes the twist by $Tw = \text{target} - Wr$ and then confirms the target is circular: it passes for *any* integer target and proves only the Calugareanu identity $SL = Wr + Tw$. Only the target-free framing (77) and the selection argument below are admissible.

[DERIVED NEGATIVE] Ordinary $U(1)$ closure gives $\chi = 0$. With positive longitudinal phase stiffness $A_\parallel > 0$, the winding energy

$$E(\chi) \;=\; E_0 + \frac{2\pi^2 A_\parallel}{L}\,\chi^2 \tag{79}$$

is minimized over ordinary (periodic) $U(1)$ windings $\chi \in \mathbb{Z}$ at $\chi_{\rm ground} = 0$, hence $SL_{\rm phys} = pq$ (even). Pure framed-curve geometry with positive stiffness therefore yields a *boson* and cannot produce the lepton $pq + 1$ ladder.

[CONDITIONAL DERIVED] Spinorial anti-periodicity gives $\chi = \pm 1$. If the core phase is a section of a spinor (rather than a $U(1)$) bundle, then under one circuit of K the spin frame rotates by 2π and the phase is anti-periodic, so the admissible windings are $\chi \in 2\mathbb{Z}+1$. Minimizing (79) over this sector gives the degenerate ground winding

$$\chi_{\text{ground}} = \pm 1 \quad \Longrightarrow \quad SL_{\text{phys}}(T(p,q)) = pq \pm 1, \qquad SL_{\text{phys}}(T(2,q)) = 2q \pm 1, \tag{80}$$

the two signs being the matter and antimatter sectors. The selection is a *superselection* effect: the boson winding $\chi = 0$ is always energetically lower (by $2\pi^2 A_\parallel/L > 0$), so it is the boundary class, not the energetics, that renders the carrier fermionic. The result is independent of $(A_\parallel, L)$ for all positive values.

Why the scalar substrate is bosonic and $\theta = \pi$ is the fermionic $\mathbb{Z}_2$ choice. The two boundary classes are not arbitrary continuous tunings. In an $S^2 \simeq \mathbb{C}P^1$ order-parameter theory they can be identified with the two values of a discrete topological/statistical sector.

[DERIVED NEGATIVE] Scalar $U(1)$ substrate. A single-component order parameter $\psi = |\psi|e^{i\varphi}$ is single-valued only if

$$\oint \mathrm{d}\varphi \in 2\pi\mathbb{Z}, \tag{81}$$

so its scalar phase is periodic:

$$e^{i\oint \mathrm{d}\varphi} = +1 \tag{82}$$

on every closed loop. This condition allows integer windings, but it does not force the longitudinal framing class to be odd. With the positive stiffness energy (79),

$$E_\chi = E_0 + \frac{2\pi^2 A_\parallel}{L}\,\chi^2, \qquad A_\parallel > 0, \tag{83}$$

minimization over the ordinary scalar class $\chi \in \mathbb{Z}$ selects

$$\chi_{\text{ground}} = 0. \tag{84}$$

Hence a purely scalar SST substrate has no intrinsic mechanism for the odd core-phase sector required by fermions. In this operational sense the scalar substrate is bosonic: it admits the trivial periodic carrier and does not generate the $pq \pm 1$ lepton ladder.

[DERIVED / ALLOWED] $S^2 \simeq \mathbb{C}P^1$ admits a $\mathbb{Z}_2$ statistical sector. The matter target is not a scalar $U(1)$ phase but the director/Hopfion target

$$M_{\text{SST}} = S^2 \simeq \mathbb{C}P^1. \tag{85}$$

The relevant homotopy data are

$$\pi_3(S^2) = \mathbb{Z}, \qquad \pi_4(S^2) = \mathbb{Z}_2. \tag{86}$$

The first class gives the static Hopf charge,

$$Q_H = SL_{\text{phys}}, \tag{87}$$

while the second allows a $\mathbb{Z}_2$ topological/statistical phase for spacetime loops of soliton configurations [31]. More carefully, the fermionic sign is not simply a local scalar factor $e^{i\theta Q_H}$ in the static Hopf charge. It is the sign of the relevant $\mathbb{Z}_2$ spacetime class,

$$\exp(iS_\theta) = (-1)^\nu, \qquad \nu \in \mathbb{Z}_2, \tag{88}$$

whose representative may be read, for the relevant 2π rotation or exchange process, as the parity class of the Hopf sector. Thus the two possible statistical sectors are

$$\theta = 0: \quad \exp(iS_\theta) = +1 \quad \Rightarrow \quad \text{bosonic / periodic sector}, \tag{89}$$

$$\theta = \pi: \quad \exp(iS_\theta) = (-1)^\nu \quad \Rightarrow \quad \text{fermionic / spinorial sector for odd Hopf parity}. \tag{90}$$

Identifying the spinorial sector with the longitudinal core-phase boundary class gives

$$\theta = \pi \quad \Longleftrightarrow \quad \chi \in 2\mathbb{Z}+1. \tag{91}$$

Then positive stiffness no longer minimizes over all integers but over the odd class:

$$\chi \in 2\mathbb{Z}+1 \quad \Rightarrow \quad \chi_{\text{ground}} = \pm 1. \tag{92}$$

Since

$$Q_H = SL_{\text{phys}} = SL_{\text{tor}} + \chi = pq + \chi, \tag{93}$$

and $pq = 2q$ is even for the lepton family $T(2,q)$, the $\theta = \pi$ sector gives

$$SL_{\text{phys}}(T(2,q)) = 2q \pm 1, \tag{94}$$

i.e. the odd Hopf/self-linking carrier required for fermionic leptons.

[POSIT, pinned] $\theta = \pi$ is selected by observation, not derived. The framed-curve geometry and the quadratic stiffness energy do not prefer $\theta = \pi$ over $\theta = 0$. The value $\theta = \pi$ is therefore a discrete superselection input pinned by the empirical existence of fermions. It is naturally identified with the same $\mathbb{Z}_2$ datum appearing in half-quantum vortices of multi-component condensates [32, 33] and in the half/full circulation assignment of the magnetic-moment sector (Route B). With that identification, fermion statistics, the $pq \pm 1$ ladder, and the Route B origin of $g = 2$ are no longer independent posits but three faces of one $\mathbb{Z}_2$ sector.

[CRITICAL NOTE] The statement "$\pi_4(S^2) = \mathbb{Z}_2$ guarantees a fermionic Hopfion" is too strong. The homotopy group shows that a $\mathbb{Z}_2$ spacetime class exists, but hopfion statistics are set by the presence and value of the corresponding topological/statistical term. In modern language this classification can involve bordism/TQFT data rather than only bare homotopy groups [34]. What is solid here is the reduction:

$$\theta = 0 \Rightarrow \text{periodic/bosonic substrate}, \qquad \theta = \pi \Rightarrow \text{spinorial/fermionic substrate}. \tag{95}$$

The remaining open problem is to derive why the SST substrate realizes the $\theta = \pi$ sector rather than the $\theta = 0$ sector.

[OPEN] Derivation of the spinorial boundary condition. The spinorial anti-periodicity of the core phase is not yet derived from the swirl-field dynamics; it is the single remaining premise of the $pq + 1$ ladder.

[CONDITIONAL] Unification with the magnetic moment. The spinor premise above is the *same* posit as the spin-$\frac{1}{2}$ assignment of the magnetic-moment sector (Route B): one spinorial lift of the core phase yields both $g = 2$ and $\chi = \pm 1 \Rightarrow pq \pm 1$. If established, fermion statistics, the gyromagnetic ratio, and the lepton self-linking ladder collapse into a single mechanism.

[DERIVED] Falsifier. An independently computed $T(2,q)$ Hopfion self-linking with $Q_H \neq 2q \pm 1$ (e.g. $T(2,5) \neq 11$) falsifies the conditional chain irrespective of the variational argument.

5.6 Spectral Structure of Locked Closed Carriers (Secondary Result)

Scope and status. This subsection records a *secondary consequence* of the kinematics of rationally locked, closed carriers. It does not introduce new primitives. All results are conditional on the existence of a well-defined phase field and a regular mode spectrum. Status: **derived under spectral assumptions; experimentally unverified**.

Kinematic reduction. Consider a closed carrier with a single central driving phase $\phi(t)$ and N_s sector variables $\theta_i(t)$ $(i = 1, \ldots, N_s)$ subject to a rational locking relation

$$\theta_i = \sigma_i \kappa_i \phi + \delta_i, \qquad \sigma_i \in \{+1, -1\}, \quad \kappa_i \in \mathbb{Q}. \tag{96}$$

In the rigid-lock limit, the dynamics reduce to a single collective coordinate $\phi(t)$ with effective Lagrangian

$$\mathcal{L}_{\text{eff}} = \frac{I_{\text{eff}}}{2}\dot{\phi}^2 - V(\phi), \tag{97}$$

where I_{eff} is an effective inertia and $V(\phi)$ is a periodic locking potential.

Distributed phase field. On a closed carrier of length L, allow small phase fluctuations

$$\phi(t) \longrightarrow \phi_0 + \eta(s,t), \qquad s \in [0,L], \tag{98}$$

with periodic boundary condition

$$\eta(s+L,t) = \eta(s,t). \tag{99}$$

A minimal quadratic field model consistent with Eq. (97) is

$$\mathcal{L}_{\text{field}} = \int_0^L \left[\frac{\mu_{\text{eff}}}{2}(\partial_t\eta)^2 - \frac{T_{\text{eff}}}{2}(\partial_s\eta)^2 - \frac{\mu_{\text{eff}}\omega_0^2}{2}\eta^2 \right] ds, \tag{100}$$

where μ_{eff} is an effective phase inertia density, T_{eff} an effective stiffness, and ω_0 the small-oscillation frequency set by $V(\phi)$.

Dimensional consistency:

$$[\mu_{\text{eff}}] = \text{kg}\,\text{m}, \qquad [T_{\text{eff}}] = \text{N}, \qquad [\omega_0] = \text{s}^{-1}.$$

Mode spectrum. Expanding in normal modes

$$\eta(s,t) = \sum_{n\in\mathbb{Z}} a_n(t)\, e^{ik_n s}, \qquad k_n = \frac{2\pi n}{L}, \tag{101}$$

yields the dispersion relation

$$\boxed{\omega_n^2 = \omega_0^2 + c_\phi^2 k_n^2, \qquad c_\phi^2 = \frac{T_{\text{eff}}}{\mu_{\text{eff}}}.} \tag{102}$$

For large n,

$$\omega_n \sim \frac{2\pi c_\phi}{L}\, n. \tag{103}$$

Cubic spectral sums and $\zeta(3)$. Consider a cubic-weighted spectral contribution of the form

$$\Delta\mathcal{E}^{(3)} = A_3 \sum_{n=1}^{\infty} \frac{1}{\omega_n^3}, \qquad [A_3] = \text{J}\,\text{s}^{-3}. \tag{104}$$

Using the asymptotic scaling (103),

$$\Delta\mathcal{E}^{(3)} \;\approx\; A_3 \left(\frac{L}{2\pi c_\phi}\right)^3 \sum_{n=1}^{\infty} \frac{1}{n^3}. \tag{105}$$

Hence

$$\boxed{\Delta\mathcal{E}^{(3)} \;\approx\; A_3 \left(\frac{L}{2\pi c_\phi}\right)^3 \zeta(3).} \tag{106}$$

Interpretation. Equation (106) shows that $\zeta(3)$ arises from the *high-mode spectral tail* of a closed carrier, not from the geometric threefold structure alone. The role of the locking relation (96) is indirect: it fixes I_{eff}, ω_0, and admissible boundary conditions, thereby determining the spectral scale (L, c_ϕ).

Universality. Different geometric realizations (e.g., distinct multi-sector carriers) that reduce to the same class of effective phase dynamics and share the asymptotic linear spectrum (103) will produce the same $\zeta(3)$ factor in Eq. (106), up to system-dependent prefactors.

$$\boxed{\zeta(3) \text{ is a spectral invariant of the mode tower, not a direct invariant of the carrier geometry.}} \tag{107}$$

Limitations and falsifiers. The result fails if:

1. the mode spectrum is not asymptotically linear in n,
2. strong damping or nonlocal coupling invalidates the expansion (100),
3. the system does not admit a well-defined periodic phase field on a closed domain.

Minimal experimental test. Measure mode frequencies ω_n for $n = 1, 2, 3, \ldots$ and verify

$$\frac{\omega_n}{n} \to \text{constant} \quad \text{as } n \to \infty. \tag{108}$$

Deviation from this scaling excludes Eq. (106).

Status. Secondary derived result. Valid under spectral assumptions; not a defining postulate. The cubic weighting $\sum_n 1/\omega_n^3$ with $\omega_n \sim n$ yields $\zeta(3)$ for *any* asymptotically linear spectrum; this is a generic calculus property [CALIBRATED], not an SST-specific fluid-dynamical or topological invariant.

5.7 Symmetry-Sensitive Dark-Knot Classification

A compact classification retained from earlier SST topology layers distinguishes visible, quasi-dark, and dark sectors by chirality, helicity, and low-order instability content.

Let $\chi(K)$ denote a chirality indicator, let

$$\mathcal{H}[K] = \int \mathbf{v} \cdot (\nabla \times \mathbf{v})\, dV \tag{109}$$

be the helicity functional, and let $N_u^{(1)}(K)$ count low-order unstable modes in a chosen perturbative stability analysis. Then the retained taxonomy is

$$\text{dark:} \quad \chi(K) = 0, \quad \mathcal{H}[K] \approx 0, \quad N_u^{(1)}(K) = 0, \tag{110}$$

$$\text{quasi-dark:} \quad |\chi(K)| \ll 1, \quad \mathcal{H}[K] \approx 0, \quad 0 < N_u^{(1)}(K) \le 2, \tag{111}$$

$$\text{visible:} \quad \chi(K) \neq 0 \quad \text{or} \quad \mathcal{H}[K] \neq 0. \tag{112}$$

[ORTHODOX]: helicity is a standard invariant of ideal vortex dynamics [27, 6].

[SPECULATIVE]: the interpretation of Eq. (112) as a matter-sector taxonomy is an SST model choice.

Heuristic balance diagnostic from the SST helicity archive. As a supporting diagnostic (not a stand-alone classifier), define

$$\Delta_H(K) = \left| \widehat{\mathcal{H}}_b(K) + \frac{1}{2} \right|, \tag{113}$$

where $\widehat{\mathcal{H}}_b(K)$ is the archived normalized helicity-like base indicator for class K. Near-balanced values with $\Delta_H(K) \ll 1$ are treated only as candidate evidence for dark or quasi-dark sectors.

[CRITICAL NOTE]: in canon usage, the diagnostic above is subordinate to Eq. (112); no single scalar is sufficient for dark-sector membership without the symmetry and instability filters.

[CANON WORKFLOW NOTE]: if SST-labelled and parallel legacy-labelled helicity exports are numerically identical, only the SST-labelled archive is required for canon workflows.

Symmetry-first refinement (v0.8.x). In the taxonomy program, candidate sectors are labeled symmetry-first, then filtered by helicity balance and low-order instability content. Amphichiral knot types furnish the preferred *candidate* pool for the dark bin in symmetry-first tables; assigning "dark" still requires the conjunction of near-zero chirality indicators, helicity balance in the sense of Eq. (112), and vanishing low-order instability count. The implication "amphichiral ⇒ dark" is a classifier proposal tested against observables and archive diagnostics, not a closed theorem.

Symmetry-metadata guard (v0.8.19+). For canon workflows, the word *amphichiral* is not sufficiently granular by itself. Candidate dark-sector rows must record the symmetry metadata used to make the assignment:

$$\mathfrak{s}(K) = \big(A_\pm(K),\ D_2(r;K),\ I_{\text{strong}}(K),\ \chi(K), \Delta_H(K),\ N_u^{(1)}(K)\big),$$

where $A_\pm$ records positive/negative/full amphichirality status, $D_2(r;K)$ records the presence or absence of a declared dihedral rotational symmetry at scale r, and I_{strong} records whether a strong inversion is part of the adopted knot-table metadata. These entries are **metadata fields**, not new dynamics. The canonical classifier remains Eq. (112); symmetry metadata only prevents the dark bin from collapsing into the oversimplified rule "amphichiral equals dark."

Ambiguity tie-break. If rule conjunctions overlap, precedence is

$$\text{symmetry gate} \;\to\; \text{helicity balance gate} \;\to\; \text{stability gate}.$$

6 Swirl–Electromagnetic Bridge

6.1 Minimal transverse field sector

A conservative bridge from the fluid sector to a radiation sector is obtained by introducing a divergence-free transverse field $\mathbf{A}_{\text{eff}}$ with

$$\nabla \cdot \mathbf{A}_{\text{eff}} = 0, \tag{114}$$

and effective Lagrangian density

$$\mathcal{L}_{\text{rad}} = \frac{\rho_f}{2} \|\partial_t \mathbf{A}_{\text{eff}}\|^2 - \frac{K_\gamma}{2} \|\nabla \times \mathbf{A}_{\text{eff}}\|^2. \tag{115}$$

Varying Eq. (115) and using Eq. (114) gives

$$\frac{1}{c_\gamma^2} \partial_t^2 \mathbf{A}_{\text{eff}} - \nabla^2 \mathbf{A}_{\text{eff}} = 0, \qquad c_\gamma^2 = \frac{K_\gamma}{\rho_f}. \tag{116}$$

Define

$$\mathbf{E}_{\text{eff}} = -\partial_t \mathbf{A}_{\text{eff}}, \qquad \mathbf{B}_{\text{eff}} = \nabla \times \mathbf{A}_{\text{eff}}. \tag{117}$$

Then one immediately recovers the source-free identities

$$\nabla \cdot \mathbf{B}_{\text{eff}} = 0, \tag{118}$$
$$\nabla \times \mathbf{E}_{\text{eff}} = -\partial_t \mathbf{B}_{\text{eff}}. \tag{119}$$

[**ORTHODOX FORM**]: Eq. (116) and Eq. (119) are the standard transverse-wave and source-free Faraday identities in vector-potential form [16].

[**SPECULATIVE IDENTIFICATION**]: SST identifies $\mathbf{A}_{\text{eff}}$ with a coarse-grained torsional or director displacement field of the medium.

6.2 Photon / unknot sector

The minimal retained radiation picture is that an R-phase excitation is an unknotted, finite-duration torsional packet propagating in the field $\mathbf{A}_{\text{eff}}$. At the Rosetta level this corresponds to a source-free transverse wave packet with dispersion inherited from Eq. (116).

[**SPECULATIVE**]: handedness of the local torsional packet is taken to encode polarization, while additional azimuthal phase winding encodes orbital angular momentum.

[**CANON / SEMANTIC RULE**]: a photon-sector R-phase torsion packet is not a loose end of a vortex filament and does not yet carry a protected closed matter-knot invariant. It is a phase/torsion support on a swirl-characteristic. A spacetime geodesic is only the orthodox comparison language for the coarse-grained propagation path.

6.3 R-to-T closure threshold and conservation bookkeeping

[SPECULATIVE SST BRIDGE]. If electron-trefoil creation is retained, the conservative bookkeeping form is

$$\gamma_{\text{R}} + \mathcal{B}_{\text{atom}} \xrightarrow{\mathcal{K}_{\text{Kairos}}} e^-_{3_1} + \mathcal{R}, \tag{120}$$

where γ_{R} is an extended R-phase torsion packet, $\mathcal{B}_{\mathrm{atom}}$ is a local boundary or trapping condition, and $\mathcal{R}$ carries recoil, momentum balance, and any required topological or charge compensation. Equation (120) is not a pure Euler reconnection event.

The Compton anchoring gives the closure energy scale

$$\omega_c = \frac{v_{\circlearrowleft}^{*}}{r_c}, \qquad \hbar\omega_c = m_e c^2. \tag{121}$$

This scale may be used as a seed/closure threshold only in the presence of a boundary reservoir or pre-existing attachment support. Free creation of a matter carrier from radiation must respect charge, momentum, and topological compensation. A lone photon-to-lone-electron channel is therefore excluded in the canon-safe reading. The minimal free pair bookkeeping is instead

$$\gamma + \gamma \longrightarrow 3_1 + \overline{3_1}, \qquad E_{\mathrm{pair}} = 2m_e c^2, \tag{122}$$

or a field-assisted version

$$\gamma + \mathcal{F} \longrightarrow 3_1 + \overline{3_1} + \mathcal{F}'. \tag{123}$$

6.4 Swirl pressure law

For steady axisymmetric swirl, Euler radial balance gives

$$\frac{1}{\rho_f}\frac{dp_{\mathrm{swirl}}}{dr} = \frac{v_\theta(r)^2}{r}. \tag{124}$$

For a locally rigid rotation profile $v_\theta(r) = \Omega r$, Eq. (124) integrates to

$$p_{\mathrm{swirl}}(r) = p_0 + \frac{1}{2}\rho_f \Omega^2 r^2, \tag{125}$$

while for an exterior irrotational profile $v_\theta(r) = \Gamma/(2\pi r)$ one finds

$$p_{\mathrm{swirl}}(r) = p_\infty - \frac{\rho_f \Gamma^2}{8\pi^2 r^2}. \tag{126}$$

In a flat rotation curve with asymptotically constant azimuthal speed $v_\theta(r) \to v_0$ as r grows in a window where the radial balance (124) remains valid, integration yields the logarithmic tail

$$p_{\mathrm{swirl}}(r) = p_0 + \rho_f v_0^2 \ln\left(\frac{r}{r_0}\right), \tag{127}$$

for suitable reference radius r_0 and offset p_0 within that window.

[ORTHODOX] Assumption block: Eq. (124) is taken in the usual incompressible, inviscid, stationary, axisymmetric regime with azimuthal-dominant flow, so that radial drift, viscosity, and stratification corrections are subleading in the domain of validity.

[ORTHODOX] Dimensional check: $\left[\rho_f{}^{-1}\, dp/dr\right] = \mathrm{m\,s^{-2}} = \left[v_\theta^2/r\right]$.

[ORTHODOX]: Eq. (124) is the ordinary radial Euler balance for steady swirl [6, 23].

[DERIVED SST ROLE]: this pressure law is the canonical bridge between localized circulation and large-scale force analogies. Interpreting p_{swirl} as a dark-sector classifier or support layer is an SST model step, logically separate from the orthodox fluid derivation.

[DERIVED] Observables / falsifiers: (i) consistency of a pressure profile reconstructed from archived $v_\theta(r)$ with the radial gradient implied by (124); (ii) null test: systematic sign or scale mismatch of the reconstructed dp_{swirl}/dr relative to v_θ^2/r across validated radial windows excludes the applicability of the balance in those windows.

6.5 Lorentz-Type Force Density as Swirl Stress

Status labels. The vector identity used in this section is **[ORTHODOX]**. The definition of the swirl-force density is **[DERIVED]** as a hydrodynamic force-density structure. The numerical SST scales are **[CALIBRATED]** with respect to the canonical constants ρ_f, r_c, and $v_{\circlearrowleft}^{*}$. No microscopic electromagnetic element-force law is canonized here.

Hydrodynamic identity. For incompressible inviscid flow,

$$\nabla \cdot \mathbf{v} = 0, \qquad \boldsymbol{\omega} = \nabla \times \mathbf{v}.$$

The convective acceleration satisfies the exact identity

$$(\mathbf{v} \cdot \nabla)\mathbf{v} = \nabla\left(\frac{1}{2}|\mathbf{v}|^2\right) - \mathbf{v} \times \boldsymbol{\omega}.$$

Equivalently,

$$\boxed{\mathbf{v} \times \boldsymbol{\omega} = \nabla\left(\frac{1}{2}|\mathbf{v}|^2\right) - (\mathbf{v} \cdot \nabla)\mathbf{v}.}$$

This provides the hydrodynamic structure behind a Lorentz-type transverse force-density term.

Swirl-force density. SST defines the local swirl-force density

$$\boxed{\mathbf{f}_{\circlearrowleft} = \rho_f \mathbf{v} \times \boldsymbol{\omega}.}$$

Its dimensional consistency is

$$[\rho_f \mathbf{v} \times \boldsymbol{\omega}] = \mathrm{kg\,m^{-3}\,m\,s^{-1}\,s^{-1}} = \mathrm{N\,m^{-3}},$$

as required for a force density.

Bernoulli relation and sign convention. The scalar quantity

$$p_{\circlearrowleft} = \frac{1}{2}\rho_f |\mathbf{v}|^2$$

is the dynamic swirl-pressure scale. It should not be confused with the static pressure, which decreases in high-swirl regions in the usual Bernoulli sense. For stationary Euler flow with constant ρ_f and no external body force,

$$\rho_f(\mathbf{v} \cdot \nabla)\mathbf{v} = -\nabla p_{\mathrm{stat}}.$$

Using the identity above gives

$$\boxed{\rho_f \mathbf{v} \times \boldsymbol{\omega} = \nabla\left(p_{\mathrm{stat}} + \frac{1}{2}\rho_f |\mathbf{v}|^2\right).}$$

Thus the swirl-force density is canonically tied to the gradient of total Euler pressure, not to an independently postulated electromagnetic force law.

Closed-loop observables. SST assigns canonical status to integrated and topological observables rather than to a unique microscopic segment-to-segment force kernel. The canonical loop observables include circulation,

$$\Gamma = \oint_C \mathbf{v} \cdot d\boldsymbol{\ell},$$

helicity,

$$H = \int_V \mathbf{v} \cdot \boldsymbol{\omega}\, d^3x,$$

and the Gauss linking number,

$$Lk(C_1, C_2) = \frac{1}{4\pi} \oint_{C_1} \oint_{C_2} \frac{(d\mathbf{x}_1 \times d\mathbf{x}_2) \cdot (\mathbf{x}_1 - \mathbf{x}_2)}{|\mathbf{x}_1 - \mathbf{x}_2|^3}.$$

Distinct local force kernels may yield identical closed-loop observables. Therefore SST canonizes circulation, helicity, linking number, pressure differences, and net forces, but not a unique microscopic electromagnetic element-force law.

Canonical scales. Using the canonical SST constants

$$v_{\circlearrowleft}^{*} = 1.09384563 \times 10^{6}\,\mathrm{m\,s^{-1}}, \qquad r_c = 1.40897017 \times 10^{-15}\,\mathrm{m}, \qquad \rho_f = 7.0 \times 10^{-7}\,\mathrm{kg\,m^{-3}},$$

the natural circulation scale is

$$\Gamma_0 = 2\pi r_c v_{\circlearrowleft}^{*} = 9.68361920 \times 10^{-9}\,\mathrm{m^2\,s^{-1}}.$$

The associated dynamic swirl-pressure scale is

$$p_{\circlearrowleft,0} = \frac{1}{2}\rho_f v_{\circlearrowleft}^{*\,2} = 4.1877439 \times 10^{5}\,\mathrm{Pa}.$$

This value is the canonical SST stress scale for Lorentz-type force-density analogies and coincides with the calibrated source-pressure scale used in the swirl-pressure sector.

Canonical conclusion. The canonized result is not a replacement of Maxwell–Lorentz electrodynamics. The canonized result is the hydrodynamic statement that a Lorentz-type transverse force-density structure has the exact SST form

$$\boxed{\mathbf{f}_{\circlearrowleft} = \rho_f \mathbf{v} \times \boldsymbol{\omega} = \rho_f \nabla\left(\frac{1}{2}|\mathbf{v}|^2\right) - \rho_f (\mathbf{v}\cdot\nabla)\mathbf{v}.}$$

Accordingly, magnetic-type force densities in SST are interpreted as effective projections of pressure, vorticity, and closed-loop topology in the substrate.

6.6 Minimal effective Lagrangian bridge

A compact field-theoretic summary retaining the clock, radiation, and matter sectors is

$$\mathcal{L}_{\mathrm{eff}} = \mathcal{L}_T + \mathcal{L}_{\mathrm{rad}} + \sum_K \bar{\Psi}_K \big(i\gamma^\mu D_\mu - m_K^{(\mathrm{sol})}\big)\Psi_K + \cdots, \tag{128}$$

where $\mathcal{L}_T$ is the clock/foliation sector introduced in Section 10, $\mathcal{L}_{\mathrm{rad}}$ is given by Eq. (115), and the ellipsis denotes higher-order self-interaction, topological, or medium-response operators.

[**SPECULATIVE BRIDGE**]: Eq. (128) is a bookkeeping layer, not yet a completed first-principles derivation.

7 Delay and Mode Selection Theory

7.1 Circulation Delay as a Physical Primitive

We introduce the circulation delay as a constitutive element of the theory. For a closed vortex filament of effective length L and characteristic propagation speed $v_{\circlearrowleft}^{*}$, the circulation time is defined as:

$$\tau_{\mathrm{circ}}(K) = \frac{L_K}{v_{\circlearrowleft}^{*}} \tag{129}$$

In the special case of a minimal closed loop associated with the neutral circulation scale, this reduces to:

$$\tau_{\mathrm{circ,c}} = \frac{2\pi r_c}{v_{\circlearrowleft}^{*}} = \frac{2\pi}{\omega_c} \tag{130}$$

This establishes a direct identification between the circulation delay and the inverse Compton frequency.

[**DERIVED**]: The intrinsic delay of the neutral circulation loop is identical to the Compton period.

7.2 Delayed Self-Interaction Dynamics

We model the phase $\phi(t)$ of a circulating excitation as a delayed self-interacting oscillator:

$$\dot{\phi}(t) = \omega_0 + \kappa \sin\big(\phi(t - \tau_{\text{circ}}) - \phi(t)\big) \tag{131}$$

This equation represents a minimal effective description of a circulating field with finite propagation time.

[**ORTHODOX**]: Delay-differential equations of this type are well-established in feedback systems [1].

7.3 Emergence of Discrete Stable Modes

We seek steady-state solutions of the form:

$$\phi(t) = \Omega t \tag{132}$$

Substituting into Eq. 131 yields $\phi(t - \tau_{\text{circ}}) - \phi(t) = -\Omega\tau_{\text{circ}}$, hence:

$$\Omega = \omega_0 - \kappa \sin(\Omega\tau_{\text{circ}}) \tag{133}$$

This transcendental equation admits a discrete set of stable solutions Ω_n determined by the delay parameter τ_{circ}. The sign convention is not physically unique—one may absorb it into κ or reverse the delay argument—but the displayed form is the direct consequence of Eq. (131) as written.

[**CONDITIONAL DERIVED**]: Given the effective DDE ansatz of Eq. (131), the delayed feedback equation admits discrete locked branches. The claim that the SST medium obeys this specific DDE remains a bridge assumption, not a first-principles Euler theorem.

7.4 Delay-Induced Quantization Mechanism

Unlike conventional quantum mechanics, where discreteness is postulated, here it emerges dynamically:

- No boundary condition required
- No operator quantization assumed
- Discreteness arises from stability selection

[**SPECULATIVE**]: Particle spectra may be interpreted as delay-selected stable circulation modes.

7.5 Extension to Nonlinear Schrödinger Dynamics

To connect with field-theoretic descriptions, we extend the delay mechanism to a nonlinear Schrödinger framework:

$$i\hbar\partial_t\psi = -\frac{\hbar^2}{2m}\nabla^2\psi + F_{\text{delay}}[\psi(t - \tau_{\text{circ}})] \tag{134}$$

where the delayed functional introduces temporal nonlocality into the field evolution.

[**SPECULATIVE**]: This provides a bridge between classical delay systems and quantum wave dynamics.

7.6 Physical Interpretation

The delay τ_{circ} acts as:

- a memory kernel
- a nonlocal temporal coupling

- a selection mechanism for stable modes

Thus:

$$\textit{Delay} \Rightarrow \textit{Stability} \Rightarrow \textit{Discreteness}$$

This mechanism forms one of the central pillars of the SST canon.

8 Atomic Bridge Model

8.1 Introduction and epistemic framing

This section records the minimal SST atomic bridge rather than a full derivation of atomic structure. Its canon role is deliberately narrow: recover the orthodox Coulomb limit when the SST-specific sector is switched off, define a branch-resolved circulation coupling fixed by canonical medium scales, and make explicit which many-electron and shell-structure claims remain open. It is therefore a benchmark layer, not yet a complete atomic theory.

8.2 Baseline Hamiltonian

We adopt the standard Coulomb Hamiltonian as the orthodox baseline:

$$H_0 = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} \tag{135}$$

[**ORTHODOX**]: This reproduces the hydrogenic spectrum.

8.3 Legacy Soft-Core Regularization

Early canon layers introduced a finite-core regularization of the hydrogenic potential. In the present document this is retained only as a controlled bridge ansatz, not as a replacement for Eq. (135).

Define the effective coupling scale

$$\Lambda_{\mathrm{SST}} = 4\pi\,\rho_{\mathrm{core}}\,{v_{\circlearrowleft}^{*}}^{2}\,{r_c}^{4} \tag{136}$$

with dimensions

$$[\Lambda_{\mathrm{SST}}] = \mathrm{J\,m} = \mathrm{N\,m}^2. \tag{137}$$

A finite-core bridge potential is then

$$V_{\mathrm{soft}}(r) = -\frac{\Lambda_{\mathrm{SST}}}{\sqrt{r^2 + r_c^2}} \tag{138}$$

satisfying the large-radius limit

$$V_{\mathrm{soft}}(r) \xrightarrow{r \gg r_c} -\frac{\Lambda_{\mathrm{SST}}}{r}. \tag{139}$$

[**ORTHODOX**] Effective-model input: finite-core regularizations of singular central potentials are standard tools.
[**DERIVED**] Canonical consequence: the SST scale in Eq. (136) is fixed by canonical constants rather than introduced phenomenologically.
[**ORTHODOX**] **Constraint**: this sector is admissible only if the large-radius limit reproduces the standard Coulomb problem to spectroscopic accuracy.

In particular, Eq. (138) is retained as a bridge ansatz only. It is not promoted here to the primary atomic potential.

8.4 SST Circulation Correction

We introduce a circulation-dependent perturbation:

$$\delta H^{(\sigma)} = -2\sigma E_R \eta_0 \gamma + \frac{E_R \eta_0 \sigma}{2} \left\{ \frac{\hat{L}_z}{\hbar}, \gamma \right\} \tag{140}$$

[BRIDGE ANSATZ]: This is an orthodox-operator perturbation template whose coefficient depends on the SST circulation amplitude $\gamma(r,t)$. Because $\hat{L}_z$ is imported from the standard quantum Hamiltonian, Eq. (140) is not a first-principles derivation of quantum angular momentum from the substrate.

[CRITICAL NOTE]: The fractional shift scales as $\delta E_n/E_n \sim \eta_0\gamma = (\alpha/2)\gamma$. With $\gamma = \mathcal{O}(1)$ this is $\mathcal{O}(\alpha)$, far larger than the orthodox fine-structure splitting $\mathcal{O}(\alpha^2)$ (fractional). Compatibility with the atomic spectroscopy bounds (Sec. 7) therefore requires γ to be α-suppressed; this suppression is not yet derived.

8.5 Circulation source chain

The atomic perturbation is not introduced as an arbitrary extra scalar, but is intended to descend from a circulation-bearing source chain,

$$\xi \to \omega_z \to u_\theta \to \Gamma_{\mathrm{loc}} \to \gamma, \tag{141}$$

where an axial displacement field ξ generates local vorticity, the vorticity sources azimuthal swirl, and the loop circulation is normalized by the canonical quantum Γ_0. This makes the circulation amplitude γ the natural canon-level coupling variable rather than a free phenomenological insertion.

8.6 Branch Structure

Two chiral sectors exist:

$$\sigma = \pm 1 \tag{142}$$

These correspond to:

- ↺ (left-handed)
- ↻ (right-handed)

8.7 Electron trefoil identity under ordinary atomic binding

[DERIVED SST CONSTRAINT]. Ordinary propagation through vacuum and ordinary atomic binding do not change the electron carrier's knot class. In the trefoil assignment, the transition is

$$3_1^{\mathrm{free}} \longrightarrow 3_1^{\mathrm{bound}}, \tag{143}$$

not a breaking or unknotting of the carrier. Binding changes the embedding, phase envelope, branch coupling, and dressing of the carrier:

$$e^-_{\mathrm{atom}} = 3_1^{\mathrm{core}} \otimes \Psi^{\mathrm{envelope}}_{n\ell m\sigma}. \tag{144}$$

[CANON RULE]. Ordinary electron capture by an ion is a binding and phase-locking process. Destruction or conversion of the separate electron trefoil requires a non-Euler process such as annihilation with an anti-trefoil or weak-sector capture, with the necessary compensation carried by the final state.

8.8 Consistency Requirement

To remain compatible with atomic physics:

$$|\delta E_n| \ll |E_n| \tag{145}$$

[ORTHODOX] Constraint: SST corrections must not violate known spectroscopy.

8.9 Interpretation

[**STATUS**]: The atomic bridge is a controlled benchmark sector. The orthodox Coulomb problem is imported as the baseline; the circulation-dependent term is the SST addition; shell counting, exclusion-like behavior, and heavy-atom extensions remain programmatic rather than derived here. The purpose of the section is therefore structural compatibility, not yet a complete replacement of atomic quantum mechanics.

9 Spectroscopic Constraints

9.1 Motivation

Any extension of atomic structure must remain consistent with high-precision spectroscopic measurements.

[**ORTHODOX**]: Atomic energy levels are experimentally verified to extremely high precision.

Requirement:

$$|\delta E_n| \ll |E_n| \tag{146}$$

where δE_n represents any SST-induced correction.

9.2 Internal Mode Hypothesis

We consider the possibility that atomic bound states possess an additional internal excitation sector associated with vortex filament dynamics.

[**SPECULATIVE**]: Each bound state carries an internal mode spectrum $\{E_{m,n}\}$.

The effective energy becomes:

$$E_n^{\text{eff}} = E_n^{(0)} + \delta E_n \tag{147}$$

9.3 Generic Coupling Structure

We parameterize the coupling between internal modes and observable energy levels:

$$|\delta E_n| \le \lambda_n \sum_m E_{m,n}\, \nu_{m,n} \tag{148}$$

where:

- λ_n is the coupling efficiency
- $\nu_{m,n}$ is the occupation factor

[**DERIVED**]: This form captures a general upper bound independent of microscopic details.

9.4 Ungapped Spectrum: Instability

If the internal spectrum is ungapped:

- arbitrarily low-energy excitations exist
- no universal suppression mechanism applies

Consequence:

$$\delta E_n \not\to 0 \quad \text{generically} \tag{149}$$

[**CRITICAL**]: An ungapped internal sector is generically incompatible with spectroscopy unless:

- $\lambda_n \ll 1$ (extremely weak coupling), or
- the density of states is highly suppressed

9.5 Gapped Spectrum: Decoupling Mechanism

We introduce an excitation gap Δ_K:

$$E_{m,n} \geq \Delta_K \tag{150}$$

Then occupation factors follow Boltzmann suppression:

$$\nu_{m,n} \sim \exp\left(-\frac{\Delta_K}{k_B T_{\text{eff}}}\right) \tag{151}$$

Substituting into Eq. 148:

$$\delta E_n \propto \exp\left(-\frac{\Delta_K}{k_B T_{\text{eff}}}\right) \tag{152}$$

[**DERIVED**]: Exponential suppression ensures compatibility with spectroscopy.

9.6 Physical Origin of the Gap

[**SPECULATIVE**]: The gap Δ_K may arise from:

- geometric constraints on filament curvature
- torsional rigidity
- topological restrictions
- core structure of the vortex

9.7 Order-of-Magnitude Estimate

Using filament parameters:

$$\Delta_K \sim \frac{\Gamma^2}{4\pi\xi L} \tag{153}$$

where:

- ξ is the core scale
- L is the loop length

For typical SST scales:

$$\Delta_K = \mathcal{O}(10^2 - 10^3\,\text{eV}) \tag{154}$$

[**DERIVED**]: This places the internal sector well above atomic energy scales ($\sim$ eV).

9.8 Consistency Condition

To preserve atomic physics:

$$\frac{\Delta_K}{k_B T_{\text{eff}}} \gg 1 \tag{155}$$

Interpretation: The internal sector is effectively frozen at atomic energies.

9.9 Observable Consequences

[**SPECULATIVE**]: Residual effects may include:

- tiny level shifts
- suppressed line broadening
- weak temperature dependence

9.10 Falsifiability

The theory is falsifiable via:

- detection of unexplained spectral shifts
- absence of expected suppression
- deviation from Boltzmann scaling

9.11 Integration with SST Framework

The spectroscopic constraint enforces:

- compatibility with the Atomic Bridge (Section 8)
- constraints on delay-induced excitations (Section 7)
- bounds on allowable vortex configurations

9.12 Canonical Statement

Any viable SST extension must include a mechanism—such as an excitation gap—that ensures exponential suppression of internal modes at atomic energy scales.

9.13 Precision electroweak bridge: lessons from the CMS W-mass measurement

Status. [**ORTHODOX**] External input: the CMS measurement is a precision constraint on any SST effective weak-sector realization. [**DERIVED**] Methodological consequence: the result supports a forward-modelling strategy for SST phenomenology. [**CRITICAL NOTE**] The paper does *not* provide direct evidence for a structured-substrate ontology of the W boson.

Empirical anchor. Let the experimentally inferred resonance mass be denoted by

$$m_W^{\text{exp}} = 80.3602 \pm 0.0099 \text{ GeV}. \tag{156}$$

In the CMS analysis this quantity is not read off from a single observable, but extracted from a profiled likelihood over finely binned charged-lepton kinematics. This motivates the following SST principle:

> *Collider observables constrain SST through forward maps from latent swirl-string states to detector-level distributions, not through direct identification of a single internal parameter with a measured mass.*

Latent-to-observable map. Introduce a latent SST parameter set for an effective spin-1 weak excitation,

$$\Theta_W^{\text{SST}} = \Big(K_W,\, \chi_W,\, \mathcal{L}_W,\, \mathcal{H}_W,\, {S_{(t)}^{\circlearrowleft}}_W,\, {\rho_f}^{\text{eff}},\, \mathbf{u}_{\circlearrowleft}^{\text{eff}}\Big), \tag{157}$$

where:

- K_W is the topology class or excitation family,
- χ_W is chirality / handedness label,
- $\mathcal{L}_W$ is an effective geometric length scale,

- $\mathcal{H}_W$ is a helicity/linkage content,
- $S^{\circlearrowleft}_{(t)\,W}$ is the local clock factor for the mode,
- $\rho_f{}^{\mathrm{eff}}$ and $\mathbf{u}_{\circlearrowleft}^{\mathrm{eff}}$ are the coarse-grained medium density and effective local swirl-velocity field seen by the excitation.

The experimentally accessible distribution is then not a bare function of m_W alone, but a projected density

$$\mathcal{P}_{\mathrm{meas}}(x) = \int dz\; R(x|z)\,\mathcal{P}_{\mathrm{prod}}(z \mid \Theta_W^{\mathrm{SST}}), \tag{158}$$

where z denotes truth-level production/decay variables and $R(x|z)$ is the detector response kernel.

Mass as an effective resonance parameter. In SST one should distinguish:

$$m_W^{\mathrm{internal}} \neq m_W^{\mathrm{eff}} \tag{159}$$

in general, where:

- m_W^{internal} is the internal excitation-energy scale of the structured mode,
- m_W^{eff} is the resonance parameter inferred from scattering distributions.

The experimentally constrained quantity is m_W^{eff}.

A minimal canon-safe matching condition is therefore

$$m_W^{\mathrm{eff}}(\Theta_W^{\mathrm{SST}}) = m_W^{\mathrm{exp}} + \delta_W, \qquad |\delta_W| \lesssim 10~\mathrm{MeV}, \tag{160}$$

with the current precision scale set by experiment.

Distribution-level fit target. The correct SST comparison object is a likelihood built from binned observables:

$$\mathcal{L}\big(\Theta_W^{\mathrm{SST}}, \nu\big) = \prod_{b=1}^{N_{\mathrm{bins}}} \mathrm{Poisson}\big(n_b \,\big|\, \mu_b(\Theta_W^{\mathrm{SST}}, \nu)\big)\; \Pi(\nu), \tag{161}$$

where:

- n_b is the observed count in bin b,
- μ_b is the SST prediction after production, decay, and detector folding,
- ν denotes nuisance parameters,
- $\Pi(\nu)$ is the nuisance prior / constraint factor.

The profile likelihood for the physically interesting SST parameters is then

$$\mathcal{L}_{\mathrm{prof}}(\Theta_W^{\mathrm{SST}}) = \max_{\nu}\, \mathcal{L}(\Theta_W^{\mathrm{SST}}, \nu). \tag{162}$$

Helicity decomposition as a falsifier. A particularly sharp SST test is not the central value of m_W alone, but whether the effective spin-1 mode reproduces the standard angular basis. Write

$$\frac{d\sigma}{d\Omega} = \sum_{i=0}^{8} A_i\, f_i(\theta, \phi) + \epsilon_{\mathrm{SST}}\, \Delta(\theta, \phi), \tag{163}$$

where the $A_i f_i$ are the standard helicity structures and $\epsilon_{\mathrm{SST}}\Delta$ is the first SST correction. Then:

- if $\epsilon_{\mathrm{SST}} = 0$ within experimental precision, SST survives this channel,
- if $\epsilon_{\mathrm{SST}} \neq 0$, the deviation is directly falsifiable through angular distributions.

Canon interpretation. This leads to the following conservative reading:

1. The weak sector must be *SM-like at the effective distribution level.*
2. Large $\mathcal{O}(10^{-4})$–$\mathcal{O}(10^{-3})$ relative distortions in the effective W mass are disfavoured.
3. The best SST discovery channels are therefore likely to be

$$\text{small angular deviations,} \quad \text{polarization asymmetries,} \quad \text{off-shell tails,} \quad \text{rare differential distortions,} \tag{164}$$

rather than a large shift in the central value of m_W.

Minimal bridge claim. The CMS result does *not* force the W boson to be fundamental in the ontological sense. It does force any SST realization of the weak sector to satisfy the stronger requirement:

$$\mathcal{P}^{\rm SST}_{\rm prod}(z \mid \Theta^{\rm SST}_W) \approx \mathcal{P}^{\rm SM}_{\rm prod}(z \mid m_W, \text{SM nuisances}) \tag{165}$$

to current experimental precision after detector folding and nuisance profiling.

Practical SST program. A precision-compatible SST weak-sector program should therefore proceed in three layers:

$$\text{(i) effective resonance matching:} \quad m^{\rm eff}_W \to m^{\rm exp}_W, \tag{166}$$

$$\text{(ii) helicity matching:} \quad A^{\rm SST}_i \to A^{\rm SM}_i, \tag{167}$$

$$\text{(iii) residual search:} \quad \Delta \neq 0 \text{ only in subleading differential structure.} \tag{168}$$

Conclusion. The CMS W-mass result is best understood in SST as a *precision filter*: it does not supply ontology, but it strongly constrains phenomenology. Any viable SST weak-sector realization must be distributionally SM-like to present precision, while leaving room only for small, structured residuals in higher-order observables.

10 Relational Time Framework

10.1 Canonical Time Ontology

Status. [CANON / SEMANTIC ORGANIZATION]. This subsection fixes the time-language of SST. It does not introduce a new dynamical postulate. Its purpose is to prevent the absolute foliation parameter, laboratory time, local proper time, carrier loop time, the Swirl-Clock factor, and Kairos transition events from being conflated.

No bare-τ convention. Because SST uses both relativistic local clock accumulation and finite circulation delay, the unqualified symbol τ is avoided in mixed contexts. Circulation delay is denoted by $\tau_{\rm circ}$, while local proper time carried by a localized process A is denoted by τ_A or $\tau_{\rm loc}$.

Absolute foliation time. The symbol T denotes the global foliation parameter of the incompressible SST substrate. Each value of T labels a constraint slice

$$\Sigma_T. \tag{169}$$

The pressure field and incompressibility constraint are reconstructed on Σ_T. The parameter T is bookkeeping time for the medium foliation; it is not by itself a local clock reading. This foliation T is distinct from the coarse-grained clock field $T(x)$ introduced later in relational observables from conserved currents.

External comparison time. The symbol t_∞ denotes asymptotic laboratory time or external comparison time. It is the time coordinate used when matching SST clock predictions to standard relativistic observables. In weakly perturbed laboratory settings one may identify t_∞ with T, but the distinction should be retained whenever local clock rates are being discussed.

Swirl-Clock factor. The local Swirl-Clock factor is the dimensionless rate-conversion field already defined in Eq. (16):

$$S_{(t)}^{\circlearrowleft}(x,T)=\sqrt{1-\frac{\|\mathbf{u}_{\circlearrowleft}(x,T)\|^2}{c^2}}. \tag{170}$$

It is not a duration. It converts the foliation rate into the local proper-clock rate of a resolved carrier or process.

Local proper time. The symbol τ_A (or τ_{loc}) denotes local proper clock time. Along a localized carrier A,

$$d\tau_A=S_{(t)}^{\circlearrowleft}[A;T]\,dT. \tag{171}$$

Thus two localized carriers A and B may share the same foliation slice Σ_T while accumulating different proper times:

$$\frac{d\tau_A}{dT}=S_{(t)}^{\circlearrowleft}[A;T],\qquad \frac{d\tau_B}{dT}=S_{(t)}^{\circlearrowleft}[B;T],\qquad \frac{d\tau_C}{dT}=S_{(t)}^{\circlearrowleft}[C;T]. \tag{172}$$

Proper loop or turnover time. The symbol τ_{circ} denotes a carrier-internal circulation or delay time. For a loop of resolved length L_K,

$$\tau_{\mathrm{circ}}(K)=\frac{L_K}{v_{\circlearrowleft}^{*}}. \tag{173}$$

At the canonical core scale this reduces to

$$\tau_{\mathrm{circ,c}}=\frac{2\pi r_c}{v_{\circlearrowleft}^{*}}=\frac{2\pi}{\omega_c}. \tag{174}$$

The turnover time τ_{circ} counts internal circulation cycles; it is therefore distinct from the dimensionless Swirl-Clock factor $S_{(t)}^{\circlearrowleft}$.

Kairos transition event. A Kairos event is not a continuous clock variable. It is a discrete topological transition marker,

$$\mathcal{K}_{\mathrm{Kairos}}:K_-\longrightarrow K_+, \tag{175}$$

used when a process changes topological sector, chirality class, closure state, or reconnection class. It may occur at some foliation value T, but it is not itself a time coordinate.

Summary schema.

Layer	Symbol	Role
Foliation slice	Σ_T	global incompressible constraint slice
Absolute foliation time	T	ordering parameter of the substrate
External comparison time	t_∞	laboratory/asymptotic comparison time
Swirl-Clock factor	$S_{(t)}^{\circlearrowleft}$	dimensionless local rate conversion
Local proper time	τ_A, τ_{loc}	accumulated local clock time
Loop/circulation delay	τ_{circ}	internal circulation period
Kairos event	$\mathcal{K}_{\mathrm{Kairos}}$	discrete topological transition

Units guard. $S_{(t)}^{\circlearrowleft}$ is dimensionless; T, t_∞, τ_A, and τ_{circ} have units of time; $\mathcal{K}_{\mathrm{Kairos}}$ is an event, not a coordinate.

10.2 Swirl-Clock and local proper time

Section 10.1 fixes the time-language of SST. The canonical time variable is not introduced as an independent primitive. It is defined relationally from the local kinematic state of the medium through the Swirl-Clock already introduced in Eq. (16). In its coarse-grained form,

$$d\tau_{\rm loc} = S_{(t)}^{\circlearrowleft}(x)\, dt_\infty, \tag{176}$$

consistent with Eq. (171). Here t_∞ denotes the asymptotic laboratory time and $\tau_{\rm loc}$ the effective proper time associated with a localized process in the medium. In SST, this factor is interpreted as a fluid-kinematic clock field rather than as a purely geometric spacetime postulate.

No-signalling research-track note. The pressure field $p_{\circlearrowleft}(\mathbf{x}, T)$ is a constraint field on Σ_T. Its elliptic reconstruction may be instantaneous in the T-bookkeeping sense, but this does not by itself define an operational superluminal signal channel. Observable clock and photon phases are read through the local Swirl–Clock projection $d\tau_A/dT = S_{(t)}^{\circlearrowleft}[A;T]$. The no-signalling status of the pressure constraint is a research-track theorem target, not an assumed device capability.

10.3 Carrier-local clocking and the turnover scale

For localized carriers the natural microscopic rate is the turnover frequency

$$\Omega_0 = \frac{v_{\circlearrowleft}^{*}}{r_c} = \omega_c, \tag{177}$$

which coincides with the Compton anchoring scale and satisfies $\tau_{\rm circ,c} = 2\pi/\omega_c$ (Eq. (130)). A complete SST clock description therefore involves two linked but distinct levels:

- the *microscopic turnover scale* Ω_0, which counts circulation cycles of the carrier,
- the *coarse-grained Swirl-Clock* $S_{(t)}$, which relates local rates to asymptotic time.

This resolves a recurring ambiguity in earlier canon versions: local cycle counting and coarse-grained time dilation are related, but they are not identical observables.

10.4 Relational observables from conserved currents

A purely operator-based notion of time is avoided in SST by defining time observables through conserved currents. Let J^μ denote a matter current and let $j^\mu_{\rm ev}$ denote a coarse-grained event current associated with structured activity of the medium. The relational clock field $T(x)$ is then recovered only after coarse graining,

$$\partial_\mu T(x) \approx \frac{1}{\nu_0} \langle j_\mu^{\rm ev} \rangle_\ell, \tag{178}$$

where ℓ is the coarse-graining scale and ν_0 a reference event frequency. The canon interpretation is that time is read from correlations between currents, not from an abstract external operator.

10.5 Time-of-arrival as a relational field observable

Let $\mathcal{W}$ be a detector world-tube with boundary $\Sigma = \partial\mathcal{W}$. Then the arrival-time distribution associated with a detector label Θ is written as

$$p(\Theta) = \frac{1}{\mathcal{N}} \int_\Sigma d\sigma_\mu\, J^\mu(x)\, \delta\big(T(x) - \Theta\big). \tag{179}$$

This formulation is canonically important because it bypasses the need for a self-adjoint time operator conjugate to a bounded Hamiltonian. The measured time is instead the value of a physical clock field at the point where matter flux crosses the detector boundary.

10.6 Clock broadening and continuum limits

If the event current has a finite variance, then the observed time-of-arrival distribution is broadened relative to the ideal classical prediction. Writing $P_{\rm cl}$ for the classical arrival distribution and σ_τ^2 for the effective clock variance, one expects a convolution structure of the form

$$P_{\rm obs}(\Theta) = \int dt\, P_{\rm cl}(t) \frac{1}{\sqrt{2\pi\sigma_\tau^2}} \exp\left[-\frac{(\Theta - t)^2}{2\sigma_\tau^2}\right]. \tag{180}$$

In the limit $\ell \gg r_c$, the event structure is smoothed and one recovers a continuum clock field. In the opposite limit $\ell \to r_c$, the underlying discreteness of carrier events becomes operationally relevant.

10.7 Connection to gravity and geometry

In the canon, the relational-time sector and the geometric-response sector are not independent. Variations in the Swirl-Clock track variations in local swirl energy density and therefore provide the time component of the effective geometric response of the medium. At this level, the SST reading is:

gravity and time are two coarse-grained aspects of the same clock-bearing medium.

This statement should be read as a canon-level organizing principle, not yet as a completed field-theoretic proof.

10.8 Clock Field and Foliation Variable

[DERIVED] EFT bridge: the scalar Swirl-Clock field is promoted from the kinematic relation of Eq. (16) to a long-wavelength foliation variable

$$\chi(x), \tag{181}$$

whose normalized gradient defines a preferred local clock frame:

$$u_\mu = \frac{\nabla_\mu \chi}{\sqrt{g^{\alpha\beta} \nabla_\alpha \chi\, \nabla_\beta \chi}}. \tag{182}$$

[ORTHODOX] Equation (182) matches the standard hypersurface-orthogonal foliation construction used in khronometric / Einstein–Æther effective field theory [7, 8].

[DERIVED] Within SST, the interpretation is different: u_μ is not taken as a fundamental spacetime structure, but as the coarse-grained clock-flow direction associated with the medium.

10.9 Clock-Sector Effective Action

[ORTHODOX] EFT template: the most general second-order covariant clock-sector action compatible with hypersurface orthogonality is

$$\begin{aligned} S_\chi = \int d^4x \sqrt{-g}\, \Big[& c_1 (\nabla_\mu u_\nu)(\nabla^\mu u^\nu) + c_2 (\nabla_\mu u^\mu)^2 \\ & + c_3 (\nabla_\mu u_\nu)(\nabla^\nu u^\mu) + c_4 u^\mu u^\nu (\nabla_\mu u_\alpha)(\nabla_\nu u^\alpha) \Big]. \end{aligned} \tag{183}$$

[SPECULATIVE] SST interpretation: Equation (183) is not itself derived from first-principles filament dynamics in the present Canon. It is recorded as the minimal orthodox EFT bridge into a relativistic clock-field description.

10.10 Poisson Limit and Effective Gravity Bridge

[DERIVED / BRIDGE NORMALIZATION] In the weak-field, slowly varying limit, the dimensionless or time-like foliation variable χ must first be mapped to a dimensionful Newtonian-potential proxy before a Poisson equation is written. Define

$$\Phi_\chi(x) := \mathcal{N}_\chi\, \delta\chi(x), \qquad [\Phi_\chi] = \mathrm{m}^2\,\mathrm{s}^{-2}, \tag{184}$$

where $\mathcal{N}_\chi$ is the coarse-graining normalization that converts the clock-field perturbation into potential units. The admissible Poisson-type closure is then

$$\nabla^2 \Phi_\chi(x) = 4\pi G_{\rm eff}\, \rho_{\rm matter}(x), \tag{185}$$

so that outside localized sources

$$\Phi_\chi(r) \propto \frac{1}{r}, \qquad \mathbf{g}_{\rm eff} = -\nabla \Phi_\chi. \tag{186}$$

[ORTHODOX] The $1/r$ scalar potential and $1/r^2$ force law are standard consequences of Eq. (185) once Φ_χ, not bare χ, is the potential variable.

[SPECULATIVE] The identification of Φ_χ with a coarse-grained Swirl-Clock perturbation and the interpretation of Eq. (185) as an emergent gravity law are SST-specific.

10.11 Observational Constraint from Tensor-Mode Speed

[ORTHODOX] Constraint: If the clock-sector EFT is used, multimessenger gravitational-wave observations require [9]

$$c_{13} \equiv c_1 + c_3 = 0, \tag{187}$$

to enforce luminal tensor-mode propagation at observational precision.

[ORTHODOX] Constraint: any SST realization employing the EFT bridge of Eq. (183) must preserve Eq. (187).

10.12 Internal Tensor-Speed Naturalness

[DERIVED, conditional] In the clock-field / Einstein–Æther comparison layer, the spin-2 tensor-mode speed is conventionally written as

$$c_T^2 = \frac{c^2}{1 - c_{13}}, \qquad c_{13} := c_1 + c_3. \tag{188}$$

Multimessenger observations require $c_T = c$ to high precision and therefore impose $c_{13} \simeq 0$ observationally [9, 42]. In SST there is a stronger conditional interpretation: if the metric tensor mode of the coarse-grained clock-field sector is identified with the transverse excitation mode of the underlying medium, and if c is the canonical transverse propagation speed of that medium, then

$$c_T = c \quad \Longrightarrow \quad c_{13} = 0. \tag{189}$$

Status. Equation (189) is not an independent derivation of the Einstein–Hilbert dynamics. It is a conditional mode-identification result inside the orthodox Einstein–Æther comparison layer [40, 41]. Its canon role is to upgrade the luminal tensor-speed condition from a purely external observational constraint to an internal naturalness condition, provided the SST tensor-mode identification is valid.

[CRITICAL NOTE] The proposition applies only to the spin-2/tensor sector. It does not by itself constrain the scalar and vector Æ-modes, nor does it prove monometricity for all matter, gauge, and knot species. The full monometricity target and the Einstein-dynamics routes are developed in the research track (Relativity Emergence Ladder).

10.13 Interpretive Role in the Canon

This section has a deliberately limited role:

- it upgrades the Swirl-Clock from a purely local kinematic factor to a candidate field variable;
- it records a minimal orthodox EFT bridge for relativistic comparison;
- it supplies the internal reference point needed by Section 11 for the time–gravity link.

Status summary.

- Eq. (182): **[ORTHODOX]** mathematical form; **[DERIVED]** role as an SST bridge variable
- Eq. (183): **[ORTHODOX]** EFT template
- Eq. (184)–Eq. (185): **[DERIVED]** bridge normalization and closure
- gravity interpretation: **[SPECULATIVE]**

10.14 Relation to the Chronos–Kelvin law

Section 2.10 provides the local transport identity of the Swirl-Clock on a material loop. The present section supplies the long-wavelength field-theoretic and analogue-metric reading of the same clock sector. The two layers are therefore complementary:

- Eq. (29) is the loop-level transport law;
- Eq. (182)–Eq. (185) provide the coarse-grained field description after the normalization of Eq. (184).

10.15 Canonical Electromagnetic–Gravity Closure

[CANONICAL BRIDGE LAW]. The electromagnetic and gravitational sectors of SST are canonically linked by a common local swirl state, but they remain distinct projections. Electromagnetic response is carried by transverse phase/flux variables and topological piercing changes. Gravitational response is carried by pressure gradients, swirl stress, and the local clock factor. Thus SST does not identify the ordinary electromagnetic field with gravity. It identifies both as different observable projections of the same incompressible, vorticity-bearing medium.

The canonical source state for the coupled EM–gravity bridge is

$$\boxed{\mathcal{X}_\mho^{\mathrm{EMG}} = \left(\mathbf{u}_\mho, p_\mho, \boldsymbol{\omega}_\mho, \varrho_\mho, \boldsymbol{\sigma}^{\mathrm{swirl}}, \rho_E, \rho_m\right), \qquad \rho_m = \frac{\rho_E}{c^2}.} \tag{190}$$

The topological piercing density $\varrho_\mho$ belongs to the electromagnetic impulse channel, while $p_\mho$, ρ_m, and $\boldsymbol{\sigma}^{\mathrm{swirl}}$ belong to the force and gravity channel. Dimensional consistency of Eq. (194) fixes $[\varrho_\mho] = \mathrm{m}^{-2}$: it is a piercing count per oriented area, not a volume density.

[CANON / PREFACTOR GUARD] The SI-normalized electromagnetic bridge, when invoked, uses the electron mass-to-charge factor

$$\mathbf{A}_{\mathrm{SI}} = \frac{m_e}{e}\,\mathbf{u}_\mho(\mathbf{x}, t), \qquad \mathbf{B}_{\mathrm{SST}} = \nabla\times\mathbf{A}_{\mathrm{SI}} = \frac{m_e}{e}\,\boldsymbol{\omega}. \tag{191}$$

The inverse factor e/m_e is rejected in this bridge because it has the wrong SI dimensions for mapping vorticity to magnetic flux density. At the Compton angular frequency this calibrated bridge gives

$$B_{\mathrm{core}}^{\mathrm{SST}} = \frac{m_e}{e}\omega_c = \frac{m_e^2 c^2}{e\hbar} = B_{\mathrm{QED}}, \tag{192}$$

while preserving the epistemic status of the result as a normalization bridge, not a derivation of QED from Euler dynamics.

Electromagnetic projection. The canonical topological electromagnetic impulse is

$$\boxed{\Delta\Phi_\mho = \Phi_0\,\Delta N, \qquad \Phi_0 = \frac{h}{2e}, \qquad \Delta N \in \mathbb{Z}.} \tag{193}$$

The choice $\Phi_0 = h/2e$ is a sector assignment, not a universal replacement of the ordinary single-charge flux scale h/e. It is used here for the paired or half-circulation Route-B normalization of the topological impulse channel; if a single-circulation branch is used, the flux quantum must be re-stated explicitly.

In local differential notation, the canonical part of the effective Faraday projection is

$$\boxed{\nabla\times\mathbf{E}_{\mathrm{eff}} = -\frac{\partial\mathbf{B}_{\mathrm{eff}}}{\partial t} - \Phi_0\frac{\partial\varrho_\mho}{\partial t}\,\hat{\mathbf{n}} - \mathbf{b}_\mho^{(\sigma)}.} \tag{194}$$

Here $\mathbf{b}_\mho^{(\sigma)}$ is a calibrated stress-projection correction. It is not a primitive canonical constant. If no stress-projection closure has been derived for a given apparatus, set

$$\mathbf{b}_\mho^{(\sigma)} = 0. \tag{195}$$

Equation (194) is an effective-medium projection. In the R-phase null limit,

$$\partial_t\varrho_\mho \to 0, \qquad \mathbf{b}_\mho^{(\sigma)} \to 0, \tag{196}$$

and the ordinary source-free Maxwell–Faraday identity is recovered.

Gravity projection. The canonical SST gravitational force density is

$$\boxed{\mathbf{f}_{\mathrm{grav},\circlearrowleft} = \rho_m \mathbf{g}_\circlearrowleft + \nabla \cdot \boldsymbol{\sigma}^{\mathrm{swirl}}.} \tag{197}$$

The bulk acceleration field is

$$\boxed{\mathbf{g}_\circlearrowleft = -\nabla \Phi_\circlearrowleft - \frac{\partial \mathbf{A}_\circlearrowleft}{\partial t}, \qquad \mathbf{B}_\circlearrowleft = \nabla \times \mathbf{A}_\circlearrowleft.} \tag{198}$$

In the weak, stationary, coarse-grained limit the scalar sector obeys the Poisson-type closure

$$\boxed{\nabla \cdot \mathbf{g}_\circlearrowleft = -4\pi G_{\mathrm{swirl}} \rho_m, \qquad \nabla^2 \Phi_\circlearrowleft = 4\pi G_{\mathrm{swirl}} \rho_m.} \tag{199}$$

This closure is valid only after coarse graining over swirl-string microstructure and only in the weak-field, slow-flow regime where pressure reconstruction is consistent with Euler balance. The stationary equality $\nabla \cdot \mathbf{g}_\circlearrowleft = -\nabla^2 \Phi_\circlearrowleft$ also assumes the gauge and regime choice $\partial_t \mathbf{A}_\circlearrowleft = 0$, equivalently a stationary transverse sector with no vector-potential contribution to $\mathbf{g}_\circlearrowleft$. In nonstationary bridge calculations use Eq. (198) directly instead of collapsing the scalar and vector sectors.

[CANON RECONCILIATION] Equation (199) is the canonical weak-gravity source closure for the EM–gravity bridge. Equation (185) is the clock-sector comparison closure after the normalization $\Phi_\chi = \mathcal{N}_\chi \delta\chi$. The two are identified only in the calibrated limit

$$\Phi_\chi = \Phi_\circlearrowleft, \qquad G_{\mathrm{eff}} = G_{\mathrm{swirl}}, \qquad \rho_{\mathrm{matter}} = \rho_m, \tag{200}$$

and should otherwise be kept as distinct bridge layers.

The canonical pressure anchor remains the incompressible Euler radial balance

$$\boxed{\frac{1}{\rho_f} \frac{dp_{\mathrm{swirl}}}{dr} = \frac{v_\theta(r)^2}{r}.} \tag{201}$$

Therefore the Newtonian-looking gravitational acceleration is not fundamental curvature in SST; it is the coarse-grained acceleration produced by pressure/stress gradients in the swirl medium.

Clock and optical closure. The clock sector supplies the shared time-rate projection:

$$\boxed{\frac{d\tau_{\mathrm{loc}}}{dt_\infty} = S^\circlearrowleft_{(t)} = \sqrt{1 - \frac{\|\mathbf{u}_\circlearrowleft\|^2}{c^2}}, \qquad n_\gamma = S^{\circlearrowleft\,-1}_{(t)}.} \tag{202}$$

Photon phase therefore probes n_γ, while bulk matter probes Eq. (197). These are coupled through $\mathbf{u}_\circlearrowleft$, but they need not have identical spatial gradients.

Electron-normalized coupling calibration. The weak gravitational coupling is written in electron-normalized SST variables as

$$\boxed{G_{\mathrm{swirl}} = \frac{v^*_\circlearrowleft c^3 t_p^2}{r_c M_e} = 6.6743013 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}.} \tag{203}$$

Thus the EM–gravity bridge is not an unlabeled free-parameter assertion. However, because t_p already contains G in orthodox units, Eq. (203) is a **[CALIBRATED]** electron-normalized consistency identity unless a future SST sector derives t_p or G_{swirl} independently from medium dynamics.

Compton–horn reduction guard. Substituting the calibrated horn closure $r_c = v^*_\circlearrowleft/\omega_c$ into Eq. (203) gives

$$G_{\mathrm{swirl}} = \frac{\omega_c c^3 t_p^2}{M_e}. \tag{204}$$

With the orthodox definitions

$$\omega_c = \frac{M_e c^2}{\hbar}, \qquad t_p^2 = \frac{\hbar G}{c^5}, \tag{205}$$

Eq. (204) reduces identically to G. This confirms that the electron-normalized bridge is algebraically closed under the v0.8.18 calibration chain, but it also fixes the epistemic status: the identity must not be cited as an independent derivation of Newton's constant unless t_p, $G_{\rm swirl}$, or the Compton–horn closure is derived without importing G through orthodox Planck units.

[CANON / STATUS GUARD] A dimensionless balance angle ϕ_G may be introduced only as a research-track diagnostic of Eq. (203). Since the calibrated chain gives $\tan^2\phi_G = 1$, the associated 45° angle is a visualization of equal radial-confinement and tangential-swirl weighting, not a physical direction of gravitational force and not an additional gravitational law. See Sec. Q.53.

[ORTHODOX GR REPARAMETERIZATION]. In orthodox units the electron Schwarzschild radius may be written as

$$r_s(e) = \frac{2Gm_e}{c^2} = \frac{4\pi\ell_P^2}{\lambda_c}, \qquad \ell_P = \sqrt{\frac{\hbar G}{c^3}}. \tag{206}$$

This is a standard GR identity and notational bridge. It does not derive G from SST medium dynamics.

Dimensional checks. The electromagnetic source term has units

$$[\Phi_0\partial_t\varrho_{\circlearrowleft}] = (\mathrm{V\,s})(\mathrm{m^{-2}\,s^{-1}}) = \mathrm{V\,m^{-2}}, \tag{207}$$

matching $[\nabla\times\mathbf{E}_{\rm eff}]$. The gravitational force density has units

$$[\rho_m\mathbf{g}_{\circlearrowleft}] = \mathrm{kg\,m^{-3}\,m\,s^{-2}} = \mathrm{N\,m^{-3}}, \qquad [\nabla\cdot\boldsymbol{\sigma}^{\rm swirl}] = \mathrm{N\,m^{-3}}. \tag{208}$$

Finally,

$$\left[\frac{\|\mathbf{v}_{\circlearrowleft}\|c^3t_p^2}{r_cM_e}\right] = \frac{(\mathrm{m\,s^{-1}})(\mathrm{m^3\,s^{-3}})(\mathrm{s^2})}{\mathrm{m\,kg}} = \mathrm{m^3\,kg^{-1}\,s^{-2}}. \tag{209}$$

Numerical hierarchy. At the canonical speed $v_{\circlearrowleft}^* = 1.09384563\times10^6\,\mathrm{m\,s^{-1}}$,

$$\frac{1}{2}\rho_f v_{\circlearrowleft}^{*\,2} = 4.1877439\times10^5\,\mathrm{Pa}, \tag{210}$$

$$S_{(t)}^{\circlearrowleft} = 0.9999933436, \tag{211}$$

$$n_\gamma - 1 = 6.6564858\times10^{-6}. \tag{212}$$

Hence pressure/stress channels can be macroscopically large at canonical scale, while single-pass optical-index shifts are small unless a resonant optical path enhances them. This hierarchy is now part of the canonical interpretation of the EM–gravity bridge.

Canonical exclusion rules. The following statements are excluded from the canonical EM–gravity bridge:

1. Ordinary electromagnetic fields are not identified with gravity: $\mathbf{B}_{\rm eff} \neq \mathbf{B}_{\circlearrowleft}$.
2. Uncalibrated vorticity terms may not be inserted directly into Maxwell source equations.
3. Device-specific gains such as Q_γ, $\mathcal{G}_{\circlearrowleft}^{(\sigma)}$, and impedance gains are research-track unless derived from a resonator or stress-closure model.
4. Apparent laboratory forces must first be decomposed into electromagnetic, thermal, acoustic, electrohydrodynamic, mechanical, and ordinary vorticity-impulse channels before any residual is interpreted as SST-specific.

Canonical summary. The promoted law is therefore

$$\boxed{\text{EM response and gravity response are distinct sectoral projections of } \mathcal{X}_{\circlearrowleft}^{\rm EMG}.} \tag{213}$$

The electromagnetic projection is controlled by topological flux impulse $\Delta\Phi_{\circlearrowleft} = \Phi_0\Delta N$. The gravitational projection is controlled by $\rho_m\mathbf{g}_{\circlearrowleft} + \nabla\cdot\boldsymbol{\sigma}^{\rm swirl}$ with weak closure $\nabla\cdot\mathbf{g}_{\circlearrowleft} = -4\pi G_{\rm swirl}\rho_m$. Both recover their orthodox limits when the SST projection channels vanish.

10.16 Triadic Gravity-Response Corollary

[DERIVED] Canonical corollary. Because the electromagnetic, gravitational, optical, and boundary-stress responses are distinct projections of the same SST state $\mathcal{X}_{\circlearrowleft}^{\mathrm{EMG}}$, a laboratory observation must not be classified as "gravity" from force response alone. The admissible coarse response separates into three limiting channels:

$$\mathcal{R}_{\mathrm{plume}} := (\Gamma_{\circlearrowleft} \neq 0, \quad \nabla n_\gamma \approx 0, \quad \mathbf{g}_{\circlearrowleft} \approx 0)\,, \tag{214}$$

$$\mathcal{R}_{\gamma} := (\nabla_\perp n_\gamma \neq 0, \quad \Gamma_{\circlearrowleft} \approx 0, \quad \mathbf{f}_{\mathrm{matter}} \approx 0)\,, \tag{215}$$

$$\mathcal{R}_{\Pi} := \left(\nabla \cdot \boldsymbol{\sigma}^{\mathrm{swirl}} \neq 0, \quad \nabla n_\gamma \approx 0\right). \tag{216}$$

Here $\mathcal{R}_{\mathrm{plume}}$ is an impedance or damping response of a plume or flame-like medium, $\mathcal{R}_\gamma$ is an optical phase/caustic response, and $\mathcal{R}_\Pi$ is a boundary-stress or elastic-shell response. The three modes may share a common source state but are not mutually identical.

Local Euler locking of pressure and optical projections. Under the additional assumptions of a passive, stationary, inviscid Euler region with constant ρ_f, the pressure and optical projections are not independent. Bernoulli reconstruction gives

$$\delta p_{\mathrm{swirl}} = -\frac{1}{2}\rho_f \|\mathbf{u}_{\circlearrowleft}\|^2, \tag{217}$$

while the clock/optical closure gives

$$n_\gamma = \left(1 - \frac{\|\mathbf{u}_{\circlearrowleft}\|^2}{c^2}\right)^{-1/2} = 1 + \frac{\|\mathbf{u}_{\circlearrowleft}\|^2}{2c^2} + O\left(\frac{\|\mathbf{u}_{\circlearrowleft}\|^4}{c^4}\right). \tag{218}$$

Eliminating $\|\mathbf{u}_{\circlearrowleft}\|^2$ yields the parameter-free leading-order locking relation

$$\boxed{\delta n_\gamma = -\frac{\delta p_{\mathrm{swirl}}}{\rho_f c^2}} \tag{219}$$

with validity restricted to the passive Euler regime. Forcing, damping, viscosity, thermal gradients, acoustic drive, plasma response, or explicit Mode-I terms such as $\Gamma_{\circlearrowleft}\mathbf{u}_{\mathrm{plume}}$ break the assumptions and move the prediction to the research-track transfer model. Thus the refined diagnostic statement is not three fully independent channels, but two locally locked projections plus one nonlocal Poisson/bulk response whenever Eq. (199) is invoked.

With $\rho_f = 7.0 \times 10^{-7}\,\mathrm{kg\,m^{-3}}$,

$$\frac{1}{\rho_f c^2} = 1.58950008 \times 10^{-11}\,\mathrm{Pa}^{-1}, \tag{220}$$

$$\frac{1}{2}\rho_f v_{\circlearrowleft}^{*\,2} = 4.1877439 \times 10^{5}\,\mathrm{Pa}, \tag{221}$$

$$\delta n_\gamma = 6.6565 \times 10^{-6} \tag{222}$$

at canonical swirl speed, consistent with Eq. (212).

Therefore

$$\boxed{\text{matter stabilization} \neq \text{photon caustic} \neq \text{boundary shell force.}} \tag{223}$$

[CRITICAL NOTE]. This corollary canonizes only the diagnostic separation of response channels. It does not canonize any specific witness claim, device claim, or anti-gravity interpretation. Device-specific excitation factors, optical gains, and shell stiffness parameters remain research-track unless derived from an explicit resonator, stress-closure, or calibrated transfer model.

11 Unified Interpretation

11.1 Overview

Swirl-String Theory (SST) proposes a unified interpretation in which fundamental physical phenomena emerge from the dynamics of a continuous medium supporting stable vortex structures.

This section synthesizes the results of the previous sections into a coherent conceptual framework.

11.2 Ontology of the Theory

[SPECULATIVE] Ontological claim: The fundamental ontology consists of:

- A continuous incompressible medium (Section 3)
- Stable vortex filament configurations (Section 5)
- Circulation as a conserved topological invariant

Interpretation: Particles are not elementary objects, but persistent dynamical structures.

11.3 Mass as Trapped Circulation Energy

From Section 2, and in particular from Eq. (42), the mass sector is organized as the product of a medium scale, a geometric gate, a topological kernel, and a local clock-impedance factor. The canon-level reading is therefore not merely that "energy is stored in rotation," but that persistent inertial scale is assigned only after geometry, topology, and local clocking have all been specified.

$$M_K(x) = \rho_m(x)\, {r_c}^3\, \Pi_K\, \Xi_K\, S^{\circlearrowleft}_{(t)}(x)^{-2}. \tag{224}$$

[SPECULATIVE] Unified interpretation: mass corresponds to localized, self-sustained circulation energy in the medium, modulated by state geometry, topology, and local clock impedance.

11.4 Time as a Relational Quantity

From Section 10 and Section 2, time is not introduced as an independent primitive but as a relational field read from the local state of the medium. The same kinematic quantity that enters the Swirl-Clock also appears in the organizing equation of the mass sector, so the canon treats proper-time modulation and inertial scaling as two projections of one shared substrate state.

$$S^{\circlearrowleft}_{(t)}(x,t) = \sqrt{1 - \frac{\|\mathbf{u}_{\circlearrowleft}(x,t)\|^2}{c^2}}. \tag{225}$$

[SPECULATIVE] Unified interpretation: time is not fundamental, but emerges from the kinematic state of the medium and is read operationally through clock-bearing processes.

11.5 Discreteness without Quantization Postulates

From Section 7:

- Delay introduces temporal nonlocality
- Stability selects discrete modes

[DERIVED] Unified consequence:

Discrete physical spectra arise dynamically from delay-induced mode selection.

This replaces operator-based quantization with a dynamical mechanism.

11.6 Charge and Circulation

[SPECULATIVE] Provisional bridge only. The canon retains circulation as a structurally important invariant, but it does *not* yet claim a completed derivation of observed gauge charges from circulation data alone. At most, the present document records a working hypothesis that orientation and magnitude of circulation may enter a later charge-assignment program.

11.7 Unification of Interactions

The canon records a *unification program*, not yet a completed unification theorem. In the present document:

- electromagnetic effects are represented by the minimal transverse-field bridge of Section 6,
- gravitational effects are represented by the clock-field and Poisson-limit bridge of Section 10,
- inertial mass is organized by the medium/topology/geometry/clock structure of Eq. (42).

[SPECULATIVE] Programmatic claim: these sectors may ultimately be read as different effective descriptions of one substrate, but their full common derivation remains open.

11.8 Atomic Structure as an Emergent Layer

From Section 8:

- Standard quantum mechanics appears as an effective description
- SST introduces circulation-dependent corrections

[SPECULATIVE] Unified interpretation:

> Quantum mechanics is an emergent effective theory of underlying vortex dynamics.

11.9 Consistency with Spectroscopy

From Section 9:

- Internal modes are suppressed by an excitation gap
- Low-energy physics remains unchanged

[ORTHODOX] Constraint: The unified picture does not violate known experimental constraints.

11.10 Hierarchy of Description

The theory admits multiple levels:

1. **Microscopic level:** vortex filament dynamics
2. **Mesoscopic level:** delay-induced mode structure
3. **Macroscopic level:** effective quantum description

11.11 Conceptual Summary

The SST framework can be summarized as:

Mass	$\to$ trapped circulation energy
Time	$\to$ relational swirl-clock field
Charge	$\to$ provisional circulation bridge
Quantization	$\to$ delay-induced stability

11.12 Canonical Statement

All observed physical structure emerges from the dynamics of a continuous medium in which stable vortex configurations, governed by quantized circulation and delay dynamics, produce the phenomena conventionally attributed to particles, fields, and quantum mechanics.

11.13 Scope and Limits

[IMPORTANT]:

- The framework is not yet a complete fundamental theory
- Several mechanisms remain to be derived rigorously
- Many results are conditional or phenomenological

11.14 Final Consistency Principle

A valid unified interpretation must preserve all experimentally verified results while providing a deeper structural explanation of their origin.

12 Integration of Prior Canon Versions

12.1 Integration policy (v0.5.12 reintegration)

This block records only material from earlier SST canon layers that can be retained inside the v0.8.1 architecture without weakening the explicit classification discipline of Section 4.

Older canon documents are therefore *not* cited as authorities. Transferable content is rewritten here as current-canon statements, each labeled according to its present status.

12.2 Retained v0.5.12 kernel sectors

The present draft now retains the following v0.5.12 kernel sectors in rewritten form:

- the Chronos–Kelvin invariant and clock–radius transport law (Section 2.10);
- the reparametrized zero-parameter corollary and coarse-graining rule (Section 2.11);
- a minimal Swirl–EM bridge and photon / unknot sector (Section 6);
- the swirl pressure law as the Euler-level force bridge (Section 6.4);
- the analogue-metric / clock-field bridge (Section 10).

12.3 Wave–particle phase rule and measurement bridge

A retained qualitative rule from earlier canon layers is that SST distinguishes two limiting phases:

$$\text{R-phase}: \quad \text{extended, unknotted, radiative,} \tag{226}$$

$$\text{T-phase}: \quad \text{localized, knotted, tangible.} \tag{227}$$

A minimal dynamical bookkeeping law for phase conversion is

$$\partial_t P_T = \Gamma_{R\to T}[\mathcal{K}]\, P_R - \Gamma_{T\to R}[\mathcal{K}]\, P_T, \qquad P_R + P_T = 1, \tag{228}$$

where $\mathcal{K}$ denotes an environment- and coherence-dependent kernel functional. The intended interpretation is that "measurement" corresponds not to an axiomatically imposed collapse, but to a medium-induced transfer of support from the extended R-phase to the localized T-phase.

[ORTHODOX LIMIT]: in weak coupling and after coarse graining, Eq. (228) should reduce to ordinary environment-induced decoherence models [28].

[SPECULATIVE]: the identification of decoherence with an actual R↔T phase transition is SST-specific.

12.4 Gauge and weak-mixing bridge

The retained gauge-level statement is that the effective low-energy algebra should reduce to

$$\mathfrak{g}_{\text{eff}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1), \tag{229}$$

with gauge potentials interpreted as holonomy data of coarse-grained multi-director transport.
A minimal bridge ansatz for the weak mixing angle is

$$\sin^2 \theta_W^{\text{eff}} = \frac{K_Y}{K_Y + K_W}, \tag{230}$$

where K_Y and K_W are effective Abelian and weak-sector stiffness scales. Eq. (230) is not a completed derivation, but it preserves the v0.5.12 idea that weak mixing should descend from a ratio of medium-response coefficients rather than appear as an unstructured free number.

[**ORTHODOX**]: θ_W is the standard electroweak mixing parameter of the SM [29, 3].

[**SPECULATIVE**]: Eq. (230) is an SST bridge ansatz only.

12.5 What the v0.5.12 reintegration does not import

The present reintegration does *not* import v0.5.12 cosmogony, engineering subsystems, cosmological-term closures, or highly application-specific protocols into the main canon body. Those remain suitable for appendices or standalone papers until they are more tightly tied to the minimal axiom system.

12.6 Purpose of this integration layer

Version 0.8.1 is not a fresh theory written from zero, but an explicit consolidation of earlier canon lines. This section records what has been retained, what has been renamed or reorganized, and what has been demoted from canon status to research-track status.

12.7 Retained from the early canon layers

From the v0.1.x–v0.3.x line, the present canon retains the primitive closure logic:

- a continuum substrate with effective density ρ_f,
- a core scale r_c fixed by Compton anchoring,
- circulation quantization through $\Gamma_0 = 2\pi r_c \, v_{\circlearrowleft}^{*}$,
- the interpretation of matter as stable organized structures rather than point primitives.

These elements remain non-negotiable because later sectors depend on them directly.

12.8 Retained from the middle canon layers

From the v0.4.x–v0.6.x line, the canon retains three durable structures:

1. the Rosetta strategy for mapping SST statements into orthodox language,
2. the photon/torsion sector as the minimal radiation bridge,
3. delay-induced mode selection as the main route to discreteness without operator postulates.

These are now distributed across Sections 7, 10, and 11 rather than treated as isolated notes.

12.9 Retained from the late canon layers

From the v0.7.x line, v0.8.1 retains the strongest structurally reusable sectors:

- the Swirl-Clock as the canonical relational-time field,
- the atomic bridge as a benchmark consistency layer rather than a complete atomic derivation,
- the spectroscopic gap logic as a hard consistency filter,
- the topology-first mass program, with topology separated from thermodynamic or radiative dressing,
- the EFT-style foliation and gauge-emergence program as a long-range development track.

12.10 Notation and density cleanup

Earlier canon versions sometimes conflated background density, mass-equivalent density, and saturated envelope-equivalent density. The present version adopts the cleaned hierarchy

$$\begin{aligned} \rho_f &\quad \text{effective fluid density,} \\ \rho_E &\quad \text{swirl energy density,} \\ \rho_m = \rho_E/c^2 &\quad \text{mass-equivalent density,} \\ \rho_{\text{horn}}^{\text{eff}} &\quad \text{canonical saturated envelope-equivalent density,} \\ \rho_{\text{core}} &\quad \text{legacy alias, deprecated unless explicitly defined.} \end{aligned} \tag{231}$$

Any older passage that identifies a single symbol with more than one of these roles is superseded by the present separation.

12.11 Topology-first reinterpretation of layers

A major consolidation from the recent lepton and knot-mass program is the separation between topology and layering. Knot type is retained as the primary identity label, while Golden-layer or thermal ladder indices are treated as dressing or coherence variables rather than species-defining labels. This prevents the canon from over-reading numerical layer fits as ontological identities.

12.12 Status of optional sectors

The following sectors are retained as important but non-primitive research layers:

- full gauge emergence and charge assignments,
- dual-vacuum phenomenology and Unruh-echo interpretation,
- complete many-electron atomic closure,
- full weak-sector and collider phenomenology,
- explicit topology-to-particle dictionary beyond the best-supported benchmark assignments.

They remain in the canon as organized programs, not as closed theorem-level results.

12.13 Integration principle

The purpose of v0.8.1 is therefore not to maximize novelty per page, but to minimize duplication and to expose the dependency graph of the theory. Whenever two earlier canon fragments make the same structural claim in different language, the present version keeps one canonical form and demotes the others to historical context.

12.14 Research-track companion

The sectors that are useful for continuity but too long or too speculative for a minimal main-canon reading are also collected in the companion file `canon-0.8.1-research-track.tex`. That excerpt includes the thermodynamic-radiation interface, the minimal-coupling QED analogy, the ribbon-invariant chirality refinement, the particle-candidate mapping layer, the hydrodynamic exchange / Pauli-barrier estimate, the numerical-benchmark policy block, the gauge-sector roadmap, the four topology/stability appendices on non-Abelian protection, reconnection decay, fractional torus-phase winding, and topological-fluid mechanics, and the *dark-sector and galactic canonization execution package* (frozen topology toolkit, benchmark protocol template, Euler-pressure anchor details, galactic branch gate, and probe–equation map). The integration subsections below retain the full in-context treatments with labels and cross-references; the companion file provides a portable standalone copy. Editorial separation does not reject these sectors; it prevents the core narrative from over-claiming closure.

12.15 Research-track screening functional

Status. [RESEARCH TRACK / NON-DERIVED SCREENING FUNCTIONAL]. The main canon does not assume a closed universal knot-energy functional beyond the calibrated constant chain and the explicitly derived hydrodynamic energy scales. For classification and numerical screening, SST may use the dimensionless surrogate

$$\frac{E_{\mathrm{screen}}[K]}{E_0} = a_R\ \mathrm{Rop}_{\mathrm{rad}}(\gamma_K) + a_C\, c_{\min}(K) + a_H\, \widehat{\mathcal{H}}(K) + a_G\, I_G(K) + \cdots\,, \tag{232}$$

where $\mathrm{Rop}_{\mathrm{rad}}$ is the orthodox radius-convention ropelength of the chosen relaxed representative, $c_{\min}$ is the minimal crossing/contact-complexity proxy used in the benchmark, and $\widehat{\mathcal{H}}$ is a nondimensional helicity or framed-linking measure.

The non-Abelian quantity $Q_G(K)$ is a sector label, not an energy. It may enter Eq. (232) only after a scalarization map has been chosen,

$$I_G : Q_G(K) \longrightarrow \mathbb{R}_{\geq 0}. \tag{233}$$

Therefore expressions of the form $\delta Q(K)$ are not canon-admissible unless Q has first been replaced by a real-valued index.

[CANON GUARD]. Equation (232) is a screening and ranking functional for candidate representatives. It is not a derived SST Hamiltonian, not a mass theorem, and not a substitute for the hydrodynamic energy functional.

12.16 Dark-sector and galactic canonization status (v0.8.x)

This block records promotion labels for the dark-sector / galactic program consistent with the internal canonization plan. Detailed freezes, the mandatory benchmark-protocol checklist, the galactic profile branch gate (Branch A: legacy-profile refine with benchmark traceability), and the probe falsification table live in `canon-0.8.1-research-track.tex` (Section titled `Dark-sector and galactic canonization execution package`).

[DERIVED] Stage 5 review: cross-item dependencies are respected (pressure anchor and benchmark layer precede full galactic closure; taxonomy strengthens other sectors). No sector is promoted on fit quality alone; each carries explicit falsifiability or reproducibility hooks as in Table 2 and the companion execution package.

12.17 Integration Policy

This section records only material from prior internal SST development that can be retained inside the v0.8.1 architecture *without* weakening the epistemic classification framework of Section 4.

Rule. Older canon documents are *not* cited as external authorities. Instead, transferable content is rewritten here as current-canon statements, each labeled as **[ORTHODOX]**, **[DERIVED]**, or **[SPECULATIVE]**.

12.18 Selective Recovery from Pre-v0.7 Canon Layers

The present draft absorbs only those legacy ingredients that can be rewritten in the current epistemic format without appealing to earlier canon versions as authorities.

Table 2: Master tracker: dark-sector and galactic program (v0.8.x).

Item	Prior state	Target	Status after Stage 0–4 freeze
Dark-sector Euler pressure anchor	BRIDGE	CANON	CANON-CANDIDATE (derivation, assumptions, limits, falsifiers frozen; see Section 6.4)
Dark taxonomy	HOLD	CANON	BRIDGE (symmetry-first classifier + filters; Section 5.7)
Galactic swirl law	BRIDGE	CANON	BRIDGE (Branch A closure; companion execution package)
Experimental probes	HOLD	CANON-BRIDGE	CANON-BRIDGE (equation / null-result map; companion)
Benchmark / reproducibility	BRIDGE	CANON	CANON-CANDIDATE (protocol template frozen; repository-wide reuse pending)
Topological reference identities	CANON-CANDIDATE	CANON	CANON-ready scope/notation (Appendix A)

Analogue line-element bridge. Legacy SST drafts recorded a schematic analogue line element and a metric–clock identification of the form

$$ds^2 = -c^2\, S^{\circlearrowleft}_{(t)}(x)^2\, dt^2 + \delta_{ij}\, dx^i\, dx^j, \tag{234}$$

$$g_{00}(x) = -S^{\circlearrowleft}_{(t)}(x)^2. \tag{235}$$

Recovered legacy sectors include:

- the finite-core atomic bridge, Eqs. (136)–(138);
- the analogue line-element bridge, Eqs. (234)–(235);
- compact research-track interfaces to thermodynamic radiation and minimal-coupling QED analogies;
- topological chirality refinements based on standard ribbon invariants.

Legacy material not imported includes version history, cosmogonic narrative, and phenomenological galaxy-scale laws that are not yet sufficiently integrated with the present minimal axiom system.

12.19 Thermodynamic Radiation Interface

A recurring legacy idea is that a local swirl-energy density may define an effective temperature field and thereby an emergent radiation sector.

Let

$$T_{\text{eff}} = T_{\text{eff}}\big(\rho_E,\, \omega,\, S_{(t)}\big) \tag{236}$$

denote an unspecified constitutive relation. A minimal legacy interface then asks whether Planck-type spectra can emerge from a medium whose local excitation scale is set by T_{eff}.

[**ORTHODOX**]: blackbody spectra are governed by Planck's law once a valid temperature variable is defined.

[**SPECULATIVE**]: SST does not yet derive Eq. (236), so this sector remains a research program rather than canonically closed physics.

12.20 Minimal Coupling Interface to Quantum Electrodynamics

Early research notes repeatedly explored whether an effective vector potential could be built from circulation variables. The most conservative placeholder takes the form

$$\nabla \to \nabla - i\,\frac{m}{\hbar}\,\mathbf{A}_{\text{eff}}, \qquad \mathbf{A}_{\text{eff}} = \chi\,\mathbf{v}, \tag{237}$$

where χ is a scalar weight and $\mathbf{v}$ is the local carrier velocity field.

[ORTHODOX]: minimal coupling is the standard gateway from free to gauge-coupled wave dynamics.

[SPECULATIVE]: Eq. (237) is only an analogy template unless a full gauge-covariant derivation is supplied.

12.21 Ribbon-Invariant Chirality Refinement

The older canon layers also suggested using the Călugăreanu–White–Fuller relation

$$Lk = Tw + Wr \tag{238}$$

as a refinement of the knot-based matter/antimatter dictionary.

[ORTHODOX]: Eq. (238) is a standard geometric identity for framed curves and ribbons.

[DERIVED BOOKKEEPING]: once a framed curve is specified, the identity separates writhe and twist contributions.

[SPECULATIVE]: the claim that writhe-dominated and twist-dominated realizations are dynamically stable physical isomers, or map to particle/antiparticle sectors, remains a model hypothesis.

12.22 Framed-Tube Ontology and the Attachment Gate

The particle carrier in SST is not a bare centerline knot alone. The canonical object is a framed, phase-bearing tube,

$$\mathcal{K}_{\mathrm{SST}} = \big(K, \mathcal{F}, \Gamma, \theta_{\mathrm{core}}\big), \tag{239}$$

where K is the centerline knot or link, $\mathcal{F}$ is a ribbon/framing, Γ is circulation, and θ_{core} is a material or order-parameter phase attached to the core.

The current Attachment Lemma is the statement that a local Compton/core cycle can be promoted to a globally closed framed-tube holonomy,

$$\frac{1}{2\pi}\oint_K d\theta_{\mathrm{core}} = \chi, \qquad \chi \in \mathbb{Z}. \tag{240}$$

A compensated form allows local swirl-clock redistribution while preserving the exact global integer,

$$\chi_{\mathrm{total}} = \chi_{\mathrm{local}} + \chi_{\mathrm{attachment}} = \chi, \qquad \delta Q = 0. \tag{241}$$

[CANON / OPEN GATE]: Eq. (240) is not yet a theorem of the Euler–SST dynamics. It is a load-bearing gate. Integer energy selection alone is insufficient; exact holonomy is required for canon status.

[CANON / NOTATION]: the framing phase follows the Compton angular cycle ω_c, while the associated local vorticity scale is $2\omega_c$. The factor of two must not be absorbed into χ.

12.23 Particle-Candidate Mapping Layer

[SPECULATIVE] Working dictionary: a minimal knot/composition mapping retained from earlier SST work is summarized in Table 3. This table is organizational rather than definitive; it records current topological assignments and mechanical analogues used by the mass and taxonomy program.

[CRITICAL NOTE]: Table 3 is a working dictionary only. Electric charge is not identified with raw Hopf helicity or pairwise linking number. The safer current placeholder is a boundary-winding charge variable q_∂, while knot/link data supply mass, chirality, confinement, and internal topological structure.

[CANON / DECISION]: The trefoil assignment is retained for the electron/positron sector only. Fractional values such as 2/3 belong, if anywhere, to the twist-knot/quark-like boundary-winding research track rather than to the trefoil-electron calibration.

Table 3: Minimal knot-class and analogue summary retained in v0.8.3, with current semantic corrections.

Carrier / composition	Candidate role	Status / interpretation
3_1 trefoil	electron / positron sector	baseline chiral lepton calibration; not used for fractional quark charge
$T(2,5)$ or equivalent higher torus class	muon	lepton hierarchy candidate
$T(2,7)$ or equivalent higher torus class	tau	lepton hierarchy candidate
twist/clasp knots, quasi-unknot loops	quark-like defects	research-track fractional boundary winding candidates
$5_2 + 5_2 + 6_1$ and nearby composites	baryon candidates	legacy mass-functional dictionary, still under audit
triple-gear locked analogue	proton-like charged baryon analogue	mechanical model for nonzero exterior collective rotation
Borromean-type analogue	neutron-like neutral baryon analogue	mechanical/topological model for zero exterior charge with internal triple linking

12.24 Hydrodynamic Exchange and the Pauli Barrier

[**ORTHODOX**] Fluid-mechanical input: for two thin filamentary loops C_1 and C_2 in an incompressible medium, the Biot–Savart interaction energy is [6]

$$E_{\text{int}} = \frac{\rho_f \Gamma_1 \Gamma_2}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{\|\mathbf{r}_1 - \mathbf{r}_2\|}. \tag{242}$$

[**ORTHODOX**] Regularized filament-energy template: if two identical electron-candidate loops are forced into the same spatial state, a short-distance cutoff a_{cut} gives the overlap penalty

$$V_{\text{Pauli}}(\mathbf{d}; a_{\text{cut}}) \approx \frac{\rho_f \Gamma_0^2}{4\pi} \oint\oint \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{\sqrt{\|\mathbf{r}_1(\theta_1) - \mathbf{r}_2(\theta_2) + \mathbf{d}\|^2 + a_{\text{cut}}^2}}, \tag{243}$$

with maximal-overlap scale

$$V_{\text{Pauli}}^{\max}(a_{\text{cut}}) \approx \frac{\rho_f \Gamma_0^2}{4\pi} \left(\frac{L}{a_{\text{cut}}}\right) \mathcal{O}(1). \tag{244}$$

Using the canonical values

$$\begin{aligned} \rho_f &= 7.0 \times 10^{-7}\ \text{kg m}^{-3}, \\ r_c &= 1.40897017 \times 10^{-15}\ \text{m}, \\ \Gamma_0 &= 2\pi r_c v_{\circlearrowleft}^{*} \approx 9.6836 \times 10^{-9}\ \text{m}^2\,\text{s}^{-1}, \end{aligned} \tag{245}$$

and $L \sim 2\pi a_0$ at the hydrogenic scale, the legacy benchmark corresponds to the cutoff calibration $a_{\text{cut}} = r_c$:

$$V_{\text{Pauli}}^{\max}(a_{\text{cut}} = r_c) \sim 1.23 \times 10^{-18}\ \text{J} \sim 7.6\ \text{eV}. \tag{246}$$

Away from this cutoff calibration the scaling is

$$V_{\text{Pauli}}^{\max}(a_{\text{cut}}) \sim V_{\text{Pauli}}^{\max}(r_c) \left(\frac{r_c}{a_{\text{cut}}}\right). \tag{247}$$

[**CRITICAL NOTE**] The cutoff a_{cut} is not the resolved physical tube radius a_{core}. The benchmark choice $a_{\text{cut}} = r_c$ is retained only as a legacy ultraviolet regularization scale. It must not be read as $a_{\text{core}} = r_c$, and it is not a theorem-level derivation of Pauli exclusion from ideal Euler dynamics.

As written, Eq. (246) is an order-of-magnitude, cutoff-calibrated benchmark attached to the canonical constants. A physical resolved-core calculation must provide a_{core}, the tube profile, and the relation between that profile and a_{cut} independently.

12.25 Numerical Benchmark Layer

[**DERIVED**] Computational-program requirement: earlier SST work contained explicit benchmark tables for particle and atomic masses. In v0.8.1, the benchmark layer is retained as a reproducibility requirement rather than as a standalone authority claim.

Table 4: Benchmark summary structure to be populated from canonical scripts and archived CSV output.

Particle	m_{exp} (MeV)	m_{SST} (MeV)	rel. error	mode	class
e	0.51099895	0.51099895	0	predictive/closure	3_1
μ	105.6583745	...	...	predictive	candidate torus class
τ	1776.86	...	...	predictive	candidate torus class
p	938.272088	...	...	predictive or closure	composite class
n	939.565420	...	...	predictive or closure	composite class

[**CRITICAL NOTE**] A future v0.8.x benchmark appendix should include:

- the exact script or package path used for generation,
- archived CSV outputs,
- mode definitions (predictive vs. exact-closure),
- uncertainty propagation on canonical constants and geometric factors,
- theorem-style pass/fail metadata (including tail-quality checks, postchecks, and extrapolation summaries),
- explicit separation between practical best-estimate closure and theorem-level closure,
- normalization of legacy constant names into SST notation (ρ_f, r_c, Γ_0) while preserving numerical values.

This requirement is included to keep the benchmark layer falsifiable and reproducible, rather than merely illustrative.

12.26 Gauge-Sector Roadmap

[**SPECULATIVE**] Structural roadmap: a retained minimal statement of the gauge program is:

$$\mathfrak{g}_{\text{eff}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1), \tag{248}$$

with effective gauge phases organized as holonomy data of multi-director transport.

A minimal phase-based representation is

$$A_\mu^a(x) = \partial_\mu \theta^a(x), \tag{249}$$

and clock coupling acts as a common phase reference,

$$\theta^a(x) \mapsto \theta^a(x) + q^a \chi(x) \quad \Rightarrow \quad A_\mu^a \mapsto A_\mu^a + q^a \partial_\mu \chi. \tag{250}$$

[**CRITICAL NOTE**] The gauge layer is retained in v0.8.1 only as a roadmap statement. Running couplings, anomaly cancellation, and realistic mixing matrices remain open work.

13 Discussion and Limits

13.1 Status of the recovered v0.5.12 sectors

The material reintroduced here does not have uniform epistemic status.

- The Chronos–Kelvin and pressure-law sectors are orthodox fluid-mechanical structures with SST reinterpretations.
- The analogue-metric and clock-field EFT layers are orthodox comparison tools, but only partially derived within SST.
- The R/T measurement bridge and weak-mixing stiffness ansatz remain explicitly speculative.

13.2 What v0.8.1 now recovers from v0.5.12

After the present patch, v0.8.1 no longer omits the main v0.5.12 kernel sectors that are structurally compatible with the current architecture:

- Chronos–Kelvin / clock transport,
- zero-parameter reparameterization discipline,
- Swirl–EM and photon / unknot bridge,
- swirl pressure law,
- analogue metric and clock-field gravity bridge,
- R/T measurement bridge,
- weak-mixing bridge ansatz.

13.3 Open items that remain outside the present closure

The following remain open and are *not* upgraded to full canonical closure in this draft:

- a first-principles derivation of the full SM gauge sector;
- a completed EWSB derivation tied to the canon constants;
- a rigorous derivation of the R/T measurement kernel from continuum dynamics;
- full CANON promotion of the dark-sector taxonomy and galactic closure beyond the current BRIDGE labels (Section 12.16).

13.4 What is established at canon level

The present document supports the following as canon-level structures:

- the primitive constant chain $(\rho_f, v_{\circlearrowleft}, \omega_c) \to R_{\mathrm{horn}} \equiv r_c \to \Gamma_0$,
- the Swirl-Clock as the coarse-grained relational time factor,
- delay-induced mode selection as the preferred discreteness mechanism,
- the atomic bridge and spectroscopic sectors as compatibility filters,
- the topology-first program for mass organization.

These claims are not equally proven, but they are the sectors on which the rest of the canon is structurally built.

13.5 What remains closure-level or conjectural

Several central SST ambitions remain open:

1. a rigorous derivation of full particle spectra from topology alone,
2. a first-principles derivation of charge assignments and gauge couplings,
3. a complete many-body derivation of shell filling and exclusion,
4. a full demonstration that delay plus topology reproduces the observed quantum rule set without importing it.

The canon records these as active programs rather than as settled conclusions.

13.6 Compatibility limits

Any viable SST realization must remain compatible with:

- atomic spectroscopy,
- relativistic effective kinematics,
- precision electroweak data where an SST bridge is claimed,
- the absence of large low-energy deviations from standard quantum mechanics.

This is why the canon repeatedly imposes suppression, weak-coupling, and low-energy recovery conditions.

13.7 Minimal falsifiers

The present canon would be materially weakened by any of the following:

- failure of the constant chain to remain dimensionally and numerically self-consistent,
- absence of any workable suppression mechanism for internal modes at atomic energies,
- inability to formulate a non-circular route from delay dynamics to discrete stable spectra,
- systematic contradiction between any SST atomic bridge and precision spectroscopy.

Conversely, strong support would come from a successful theorem ladder showing how discrete spectra, topological stabilization, and low-energy effective quantum structure arise from one shared substrate model.

13.8 Methodological limit

The canon is deliberately hybrid in style. Some sectors are written as exact closures, some as effective models, and some as programmatic roadmaps. This is a feature rather than a flaw, provided the epistemic status of each claim remains explicit. The document should therefore not be judged as though every section were intended to be a stand-alone journal theorem.

13.9 Near-term priorities

The next canon-strengthening tasks are clear:

1. complete the theorem ladder for the mass and clock sectors,
2. expand numerical validation blocks using the canonical constants,
3. refine the atomic bridge into sharper benchmark observables,
4. decide which optional sectors belong in the core canon and which should migrate to separate technical notes or the research-track companion.

13.10 Final limit statement

SST is presently best understood as a structured unification program with several strong internal bridges and several open derivational gaps. The canon is therefore mature enough to organize the theory coherently, but not yet so mature that every phenomenological sector should be presented as theorem-complete.

13.11 Status of Legacy Recoveries

The additional material imported here from older canon layers does not have uniform status.

- The analogue-metric bridge and ribbon identity are mathematically orthodox tools.
- Their SST interpretation is partially derived but not fully closed dynamically.
- The thermal-radiation and QED-interface sectors remain explicitly research-track.

This section therefore serves as a controlled retention layer, not as a blanket validation of all legacy material.

Accordingly, these recoveries should be read as structured extensions of the minimal core, not as fully promoted final doctrine.

13.12 What v0.8.1 Now Retains

After integration, the present Canon retains the following prior-development sectors in current-canon form:

- a particle-candidate topological dictionary (Section 12.23);
- a hydrodynamic exchange-energy estimate for fermionic exclusion (Section 12.24);
- a benchmark / reproducibility layer (Section 12.25);
- a minimal gauge-sector roadmap (Section 12.26);
- a clock-field EFT bridge and Poisson gravity limit (Section 10).

13.13 What is Deliberately Not Claimed

The present document still does *not* claim:

- a completed first-principles derivation of the Standard Model;
- a unique finalized knot-to-particle dictionary;
- a closed proof of the exclusion barrier beyond the current scaling benchmark;
- a fully renormalized relativistic field theory for the gauge and clock sectors.

These omissions are deliberate. v0.8.1 is intended to remain logically strict even when phenomenological sectors are reintroduced.

13.14 Canon Rule Going Forward

Any further reintegrated material must satisfy all of the following:

1. it can be written without citing older internal canon versions as authorities;
2. it can be assigned an explicit epistemic label;
3. it includes either an internal derivation path or a reproducible benchmark route;
4. it preserves compatibility with the orthodox limits recorded earlier in the document;
5. it states whether a displayed energy expression is a derived Hamiltonian term, a benchmark branch, or a screening surrogate;
6. it records the radius/diameter ropelength convention whenever finite-thickness knot data are used.

Appendices

A Topological reference toolkit

The following orthodox items form the mandatory internal mathematical toolkit cited by topology-first SST sectors. Statements are standard; SST-specific mappings appear in the main text with explicit status labels.

1. Gauss linking integral and linking number for disjoint closed curves.
2. Helicity $\mathcal{H} = \int \mathbf{v} \cdot (\nabla \times \mathbf{v})\, dV$ in ideal (inviscid) fluid flow and its conservation in the appropriate limit.
3. Thin-tube reduction of helicity to flux–linkage form in the standard filament limit.
4. Hopf invariant for localized field configurations (degree–type integral) where used as a topological density.
5. Basic torus-knot invariants as needed in the mass program (e.g. minimal crossing number, braid index, genus) with fixed naming consistent with Rolfsen notation in the main text.
6. Călugăreanu–White–Fuller relation $Lk = Tw + Wr$ for framed ribbons (already recalled as Eq. (238)) in the integration layer).

[ORTHODOX]: each item is classical mathematics [27, 22, 6]. **[DERIVED]**: scope and notation for v0.8.x are frozen to this list plus house symbols $(\rho_f, \Gamma, r_c, \mathcal{H})$ without parallel aliases.

B Full Derivations

B.1 Canonical horn-radius constant chain

The present canon uses the closure chain

$$r_e = \frac{\alpha\hbar}{m_e c}, \tag{251}$$

$$r_c = \frac{r_e}{2}, \tag{252}$$

$$\omega_c = \frac{m_e c^2}{\hbar}, \tag{253}$$

$$v_{\circlearrowleft} = \omega_c r_c, \tag{254}$$

$$\Gamma_0 = 2\pi r_c v_{\circlearrowleft}. \tag{255}$$

Substituting $r_c = \alpha\hbar/(2m_e c)$ into $v_{\circlearrowleft} = \omega_c r_c$ gives

$$v_{\circlearrowleft} = \left(\frac{m_e c^2}{\hbar}\right)\left(\frac{\alpha\hbar}{2m_e c}\right) = \frac{\alpha}{2}c, \tag{256}$$

hence the canonical ratio

$$\eta_0 = \frac{v_{\circlearrowleft}}{c} = \frac{\alpha}{2}. \tag{257}$$

B.2 Horn-Envelope Density Normalization

The electron rest-energy normalization may be written as an effective horn-envelope density relation,

$$\rho_{\mathrm{horn}}^{\mathrm{eff}} \equiv \frac{m_e c^2}{2\pi v_{\circlearrowleft}^{*\,2} R_{\mathrm{horn}}^3}. \tag{258}$$

This relation is dimensionally exact,

$$\left[\frac{m_e c^2}{v^2 r^3}\right] = \mathrm{kg\,m^{-3}}, \tag{259}$$

but its interpretation is now restricted.

[CANON / SEMANTIC RULE]: Eq. (258) is an effective horn-envelope or normalization density. It is not a measured local material density of the resolved vortex tube. Any resolved-tube model must use $a_{\rm core}$, a profile function, or ropelength/finite-thickness data rather than silently identifying $a_{\rm core} = r_c$.

[CANON / MASS-FUNCTIONAL NOTE]: Existing mass-functionals that use $\rho_{\rm core} {r_c}^3$ should be read as horn-envelope normalizations unless a separate local-core profile is explicitly introduced.

B.3 Geometric gate identity

Using the Compton wavelength $\lambda_c = h/(m_e c)$ and the horn-radius closure $r_c = R_{\rm horn} = \alpha\hbar/(2m_e c)$, one obtains

$$\frac{\lambda_c}{\pi r_c} = \frac{h/(m_e c)}{\pi\alpha\hbar/(2m_e c)} = \frac{2h}{\pi\alpha\hbar} = \frac{4}{\alpha}. \tag{260}$$

This is the canonical geometric shielding factor used throughout the late mass-functional papers.

B.4 Useful numerical anchors

With the canonical values

$$v_{\circlearrowleft} \approx 1.09384563 \times 10^6 \ \mathrm{m\,s^{-1}}, \tag{261}$$

$$r_c \approx 1.40897017 \times 10^{-15} \ \mathrm{m}, \tag{262}$$

$$\Gamma_0 \approx 9.6836 \times 10^{-9} \ \mathrm{m^2\,s^{-1}}, \tag{263}$$

one has the Compton turnover time

$$\tau_{\rm circ,c} = \frac{2\pi r_c}{v_{\circlearrowleft}^*} = \frac{2\pi}{\omega_c}, \tag{264}$$

and the geometric gate

$$\frac{\lambda_c}{\pi r_c} \approx 5.48 \times 10^2. \tag{265}$$

C Delay Mathematics

C.1 Fixed-point condition for locked modes

Starting from the delayed phase equation

$$\dot\phi(t) = \omega_0 + \kappa \sin\big(\phi(t - \tau_{\rm circ}) - \phi(t)\big), \tag{266}$$

seek a rigidly rotating solution

$$\phi(t) = \Omega t + \phi_\star. \tag{267}$$

Substitution gives the algebraic mode condition

$$\Omega = \omega_0 - \kappa \sin(\Omega\tau_{\rm circ}), \tag{268}$$

where the minus sign follows from $\sin[-\Omega\tau_{\rm circ}] = -\sin(\Omega\tau_{\rm circ})$. This admits multiple branches labelled by the effective phase winding across the delay interval.

C.2 Linear stability of a locked branch

Let

$$\phi(t) = \Omega t + \epsilon(t), \qquad |\epsilon| \ll 1. \tag{269}$$

Linearizing yields

$$\dot\epsilon(t) = \kappa\cos(\Omega\tau_{\rm circ})\big(\epsilon(t - \tau_{\rm circ}) - \epsilon(t)\big). \tag{270}$$

Using the exponential ansatz $\epsilon(t) \propto e^{\lambda t}$ gives the characteristic equation

$$\lambda = \kappa\cos(\Omega\tau_{\rm circ})\left(e^{-\lambda\tau_{\rm circ}} - 1\right). \tag{271}$$

Stability requires $\mathrm{Re}(\lambda) < 0$ for all nontrivial roots. Thus discrete spectra in SST are not arbitrary: only those branches satisfying both the fixed-point equation and the stability inequality survive dynamically.

C.3 Small-delay expansion

For $|\lambda\tau_{\rm circ}| \ll 1$,

$$e^{-\lambda\tau_{\rm circ}} = 1 - \lambda\tau_{\rm circ} + \mathcal{O}((\lambda\tau_{\rm circ})^2), \tag{272}$$

so the characteristic equation becomes

$$\lambda \approx -\kappa\tau_{\rm circ}\cos(\Omega\tau_{\rm circ})\,\lambda. \tag{273}$$

Nontrivial damping therefore requires the effective factor $\kappa\tau_{\rm circ}\cos(\Omega\tau_{\rm circ})$ to have the correct sign. This provides a simple branch-selection criterion at weak delay.

C.4 Interpretive consequence

The important canon point is that discreteness is produced in two stages:

$$\text{delay} \longrightarrow \text{multiple algebraic branches} \longrightarrow \text{stability filter}. \tag{274}$$

Quantization is therefore not inserted by hand; it is the residue of a dynamical selection problem on a delayed closed carrier.

D Conversation-Derived Insights

D.1 Role of this appendix

This appendix records recurrent insights that emerged during the project-level development of the canon and are useful for guiding future work, even when they do not yet qualify as theorem-level results.

D.2 Topology first, layers second

A central refinement of the recent canon is the separation

$$\text{particle identity} \longleftrightarrow \text{topology}, \qquad \text{excitation or dressing state} \longleftrightarrow \text{layer index}. \tag{275}$$

This prevents Golden-layer fits from being mistaken for a species ontology and keeps the topology program structurally cleaner.

D.3 Spectral origin of $\zeta(3)$

The project discussions repeatedly converged on the conclusion already encoded in Section 5.6: the appearance of $\zeta(3)$ is best interpreted as a spectral invariant of a cubic-weighted closed-carrier mode tower, not as a direct invariant of threefold geometry by itself.

D.4 Atomic bridge status discipline

Another recurring project insight is that the atomic bridge is strongest when treated as a benchmark layer with explicit status tags. Hydrogenic consistency, branch-sensitive couplings, and the spectroscopic gap form a robust bridge; shell filling, exclusion, and full heavy-atom structure remain open programs rather than completed derivations.

D.5 Master-equation viewpoint

A major organizational gain of the recent conversations is the recognition that many SST sectors can be read as different projections of one shared structure:

$$\text{medium scale} \times \text{topology} \times \text{geometry} \times \text{clocking}. \tag{276}$$

Even when this is not written as a single canonical equation inside the present draft, it remains a useful guide for reducing duplication across the mass, time, and thermodynamic sectors.

D.6 State-transition taxonomy from the trefoil-closure discussion

The present conversation also fixes a useful state-transition taxonomy:

$$\text{smooth deformation/binding} : K = 3_1 \text{ preserved,} \tag{277}$$

$$\text{R-to-T closure/nucleation} : K \text{ created at a boundary event,} \tag{278}$$

$$\text{annihilation or weak capture} : K \text{ converted with compensation.} \tag{279}$$

This taxonomy prevents ordinary atomic binding, non-Euler closure, and true particle conversion from being conflated.

D.7 Dual-vacuum and photon sector status

The project-level discussions also clarified a canon policy: the photon/torsion bridge and the dual-vacuum phenomenology are valuable and fertile, but they should not silently override the simpler core canon. They are best treated as phenomenological extensions built on top of the primitive constant chain and Swirl-Clock structure.

D.8 Canon edition notes

D.8.1 v0.8.19

v0.8.19 adds an orthodox radius-convention ropelength/thickness foundation, a main-text guard that knot type alone does not determine a unique physical representative, a scalarized research-track screening functional E_{screen} with Q_G retained only as a sector label unless mapped to I_G, a derived leading thin-filament length/log energy anchor, a density and dimensional audit for geometric mass branches, a geometric impedance bridge demoted to research-track pointer form, an explicit T_{core} free-symbol guard, symmetry-metadata guards for dark-knot taxonomy, research-track operator-valued Swirl-Clock diagnostics, and stricter reintegration rules for derived Hamiltonian terms versus screening surrogates on top of v0.8.18. **Audit bundle (in-place):** [CALIBRATED]/[CONDITIONAL] reading-guide rows; unified $S^{\circlearrowleft}_{(t)}(x,t) = \sqrt{1 - \|\mathbf{u}_{\circlearrowleft}\|^2/c^2}$ in Eq. 16 and Axiom 7; $\rho^{\text{eff}}_{\text{horn}}$ and $16\pi^2$ F_max hygiene; Poisson–clock bridge (Eqs. 184, 200); EMG sector guards; trefoil ropelength precision; Eq. 299. **Foucault RT:** rotation-to-clock inference protocol (§Q.64; Eqs. 653–662). **Projected swirl-vorticity frequency:** research-track diagnostics (Eqs. 664, 669, 671); Foucault guardrail item 4 tightened against $\rho_f = K\Omega$ identifications. **Dark taxonomy + operator clock:** benchmark CSV schema hardening; symmetry metadata vector (Eq. 362); main-canon symmetry-metadata guard; operator-valued Swirl-Clock diagnostics (§Q.11; Eqs. 523–527). **Claude audit cleanup:** geometric impedance bridge demoted to research-track (§K); main appendix pointer guard; explicit T_{core} free-symbol guard; non-Abelian screening energy aligned to E_{screen} notation.

D.8.2 v0.8.18

v0.8.18 adds calibration-chain guardrails (rc/α gates, kernel normalization, EM prefactor, $\rho^{\text{eff}}_{\text{horn}}$ baseline fix), research-track observational falsifier bounds (GW170817, pulsar preferred-frame limits), orthodox resolved-tube reach/thickness definitions, the four topology/stability research appendices (non-Abelian protection, reconnection decay, fractional torus-phase winding, and topological-fluid mechanics), the research-track knot-energy normal form with I_G scalarization, Euler/Magnus probe-transport polish, canonical time ontology with no-bare-τ policy, golden-layer hyperbolic rapidity guards, Planck-suppressed $F^{\max}_{\text{swirl}}$ reparameterization, Compton–horn G_{swirl} reduction guard and balance-angle diagnostic, hybrid density-source Swirl-Clock benchmark, and the contact-stress geometry research appendix on top of v0.8.17.

D.8.3 v0.8.17

v0.8.17 adds the T-foliation versus local Swirl-Clock remark (Eq. 172), the ropelength/trefoil-α convention guard, Route-I SST-63/23/56 integration in the research track, and aligned updates in SST-73, SST-63, SST-64 v2, and SST-34 on top of v0.8.16. RC2 release-clean polish: p.119 summary quote, ‘Æ‘-modes fix, ‘tabularx‘ particle-candidate tables, and targeted version-phrase cleanup.

D.8.4 v0.8.16

v0.8.16 adds the pressure–optical locking relation (Eq. 219) in the Triadic Gravity-Response Corollary, the research-track no-monopole transport-stress audit (§Q.35), and the relativity falsifier ladder on top of v0.8.15. SST-73 receives matching critical notes on Candidate C and $[\alpha_{\text{grav}}] = \mathrm{s}^2$.

D.8.5 v0.8.15

v0.8.15 adds the canon hydrodynamic Lorentz-type swirl-force density (§6.5: Bernoulli tie, loop observables, calibrated scales) and the research-track EM-to-swirl correspondence with $\lambda_{\text{EM}\to\circlearrowleft}$, SST-44 stress closure, and pending $\mathbf{f}_{\text{link}}$ (§Q.63.1) on top of v0.8.14.

D.8.6 v0.8.14

v0.8.14 adds the two-speed clock discipline (§2.7: $v^*_{\circlearrowleft}$ vs. c, NLSE core sound speed $c_s = v^*_{\circlearrowleft}/\sqrt{2}$, research-track gate), the α-gate guard on $\lambda_c/(\pi r_c) = 4/\alpha$, and the research-track core–torsion impedance-matching lemma (§Q.58) on top of v0.8.13.

D.8.7 v0.8.13

v0.8.13 adds the relativity-emergence audit: the internal tensor-speed naturalness proposition (§10.12, conditional $c_{13} = 0$) in the main canon, the research-track Relativity Emergence Ladder (§Q.45) separating derivable SR/GR kinematics from the open Einstein-dynamics program, and relativity/induced-gravity/cosmology bibliography on top of v0.8.12.

D.8.8 v0.8.12

v0.8.12 applies the Gemini round-3 epistemic audit: $\mathcal{P}_{\text{cal}}$-aligned identity relabels, coarse-grain density caveat, Pauli a_{cut} disambiguation, galaxy-scale ρ_f note, and research-track EM-bridge limits on top of v0.8.11.

D.8.9 v0.8.11

v0.8.11 completes the final hygiene pass: consistent $\mathcal{P}_{\text{cal}}$, $v^*_{\circlearrowleft}$ vs. $\mathbf{u}_{\circlearrowleft}$ usage in EMG, framed self-linking, W-boson sector, and research-track numerical bridges on top of v0.8.10.

D.8.10 v0.8.10

v0.8.10 applies the Gemini round-2 audit: canonical speed $v^*_{\circlearrowleft}$ vs. local $\mathbf{u}_{\circlearrowleft}(x,t)$ discipline, η_0 algebraic-within-calibration status, delay-equation sign convention, Rydberg/finite-cell relabeling, and research-track v_K disambiguation on top of v0.8.9.

D.8.11 v0.8.9

v0.8.9 adds the triadic gravity-response corollary (§10.16) and research-track flame/caustic/shell diagnostics plus atomic condensate closure notes on top of v0.8.8.

D.8.12 v0.8.8

v0.8.8 applies the Gemini audit: primitive calibration notation ($\mathcal{P}_{\text{cal}}$, $v^*_{\circlearrowleft}$, $\mathbf{u}_{\circlearrowleft}$), conditional Kelvin/DDE labels, Pauli a_{cut} disambiguation, and research-track epistemic relabeling on top of v0.8.7.

D.8.13 v0.8.7

v0.8.7 adds the scalar-vs-$\mathbb{C}P^1$ substrate argument, the $\theta = \pi$ fermionic $\mathbb{Z}_2$ sector, and supporting bibliography on top of v0.8.6.

D.8.14 v0.8.6

v0.8.6 adds framed-tube self-linking ($SL = pq$), spinorial $\chi = \pm 1$ selection, and the lepton ladder (§5.5) on top of v0.8.5.

D.8.15 v0.8.5

v0.8.5 adds the highres conversation audit (Euler discipline, R-to-T bookkeeping, twist-ladder research track, horn/envelope wording) and the CALIBRATED circularity-honesty relabeling on top of v0.8.4.

D.8.16 v0.8.4

v0.8.4 integrates the canonical EM–gravity bridge, finite-cell α obstruction summary, and extended research-track taxonomy blocks.

D.8.17 v0.8.3

v0.8.3 integrates framed-tube ontology, attachment gate, trefoil closure discipline, updated particle dictionary, and Pauli a_{core} regularization per conversation, highres, and trefoil diffs.

D.8.18 v0.8.2

v0.8.2 integrates radius/density semantic correction: $r_c \equiv R_{\text{horn}}$, separate a_{core}, and horn-envelope density normalization per conversation and highres diffs.

D.9 Trefoil closure and persistence discipline

The project-level conclusion is that electron-trefoil creation, if retained, must be formulated as a boundary-supported R-to-T closure event rather than as ordinary reconnection of an already closed ideal-Euler vortex tube. The canon distinction is therefore

$$\text{pre-closure R-phase support} \xrightarrow{\mathcal{K}_{\text{Kairos}}} \text{closed } 3_1 \text{ carrier}, \tag{280}$$

$$\text{closed } 3_1 \text{ carrier} \xrightarrow{\text{smooth Euler}} \text{same knot class } 3_1. \tag{281}$$

This keeps the orthodox frozen-in topology constraint intact while preserving the SST closure program as a clearly labeled bridge.

The radius notation is correspondingly split into the local tube radius a_{core}, the neutral circulation radius r_c, and the classical electron envelope radius $r_e = 2r_c$:

$$a_{\text{core}} \neq r_c, \qquad r_e = 2r_c. \tag{282}$$

Ordinary atomic binding changes the phase envelope of the carrier, not its knot class:

$$e^-_{\text{atom}} = 3_1^{\text{core}} \otimes \Psi^{\text{envelope}}_{n\ell m\sigma}. \tag{283}$$

Ropelength representative and thickness-constrained ground state. For a tame knot or link type K, orthodox ropelength theory proves the existence of a ropelength-minimizing representative [45, 48]. With the radius convention for thickness,

$$\text{Rop}_{\text{rad}}(\gamma) = \frac{\text{Len}(\gamma)}{\text{Thi}_{\text{rad}}(\gamma)}, \tag{284}$$

and the minimizer is $C^{1,1}$, with bounded curvature, but need not be C^2 [45].

[ORTHODOX GEOMETRY / RESOLVED-TUBE DEFINITION]. For a C^1 embedded centerline, orthodox ropelength theory identifies thickness with both reach and normal injectivity radius [45]. Therefore the SST resolved-tube radius is, when an ideal-tube representative is being used,

$$a_{\text{core}}(K) \equiv \text{Thi}_{\text{rad}}(\gamma_K) = \text{reach}(\gamma_K) = \text{NIR}(\gamma_K), \tag{285}$$

$$\text{Thi}_{\text{rad}}(\gamma_K) = \min\left(\inf_s \frac{1}{\kappa(s)}, \frac{1}{2} d_{\text{dcsd}}(\gamma_K)\right), \tag{286}$$

where d_{dcsd} is the doubly-critical self-distance. Equations (285)–(286) are geometric definitions of the resolved tube; they do not license the identification $a_{\text{core}} = r_c$. In an SST sector where the resolved vortex-string energy is tension dominated, with an orthodox knotted flux-tube analogue in the glueball literature [49],

$$E_{\mathrm{SST}}[\gamma] = T_{\mathrm{eff}}\,\mathrm{Len}(\gamma) + E_{\mathrm{sub}}[\gamma], \qquad \mathrm{Thi}_{\mathrm{rad}}(\gamma) \geq r_c, \tag{287}$$

the leading-order ground-state problem reduces to ropelength minimization:

$$E^{(0)}_{\mathrm{SST}}(K) \approx T_{\mathrm{eff}} r_c\,\mathrm{Rop}_{\mathrm{rad}}(K). \tag{288}$$

This is a derived-conditional reduction, not an independent proof that all shape-dependent observables are ropelength functions.

Critical note: degeneracy and subleading splitting. Ropelength minimizers need not be unique. Therefore, while the scalar mass proxy $M_K \propto \mathrm{Rop}(K)$ may remain well-defined, shape-sensitive observables such as dipole structure, magnetic moment, moment of inertia, writhe distribution, or local swirl-stress maps may require subleading splitting terms:

$$E_{\mathrm{sub}} = E_{\mathrm{bend}} + E_{\mathrm{twist}} + E_{\text{Biot-Savart}} + E_{\mathrm{core}} + \cdots . \tag{289}$$

Convention guard for trefoil alpha comparisons. The mathematical literature uses both radius- and diameter-based thickness conventions. If

$$\mathrm{Thi}_{\mathrm{diam}} = 2\,\mathrm{Thi}_{\mathrm{rad}}, \tag{290}$$

then

$$\mathrm{Rop}_{\mathrm{diam}} = \frac{1}{2}\,\mathrm{Rop}_{\mathrm{rad}}\,. \tag{291}$$

For the trefoil, using certified ideal-knot data [46],

$$\mathrm{Rop}_{\mathrm{rad}}(3_1) \approx 32.743274, \qquad \mathrm{Rop}_{\mathrm{diam}}(3_1) = \mathcal{L}_{3_1} \approx 16.371637. \tag{292}$$

The leading coincidence

$$\alpha^{-1}_{\mathrm{lead}} = \frac{8\pi}{3}\,\mathrm{Rop}_{\mathrm{diam}}(3_1) \approx 137.15471 \tag{293}$$

is therefore a robust sub-promille numerical coincidence, not an exact or ppm-certified derivation of the fine-structure constant.

D.10 Finite-cell α coincidence and obstruction summary

[BRIDGE ANSATZ / SPECULATIVE FIT]. A parameter-light finite-cell variational construction for an ideal trefoil filament, containing partially open modelling choices, yields a leading dimensionless scale

$$\alpha^{-1}_{\mathrm{lead}} = \frac{8\pi}{3}\,\mathcal{L}_{3_1} \approx 137.15471, \tag{294}$$

using the numerically computed ideal trefoil ropelength $\mathcal{L}_{3_1} = 16.371637$ [46] as a *benchmark* topological-geometric input. This lies within 866 ppm of CODATA $\alpha^{-1} = 137.035999$ and is an α-free sub-per-mille coincidence, not a part-per-million derivation.

The prefactor $8\pi/3$ contains a mode-count factor $N_p = 4$ derived from the soft sector of the spherical finite-cell surface spectrum $k_\ell = \ell(\ell+1) - 2$ (zero and unstable modes only). The sector volume $4\pi/3$ and overall normalization remain partially open modelling choices.

[NEGATIVE / OBSTRUCTION RESULT]. Closing the residual to ppm accuracy is underdetermined: at fixed cell-radius gate the required transverse shell weight is $w_\perp = 1.0675$, not the round value 1 (which reaches only 10.5 ppm). An independent Gross–Pitaevskii core-profile proxy returns $w_\perp \approx 2.076$, overshooting to -157 ppm. The 10.5 ppm residual can be absorbed independently by any of five sub-percent coefficients (shell weight, cell-radius gate, sector volume, ropelength value, or a third-order term), so ppm closure carries no discriminating power without pinning all inputs to $\sim 10^{-6}$ relative precision—beyond present ideal-knot ropelength certification [46, 45].

The gate hierarchy is therefore

$$G_0 \text{ (input provenance)} \longrightarrow G_1 \text{ (identifiability)} \longrightarrow G_2 \text{ (operator Hessians)} \longrightarrow G_3 \text{ (ppm closure)}, \tag{295}$$

with G_0 and G_1 blocking closure before Hessian-level work becomes relevant.

The ring-constant proximity $\alpha_{\text{ring}} \approx \varphi$ is treated in Subsection I.14 of the research track. It is not a canon-level trefoil curvature closure.

[CANON / NON-CLAIM]. This block does not assert a first-principles derivation of the fine-structure constant. The defensible content is the α-free sub-per-mille leading coincidence with partially derived $N_p = 4$, plus a localized obstruction to ppm closure. Full scripts, appendices, and audit details are in `finite_cell_obstruction.tex` and `Derive_Constants/`.

D.11 Open theorem ladder

The most important unfinished task identified across the project is not adding new sectors, but tightening the derivational ladder between existing ones. In practice this means:

1. turning heuristic closures into explicit lemmas,
2. attaching numerical validation to each nontrivial scaling law,
3. separating benchmark results from full ontological claims.

This appendix therefore serves as a reminder that canon quality increases more by sharpening structure than by increasing conceptual breadth.

E Explicit Topological Mass Closures

E.1 Purpose and status

This appendix supplies an explicit closure layer for the mass program without replacing the canonical master equation of Eq. (42). Its role is to make concrete those invariant choices that were historically used in pre-v0.8 layers whenever actual particle-mass calculations were attempted.

[DERIVED-CONDITIONAL / BENCHMARK TARGET] Canon rule: the abstract kernel Ξ_K of the main text may be specialized to more explicit invariant combinations provided the resulting branch remains dimensionless, benchmarkable, and subordinate to the topology-first hierarchy.

[CRITICAL NOTE] Nothing in this appendix upgrades the explicit kernel to a theorem of first-principles filament dynamics. The present role is closure discipline and reproducible benchmarking.

E.2 Density and dimensional audit

[CANON GUARD]. The mass program must keep energy density and mass-equivalent density separate. If ρ_E denotes an energy density in $\mathrm{J\,m^{-3}}$, then its mass-equivalent density is

$$\rho_m^{(E)} = \frac{\rho_E}{c^2} \qquad [\rho_m^{(E)}] = \mathrm{kg\,m^{-3}}. \tag{296}$$

A baseline mass expression may use either a mass density directly, or an energy density divided by c^2, but it must not multiply an already energy-normalized density by an additional hidden c^2.

For the geometric baseline branch below, dimensional closure requires

$$\left[\rho_{\text{horn}}^{\text{eff}} \, \frac{{r_c}^5}{\lambda_c^2}\right] = \mathrm{kg}. \tag{297}$$

This is the audit that distinguishes a mass closure from an energy closure.

E.3 Pure geometric baseline branch

[DERIVED-CONDITIONAL / NORMALIZATION BRANCH]. A pure geometric baseline mass scale for topology class T is

$$M_0(T) = 2\pi^3 \, \rho_{\text{horn}}^{\text{eff}} \, \frac{{r_c}^5}{\lambda_c^2} \, \mathcal{L}_{\text{tot}}(T). \tag{298}$$

This branch is subordinate to Eq. (42). It records a normalization-level geometric closure used in historical mass-program benchmarks; it does not replace the canonical master functional or the explicit kernel branch below. The density in Eq. (298) is the horn-envelope normalization density of Eq. (25); using the background mass-equivalent density ρ_m here would move the benchmark down by the factor $\rho_m/\rho_{\text{horn}}^{\text{eff}} \simeq 1.20 \times 10^{-30}$ and is therefore a symbol-collision error, not a physical branch.

[CRITICAL NOTE / CALIBRATION COLLAPSE] Under the canonical horn closure of Eq. (25) and the Compton–core identity $\lambda_c = 2\pi c r_c / v_{\circlearrowleft}^*$, Eq. (298) reduces identically to

$$M_0(T) = \frac{m_e}{4} \, \mathcal{L}_{\text{tot}}(T). \tag{299}$$

Therefore this branch is a calibrated electron-mass rescaling of ropelength, not an independent medium-derived mass scale. In particular, an electron-normalized trefoil benchmark would require $\mathcal{L}_{\text{tot}} = 4$ in this branch, whereas certified ideal-trefoil ropelength values are of order 16.37 in diameter convention or 32.74 in radius convention. The branch remains useful only as a normalization diagnostic subordinate to the canonical master functional.

E.4 Geometric shielding gate and exposure branch

[CALIBRATED CLOSURE / NOT AN INDEPENDENT α-DERIVATION]. The geometric gate may be written as

$$\mathcal{G}_\alpha := \frac{\lambda_c}{\pi r_c} = \frac{4}{\alpha} = \frac{2c}{v_{\circlearrowleft}^*}, \tag{300}$$

inside the calibrated SST constant chain. Equation (300) is a closure identity and normalization guard. It must not be cited as an independent derivation of the electromagnetic fine-structure constant.

With an exposure flag $g_{\rm exp}(T) \in \{0, 1\}$ and a dimensionless topology kernel $K_{\rm top}(T)$, the geometric benchmark branch is

$$M_{\rm geom}(T) = \mathcal{G}_\alpha^{g_{\rm exp}(T)} K_{\rm top}(T) M_0(T). \tag{301}$$

This is a benchmark branch of the mass program. It becomes canonically stronger only after a fixed particle table, fixed representatives γ_K, and fixed residual tolerances are supplied.

E.5 Explicit invariant branch of the canonical kernel

A conservative explicit specialization is obtained by resolving the topological kernel into ropelength, braid complexity, and Seifert-genus data:

$$\Xi_K^{(\rm exp)} = \left[\alpha_{\mathcal{L}} \frac{\mathcal{L}_{\rm tot}(K)}{\mathcal{L}_{\rm tot}(3_1)} + \alpha_b\big(b(K) - 1\big) + \alpha_g\, g(K)\right] \varphi^{-2k(K)}. \tag{302}$$

Here:

- $\mathcal{L}_{\rm tot}(K)$ is the total ropelength-like invariant used for the chosen embedding or ideal representative,
- $b(K)$ is the braid index,
- $g(K)$ is the Seifert genus,
- $k(K)$ is the canonical layer / dressing index already used in the main text.

[ORTHODOX] The invariants $\mathcal{L}_{\rm tot}(K)$, $b(K)$, and $g(K)$ are standard topological or geometric knot descriptors.

[DERIVED / MATCHING-ANSATZ GUARD] The dimensional admissibility of Eq. (302) is derived. The displayed three-coefficient form and the $\varphi^{-2k(K)}$ dressing are a benchmark matching ansatz; the coefficients $\alpha_{\mathcal{L}}$, α_b, and α_g remain calibrated unless a separate variational or statistical-mechanical calculation derives them.

Substituting Eq. (302) into Eq. (42) yields the explicit benchmark mass branch

$$M_K^{(\rm exp)}(x) = \rho_m(x)\, {r_c}^3\, \Pi_K\, \Xi_K^{(\rm exp)}\, S_{(t)}^{\circlearrowleft}(x)^{-2}. \tag{303}$$

E.6 Baryon benchmark closure

For composite baryons built from constituent topology classes K_u and K_d, a conservative additive benchmark branch is

$$M_B^{(\rm add)} = \Pi_B\, M_0 \left(n_u\, \Xi_{K_u}^{(\rm exp)} + n_d\, \Xi_{K_d}^{(\rm exp)}\right), \tag{304}$$

where n_u and n_d are constituent counts and M_0 is the common medium scale extracted from the main mass functional.

A retained legacy golden-closure benchmark is

$$M_p^{(\varphi)} = \varphi^{-2} 3^{-1/\varphi} \left(2M_u + M_d\right), \tag{305}$$

$$M_n^{(\varphi)} = \varphi^{-2} 3^{-1/\varphi} \left(M_u + 2M_d\right). \tag{306}$$

[SPECULATIVE] Equations (305)–(306) are retained as baryon-sector closure benchmarks only. They are not promoted to universal mass laws.

E.7 Benchmark promotion requirement

A geometric mass branch is promotable only after the following fields are fixed for every benchmark state:

$$\{K,\, \gamma_K,\, {\rm Rop}_{\rm rad}(\gamma_K),\, g_{\rm exp}(T),\, K_{\rm top}(T),\, M_{\rm pred},\, M_{\rm obs},\, \epsilon_M\}\,. \tag{307}$$

Until this table is reproducible, Eq. (301) remains a geometric benchmark branch rather than a universal mass theorem.

E.8 Canonical use policy

The explicit closures above are intended for:

1. theorem-ladder benchmarking,
2. archived CSV reproduction,
3. side-by-side comparison of topology branches,
4. explicit baryon fits.

They are *not* intended to replace the abstract organizing form of the main text.

F Effective Quantum Bridge

F.1 Purpose and status

The main canon states that quantum mechanics should appear as an effective low-energy description of the medium. The present appendix supplies the minimal bridge formulas needed to make that statement mathematically legible without claiming a completed derivation of the full quantum rule set.

[**ORTHODOX**] The Madelung decomposition and Schrödinger-form continuum limit are standard.

[**DERIVED**] SST may use these formulas as a coarse-grained field representation once a local amplitude density and phase have been specified.

[**SPECULATIVE**] The identification of the resulting fields with branch-resolved swirl carriers remains an SST interpretation.

[**CRITICAL NOTE**]: ρ_ψ is a coarse-grained mode-support density of the emergent carrier field, *not* a compression of the incompressible substrate ρ_f (Axiom 1, $\nabla \cdot \mathbf{v} = 0$). The two densities are distinct objects, so the Madelung amplitude does not violate substrate incompressibility.

F.2 Amplitude–phase decomposition

Introduce an effective complex mode field

$$\psi(x,t) = \sqrt{\rho_\psi(x,t)}\, e^{i\theta(x,t)}, \tag{308}$$

with $\rho_\psi \geq 0$ and real phase θ. The associated effective transport velocity is

$$\mathbf{u}_\psi = \frac{\hbar}{m_{\text{eff}}} \nabla\theta. \tag{309}$$

Assuming a Schrödinger-form bridge equation

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m_{\text{eff}}}\nabla^2 + V_{\text{eff}}\right)\psi, \tag{310}$$

one obtains the equivalent real system

$$\partial_t\rho_\psi + \nabla \cdot \left(\rho_\psi \mathbf{u}_\psi\right) = 0, \tag{311}$$

$$\partial_t\theta + \frac{m_{\text{eff}}}{2}\|\mathbf{u}_\psi\|^2 + V_{\text{eff}} + Q_{\text{eff}} = 0, \tag{312}$$

with effective quantum potential

$$Q_{\text{eff}} = -\frac{\hbar^2}{2m_{\text{eff}}}\frac{\nabla^2\sqrt{\rho_\psi}}{\sqrt{\rho_\psi}}. \tag{313}$$

[**ORTHODOX**] Equations (311)–(313) are the standard Madelung system associated with Eq. (310) [14, 15].

F.3 SST reading of the bridge

The conservative SST interpretation is:

$$\rho_\psi \leftrightarrow \text{coarse-grained mode-support density,} \tag{314}$$
$$\theta \leftrightarrow \text{effective circulation phase,} \tag{315}$$
$$\mathbf{u}_\psi \leftrightarrow \text{phase-transport velocity of the low-energy envelope.} \tag{316}$$

At this level, the branch-resolved atomic sector of Section 8 is read not as a replacement of the orthodox wavefunction, but as a candidate microscopic underpinning for the effective fields ρ_ψ and θ.

F.4 Two-component branch bookkeeping

A minimal two-branch extension consistent with the $\sigma = \pm 1$ structure of the atomic sector is

$$\Psi = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}, \qquad \psi_\pm = \sqrt{\rho_\pm}\, e^{i\theta_\pm}, \tag{317}$$

with total coarse-grained density

$$\rho_{\text{tot}} = \rho_+ + \rho_-. \tag{318}$$

[**SPECULATIVE**] This two-component structure is the minimal SST bridge to spinor-like bookkeeping. It is not yet a first-principles derivation of fermionic spin statistics.

F.5 Topological reading of the bridge

The most conservative topology statement retained at appendix level is that nontrivial phase maps may encode nontrivial carrier organization. In particular, the effective phase field may carry nontrivial winding or linking data even when the coarse-grained description is written in wave language. This is the appendix-level reason why the canon treats topology and wave structure as compatible rather than as competing descriptions.

G Geometric Impedance Bridge Status Guard

[**RESEARCH-TRACK POINTER / MAIN-CANON GUARD**]**.** The geometric impedance bridge is not part of the closed main-canon derivation chain. It is retained as a research-track reparameterization lemma in the companion file, Sec. K. It may be used for comparison, normalization, and falsifier design, but it must not be used as a derivation of Einstein dynamics or as a replacement for the retained Poisson/clock-field bridge.

H Dark-Sector and Galactic Continuum Law

H.1 Purpose and status

The main canon already retains the dark-knot taxonomy and the radial swirl pressure law. This appendix collects the next phenomenological layer: the galaxy-scale force law and the simplest rotation-curve ansatz inherited from earlier SST development.

[**ORTHODOX**] The underlying force law is ordinary Euler balance in steady swirl.

[**DERIVED**] SST reinterprets that balance as a candidate large-scale acceleration source.

[**SPECULATIVE**] The explicit galaxy-fit profile below is a phenomenological closure ansatz, not a theorem.

[**CRITICAL NOTE**] The canonical $\rho_f = 7.0 \times 10^{-7}\,\mathrm{kg\,m^{-3}}$ is an effective medium parameter, not an ordinary gravitating baryonic/cosmological rest-mass density. Applying it on galactic scales requires a separate coupling and screening law. Without that law, the galaxy-scale pressure profile is a fit ansatz only and must not be interpreted as a literal dense fluid through which stars experience ordinary hydrodynamic drag.

H.2 Effective dark acceleration law

Starting from the steady radial pressure law of Eq. (124),

$$\frac{1}{\rho_f}\frac{dp_{\text{swirl}}}{dr} = \frac{v_\theta(r)^2}{r}, \tag{319}$$

define the effective SST acceleration by

$$a_{\text{SST}}(r) := \frac{1}{\rho_f}\frac{dp_{\text{swirl}}}{dr}. \tag{320}$$

Then

$$a_{\text{SST}}(r) = \frac{v_\theta(r)^2}{r}. \tag{321}$$

[DERIVED] Equations (320)–(321) are immediate consequences of the retained pressure law.

H.3 Two-component galactic rotation ansatz

A retained phenomenological velocity profile is

$$v_{\text{swirl}}(r) = \frac{C_{\text{core}}}{\sqrt{1+\left(r_c^{\text{gal}}/r\right)^2}} + C_{\text{tail}}\left(1-e^{-r/r_c^{\text{gal}}}\right), \tag{322}$$

with corresponding total circular speed

$$v_{\text{tot}}^2(r) = v_{\text{bar}}^2(r) + v_{\text{swirl}}^2(r). \tag{323}$$

[SPECULATIVE] Equation (322) is a phenomenological fit ansatz and should be used only at the appendix / research-track level.

H.4 Relation to dark-knot taxonomy

The taxonomy of Eq. (112) classifies low-helicity, low-instability knot sectors as dark or quasi-dark candidates. The present galactic law does *not* assume that all galactic anomalies are produced by localized dark knots alone. Rather, it provides the complementary continuum statement: long-range swirl pressure gradients may produce effective extra acceleration even when the underlying microscopic carriers are topologically sparse or weakly luminous.

H.5 Use policy

The galactic law belongs in the canon only as:

1. an appendix-level phenomenological sector,
2. a separate astrophysics / dark-sector track,
3. a fit target for future rotation-curve studies.

It is deliberately not promoted in v0.8.18 to the same status as the primitive constant chain, the Swirl-Clock, or the delay-selection sector.

I Research-track sectors migrated from the main canon

This companion document collects those sectors that remain useful for continuity and future development but are too long, too provisional, or too phenomenology-specific for the core body of *Swirl-String-Theory_Canon-v0.8.19*. Their migration out of the main canon is editorial rather than negative: each subsection remains part of the broader SST research program, but none is required for the minimal logical closure of the core canon.

Editorial note (v0.8.19): the maximal swirl tension sector ($F^{\max}_{\text{swirl}}$, Rydberg bridge identities, Gibbons-class maximum-tension comparison, and the threefold interpretation table) is developed in the core document under `been_processed/v0.8.19/SST_CANON-v0.8.19.tex`, subsection `Maximal Swirl Tension` (label `sec:Fmax`); it is not duplicated here.

I.1 Thermodynamic Radiation Interface

A recurring legacy idea is that a local swirl-energy density may define an effective temperature field and thereby an emergent radiation sector. Let

$$T_{\text{eff}} = T_{\text{eff}}\big(\rho_E,\, \omega,\, S^{\circlearrowleft}_{(t)}\big) \tag{324}$$

denote an unspecified constitutive relation. A minimal legacy interface then asks whether Planck-type spectra can emerge from a medium whose local excitation scale is set by T_{eff}.

[**ORTHODOX**]: blackbody spectra are governed by Planck's law once a valid temperature variable is defined.

[**SPECULATIVE**]: SST does not yet derive the constitutive relation above, so this sector remains a research program rather than canonically closed physics.

I.2 Minimal Coupling Interface to QED

Early research notes repeatedly explored whether an effective vector potential could be built from circulation variables. The most conservative placeholder takes the form

$$\nabla \to \nabla - i\frac{m}{\hbar}A_{\text{eff}}, \qquad A_{\text{eff}} = \chi\, v, \tag{325}$$

where χ is a scalar weight and v is the local carrier velocity field.

[**ORTHODOX**]: minimal coupling is the standard gateway from free to gauge-coupled wave dynamics.

[**SPECULATIVE**]: the above substitution is only an analogy template unless a full gauge-covariant derivation is supplied.

I.3 Ribbon-Invariant Chirality Refinement

The older canon layers also suggested using the Calugareanu–White–Fuller relation

$$Lk = Tw + Wr \tag{326}$$

as a refinement of the knot-based matter/antimatter dictionary.

[**ORTHODOX**]: this is a standard geometric identity for framed curves and ribbons.

[**DERIVED BOOKKEEPING**]: once a framed curve is specified, the identity separates writhe and twist contributions.

[**SPECULATIVE**]: the dynamical stability of writhe-dominated or twist-dominated realizations, and the mapping from $\text{sign}(Tw)$ or related chirality measures to particle/antiparticle sectors, remains a model hypothesis.

I.4 Particle-Candidate Mapping Layer

[**SPECULATIVE**] **Working dictionary:** a minimal knot/composition mapping retained from earlier SST work is summarized in Table 5. This table is organizational rather than definitive; it records the current topological assignments used by the mass and taxonomy program.

[**CRITICAL NOTE**]: Table 5 is a working dictionary only. A full canonical assignment requires an explicit invariant set and a mass-functional comparison program.

Knot class / composition	Candidate	Status / role
3_1 trefoil	electron / positron sector	baseline chiral lepton calibration
$T(2,5)$ or equivalent higher torus class	muon	lepton hierarchy candidate
$T(2,7)$ or equivalent higher torus class	tau	lepton hierarchy candidate
twist/clasp knots, quasi-unknot loops	quark-like defects	research-track fractional boundary winding candidates
$5_2 + 5_2 + 6_1$ and nearby composites	baryon candidates	legacy mass-functional dictionary, still under audit
triple-gear locked analogue	proton-like charged baryon analogue	mechanical model for nonzero exterior collective rotation
Borromean-type analogue	neutron-like neutral baryon analogue	mechanical/topological model for zero exterior charge with internal triple linking

Table 5: Minimal knot-class and analogue summary (v0.8.19 research-track), aligned with main canon semantic corrections.

I.5 Quark-Like Twist Knots as Quasi-Unknot Defects

[**RESEARCH TRACK / CANON CANDIDATE**] Twist knots and related clasp knots may be interpreted as quasi-unknot carrier loops with a localized twist/clasp defect. Geometrically such a knot can be stretched so that it resembles an unknot with a central aperture, while topologically remaining nontrivial until a crossing change or reconnection occurs.

A useful diagnostic decomposition is

$$K_{\mathrm{twist}} \simeq 0_1 + \Delta_{\mathrm{clasp}}, \tag{327}$$

where Δ_{clasp} denotes a localized topological defect region. Candidate audit quantities are

$$\mathcal{U}_{\mathrm{quasi}}, \qquad A_{\mathrm{aperture}}, \qquad f_{\mathrm{clasp}}, \qquad f_{\kappa,\mathrm{localized}}, \tag{328}$$

standing for quasi-unknot score, central aperture, clasp-localization fraction, and curvature-concentration fraction.

[**CHARGE PLACEHOLDER**] Fractional quark-like charge should not be assigned to the trefoil. If retained, it is represented as fractional exterior boundary winding,

$$\frac{Q}{e} = q_\partial, \qquad q_u = +\frac{2}{3}, \qquad q_d = -\frac{1}{3}. \tag{329}$$

The topological knot supplies the defect carrier; q_∂ supplies the charge-sector bookkeeping.

I.6 Twist-Ladder Closure from a Folded Clasp

[**ORTHODOX / RESEARCH TRACK**]. The rope experiment described in the project discussion is naturally modeled as a twist box plus a single clasp closure. If m denotes the number of signed half-twists retained in the folded region at closure, the resulting twist-knot ladder is

$$K_m = \text{clasp} + m\text{-half-twist box}, \qquad m \in \mathbb{Z}. \tag{330}$$

For the positive convention used here,

$$K_1 = 3_1, \qquad K_2 = 4_1, \qquad K_3 = 5_2, \qquad K_4 = 6_1, \qquad K_5 = 7_2, \qquad K_6 = 8_1. \tag{331}$$

The standard invariant checks are

$$c_{\min}(K_m) = m + 2, \qquad \det(K_m) = 2m + 1 \quad (m \geq 1). \tag{332}$$

Changing the twist sign gives the mirror sector,

$$K_{-m} = \overline{K_m}. \tag{333}$$

[CANON-COMPATIBLE INTERPRETATION]. End-rotation and plectonemic folding redistribute twist and writhe according to the framed-ribbon bookkeeping $Lk = Tw + Wr$, but they do not change the topological class of an already closed ideal carrier. The knot class is fixed only when the clasp/boundary closure is performed. This module is therefore a research-track generator for twist/clasp candidates and for the $m = 1$ trefoil closure picture; it is not yet a dynamical theorem of Euler–SST.

I.7 Triple-Gear and Borromean Baryon Analogues

[**MECHANICAL ANALOGUE / NOT A PROOF**] The triple-gear print and Borromean-type links are retained as baryon analogues, not as final particle assignments.

The proton-like analogue is a three-component locked system with nonzero exterior collective rotation,

$$\mathcal{B}_p^{\text{gear}} = (C_1, C_2, C_3;\ q_\partial = +1,\ \Theta_{\text{ext}} \neq 0), \tag{334}$$

where the charge proxy is the boundary winding q_∂, not raw pairwise helicity.

The neutron-like analogue is a Borromean or Borromean-adjacent system with no net exterior boundary winding but nontrivial three-component topology,

$$\mathcal{B}_n^{\text{Bor}} = (C_1, C_2, C_3;\ q_\partial = 0,\ \mu_{123} \neq 0). \tag{335}$$

This preserves the intuition that the neutron is neutral externally while still carrying internal topological structure.

[**FALSIFIERS**] The proton analogue fails if the locked triple system has no stable exterior collective phase. The neutron analogue fails if the Borromean sector necessarily induces nonzero exterior boundary winding.

I.8 Trefoil–Gear Kinematic Non-Equivalence

[**RESEARCH TRACK**] The triple gear and segmented trefoil are not kinematically identical even if they use related printed parts. The gear model measures rotation about a straight axle; the trefoil model measures phase transport along a curved framed tube.

For the gear analogue,

$$\omega_{\text{gear}} = \frac{1}{5}\,\omega_{\text{core}} \tag{336}$$

if five full core rotations produce one full gear rotation.

For the segmented trefoil Blender model, the shell/blade winding budget is

$$N_{\text{shell,total}} = 10, \qquad N_{\text{shell,segment}} = \frac{10}{3}. \tag{337}$$

If the same core-shaft mechanism gives five core rotations per segment, then

$$\frac{N_{\text{shell,segment}}}{N_{\text{core,segment}}} = \frac{10/3}{5} = \frac{2}{3}. \tag{338}$$

This 2/3 is therefore a conditional shell/core kinematic ratio, not a trefoil-electron charge assignment.

I.9 Higgs-as-Trefoil High-Temperature Hypothesis

[**PARKED / SPECULATIVE**] The idea that a Higgs-like state is a thermally activated trefoil phase is not canon-ready. A single trefoil is chiral and lepton-like in the current taxonomy, whereas a Higgs-like excitation must behave effectively as a scalar sector.

A viable future version would need at least one of the following mechanisms:

1. a trefoil–anti-trefoil paired condensate with scalar effective symmetry,
2. a high-temperature unframed phase where chirality averages out,
3. an independent scalar amplitude mode coupled to trefoil attachment without identifying the Higgs with the trefoil itself.

Until such a mechanism is derived, the hypothesis remains outside the canon.

I.10 Hydrodynamic Exchange and the Pauli Barrier

[ORTHODOX]: for two thin filamentary loops C_1 and C_2 in an incompressible medium, the Biot–Savart interaction energy is

$$E_{\mathrm{int}} = \frac{\rho_f \Gamma_1 \Gamma_2}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\ell_1 \cdot d\ell_2}{\|r_1 - r_2\|}. \tag{339}$$

[DERIVED]: if two identical electron-candidate loops are forced into the same spatial state, regularization at a short-distance cutoff a_{cut} yields the overlap penalty

$$V_{\mathrm{Pauli}}(d) \approx \frac{\rho_f \Gamma_0^2}{4\pi} \iint \frac{d\ell_1 \cdot d\ell_2}{\sqrt{\|r_1(\theta_1) - r_2(\theta_2) + d\|^2 + a_{\mathrm{cut}}^2}}, \tag{340}$$

with maximal-overlap scale

$$V_{\mathrm{Pauli}}^{\max} \approx \frac{\rho_f \Gamma_0^2}{4\pi} \left(\frac{L}{a_{\mathrm{cut}}} \right) \mathcal{O}(1). \tag{341}$$

Using the canonical values

$$\rho_f = 7.0 \times 10^{-7}\ \mathrm{kg\,m^{-3}}, \tag{342}$$

$$r_c = 1.40897017 \times 10^{-15}\ \mathrm{m}, \tag{343}$$

$$\Gamma_0 = 2\pi r_c \,\|\mathbf{v}_{\circlearrowleft}\| \approx 9.6836 \times 10^{-9}\ \mathrm{m^2\,s^{-1}}, \tag{344}$$

and $L \sim 2\pi a_0$ at the hydrogenic scale, the legacy benchmark corresponds to the cutoff calibration $a_{\mathrm{cut}} = r_c$. This does *not* identify the resolved tube radius a_{core} with r_c; it only fixes the ultraviolet cutoff used in this benchmark. One obtains

$$V_{\mathrm{Pauli}}^{\max} \sim 1.23 \times 10^{-18}\ \mathrm{J} \sim 7.6\ \mathrm{eV}. \tag{345}$$

[CRITICAL NOTE]: this is retained as a cutoff-calibrated scale benchmark, not yet as a closed theorem. Its role is to show that the exclusion-energy estimate falls in an atomically relevant window rather than at obviously pathological scales. A physical resolved-core calculation must use an independently specified a_{core} or profile model.

I.11 Numerical Benchmark Layer

[DERIVED] Computational-program requirement: earlier SST work contained explicit benchmark tables for particle and atomic masses. In the present canon, the benchmark layer is retained as a reproducibility requirement rather than as a stand-alone authority claim.

Particle	m_{exp} (MeV)	m_{SST} (MeV)	rel. error	mode class
e	0.51099895	0.51099895	0	predictive/closure
μ	105.6583745	...	...	predictive candidate
τ	1776.86	...	...	predictive candidate
p	938.272088	...	...	predictive or closure
n	939.565420	...	...	predictive or closure

Table 6: Benchmark summary structure to be populated from canonical scripts and archived CSV output.

[CRITICAL NOTE]: a future v0.8.x benchmark appendix should include the exact script or package path used for generation, archived CSV outputs, mode definitions (predictive vs. exact-closure), and uncertainty propagation on canonical constants and geometric factors.

I.12 Helicity-by-base archive (diagnostic layer)

[**DERIVED**] **Archive role:** the SST helicity-by-base outputs are retained as a measurable balance diagnostic for taxonomy support, not as an independent law. Define the midpoint-distance diagnostic

$$\Delta_H(K) = \left|\widehat{\mathcal{H}}_b(K) + \frac{1}{2}\right|, \tag{346}$$

where $\widehat{\mathcal{H}}_b(K)$ is the archived normalized helicity-like base indicator for class K.

Near-balanced values with $\Delta_H(K) \ll 1$ indicate candidate dark or quasi-dark sectors only when symmetry and instability filters are also satisfied.

[**CRITICAL NOTE**]: no single helicity-like scalar is sufficient for sector assignment; classifier decisions remain conjunction rules over symmetry, helicity balance, and low-order stability content.

[**CANON WORKFLOW NOTE**]: if SST-labelled and parallel VAM-labelled helicity outputs are numerically identical, the SST-labelled archive is the only required source for canon workflows.

I.13 Trefoil-closure archive (benchmark discipline)

[**DERIVED**] **Benchmark role:** ideal-trefoil robustness sweeps, final-best-estimate tables, and theorem-style reports are retained as benchmark-discipline infrastructure.

[**CRITICAL NOTE**]: practical best-estimate closure and theorem-level closure are distinct. A result may remain useful for calibration while still failing strict theorem-level tail or postcheck requirements.

[**REQUIRED METADATA**]: benchmark reports should include explicit pass/fail logic, postchecks, extrapolation summaries, and local-curve diagnostics so that closure claims are auditable.

I.14 G7: Ring Constant versus Trefoil-Geometric Closure

[**NUMERICAL OBSERVATION / OPEN GATE / NOT CANONICAL**]. Track-B calculations give the numerical vortex-ring constant

$$\alpha_{\text{ring}} = 1.61935\ldots = \varphi + 0.00132\ldots, \tag{347}$$

with

$$\frac{\alpha_{\text{ring}} - \varphi}{\varphi} \approx 8.16 \times 10^{-4} = 0.0816\%. \tag{348}$$

This is retained as a genuine sub-per-mille numerical proximity between the GP/NLSE ring-energy constant and the golden ratio.

However, the identification

$$\alpha_{\text{ring}} \stackrel{?}{=} \mathcal{C}_{\text{trefoil}} = \frac{R_{\text{eff}}}{a_{\text{eff}}} \tag{349}$$

is not derived. The quantity α_{ring} is a vortex-ring energy constant, whereas $\mathcal{C}_{\text{trefoil}}$ is a geometric centerline-thickness observable of an ideal trefoil.

The recursive closure relation

$$\mathcal{C}_{\text{cl}} = 1 + \frac{1}{\mathcal{C}_{\text{cl}}} \tag{350}$$

implies

$$\mathcal{C}_{\text{cl}} = \varphi, \tag{351}$$

but this proves only the consequence of the ansatz. It does not prove that the ideal trefoil realizes the ansatz.

We therefore split G7 into three independent subgates:

$$\text{G7a}: \quad \alpha_{\text{ring}} \to \varphi, \tag{352}$$

$$\text{G7b}: \quad \mathcal{C}_{\text{trefoil}} = \frac{R_{\text{eff}}}{a_{\text{eff}}} \to \varphi, \tag{353}$$

$$\text{G7c}: \quad \xi = r_c. \tag{354}$$

At present all three remain open. In particular, $\xi = r_c$ is an imposed core-matching condition in the NLSE/GP diagnostic layer, not a derived identity.

Canon rule. The canon may cite

$$\alpha_{\mathrm{ring}} \approx \varphi \tag{355}$$

as a numerical observation, but it must not cite this as evidence that

$$\mathcal{C}_{\mathrm{trefoil}} \approx \varphi \tag{356}$$

without an independent geometric convergence test on ideal-trefoil data.

I.15 ϕ_{3_1} transform archive (deferred)

[SPECULATIVE] Status: retained as research-track spectral evidence for later zeta/spectral programs.

[**SCOPE RULE**]: this archive is not part of the current dark-sector or galactic canonization closure path in v0.8.x.

I.16 Gauge-Sector Roadmap

[SPECULATIVE] Structural roadmap: a retained minimal statement of the gauge program is

$$g_{\mathrm{eff}} = su(3) \oplus su(2) \oplus u(1), \tag{357}$$

with effective gauge phases organized as holonomy data of multi-director transport.

A minimal phase-based representation is

$$A^a_\mu(x) = \partial_\mu \theta^a(x), \tag{358}$$

and clock coupling acts as a common phase reference,

$$\theta^a(x) \mapsto \theta^a(x) + q^a \chi(x) \quad \Rightarrow \quad A^a_\mu \mapsto A^a_\mu + q^a \partial_\mu \chi. \tag{359}$$

[**CRITICAL NOTE**]: the gauge layer is retained only as a roadmap statement. Running couplings, anomaly cancellation, and realistic mixing matrices remain open work.

J Dark-sector and galactic canonization execution package (v0.8.x)

J.1 Stage-0 freeze: topology toolkit and benchmark protocol

[DERIVED] Scope freeze: the mandatory topology toolkit for v0.8.x canonization is fixed to:

1. Gauss linking integral,
2. helicity in ideal fluids,
3. thin-tube helicity reduction,
4. Hopf invariant,
5. basic torus-knot invariants (c, b, g),
6. Calugareanu–White–Fuller relation $Lk = Tw + Wr$.

[DERIVED] Notation freeze: one notation family is used in this package:

$$\rho_f, \quad v_\theta(r), \quad p_{\mathrm{swirl}}(r), \quad \mathcal{H}(K), \quad N_u^{(1)}(K), \quad \Gamma, \quad \kappa, \quad S^{\circlearrowleft}_{(t)}.$$

No parallel symbol aliases are introduced for the same quantity.

[DERIVED] Mandatory benchmark protocol template:

1. **Inputs:** canonical constants, topology identifiers, profile parameters, and dataset/version tags.
2. **Procedure:** exact script or package path, run command, and branch/commit hash.

3. **Outputs:** archived CSV files with fixed column schema and an attached metadata note.
4. **Uncertainty:** first-order propagation for calibrated constants and sensitivity sweep on profile parameters.
5. **Rerun:** deterministic rerun steps and acceptance tolerances.

[**CANON WORKFLOW NOTE**]: every promoted quantitative claim in this package is invalid without an explicit input-to-CSV trace.

[**CANON WORKFLOW NOTE**] **Benchmark schema hardening (v0.8.19+):** Every dark-sector or galactic benchmark table must include, at minimum, the following machine-readable columns:

`run_id`,	`canon_version`,	`script_path`,	`commit_hash`,
`knot_id`,	`symmetry_metadata_id`,	`branch_id`,	`dataset_id`,
`input_constants_id`,	`profile_parameters`,	`observable_name`,	`observable_value`,
`observable_unit`,	`uncertainty_model`,	`tolerance_rule`,	`pass_fail`.

The corresponding metadata note must state whether each quantity is [**ORTHODOX INPUT**], [**DERIVED**], [**CALIBRATED**], [**CONDITIONAL**], or [**SPECULATIVE**]. A benchmark cannot promote a dark-sector rule unless the CSV row identifies both the topology metadata and the dynamical branch used to compute the observable.

J.2 Stage-1 freeze: dark-sector Euler pressure anchor

[**ORTHODOX**] **Euler radial balance (steady, axisymmetric, azimuthal-dominant):**

$$\frac{1}{\rho_f}\frac{dp_{\mathrm{swirl}}}{dr} = \frac{v_\theta(r)^2}{r}. \tag{360}$$

[**DERIVED**] **Flat-tail limit (constant tail speed):**

$$v_\theta(r) \to v_0 \quad \Longrightarrow \quad p_{\mathrm{swirl}}(r) = p_0 + \rho_f v_0^2 \ln\left(\frac{r}{r_0}\right). \tag{361}$$

[**DERIVED**] **Assumption block:** incompressible, inviscid, stationary, axisymmetric, azimuthal-dominant flow, with validity restricted to regions where radial drift and stratification corrections are subleading.

[**DERIVED**] **Dimensional check:**

$$\left[\frac{1}{\rho_f}\frac{dp}{dr}\right] = \frac{\mathrm{kg\,m^{-2}\,s^{-2}}}{\mathrm{kg\,m^{-3}}} = \mathrm{m\,s^{-2}} = \left[\frac{v^2}{r}\right].$$

[**SPECULATIVE**] **SST interpretation:** Eq. (360) is interpreted as a classifier/driver layer for dark-sector support. This interpretation is model-level and separate from the orthodox derivation.

[**CRITICAL NOTE**]: Applying Eq. (360) unscreened with the microscopic $\rho_f = 7.0\times10^{-7}\,\mathrm{kg\,m^{-3}}$ at galactic scales ($v_\theta \sim 200\,\mathrm{km\,s^{-1}}$, $r \sim 10\,\mathrm{kpc}$) gives $dp/dr \sim 10^{-16}\,\mathrm{Pa\,m^{-1}}$ and $\Delta p \sim 2.8\times10^{3}\,\mathrm{Pa}$ over 1 kpc, far above the observed ISM. The swirl pressure is not the thermal ISM pressure, but no screening/scaling law for ρ_f across scales is yet derived; this sector stays research-track and is falsifiable on exactly this point.

[**DERIVED**] **Attached observables/falsifiers:**

1. Rotation-curve consistency after pressure reconstruction from archived $v_\theta(r)$ data.
2. Null falsifier: systematic mismatch in the reconstructed pressure gradient sign or scale across validated radial windows.

J.3 Stage-2 freeze: measurable dark taxonomy

[**SPECULATIVE**] **Classifier variables:** symmetry class (including amphichirality status), chirality sign/state, helicity-balance indicator $\Delta_H(K)$, and low-order instability count $N_u^{(1)}(K)$.

[SPECULATIVE / TAXONOMY METADATA] Symmetry refinement fields: For v0.8.19+ benchmarks, the symmetry gate is not a single Boolean. The taxonomy row for a candidate knot state must record

$$\mathfrak{s}(K) = \big(A_\pm(K),\ D_2(r;K),\ I_{\rm strong}(K),\ \chi(K), \Delta_H(K),\ N_u^{(1)}(K)\big), \tag{362}$$

where $A_\pm$ distinguishes none/positive/negative/full amphichirality status, $D_2(r;K)$ records declared dihedral rotational symmetry metadata when available, and $I_{\rm strong}$ records whether the knot-table entry is strongly invertible. Unknown entries are reported as `unknown`; they must not be silently treated as false.

Candidate	Symmetry role	Required extra metadata	Allowed taxonomy use
3_1	chiral visible baseline	chirality sign, helicity sign, $N_u^{(1)}$	visible calibration / control row
4_1	amphichiral baseline candidate	$A_\pm$, strong-inversion flag, Δ_H, $N_u^{(1)}$	dark candidate only after stability gate
8_{17}	high-symmetry candidate row	verified amphichirality type, $D_2(r)$ flag, Δ_H, $N_u^{(1)}$	candidate benchmark row; not canon by name alone
15_{331}	deep-table candidate row	external knot-table verification, $D_2(r)$ flag, Δ_H, $N_u^{(1)}$	archive/audit row until metadata is complete
Borromean-type link	quasi-dark control	link invariants, weak-coupling branch, $N_u^{(1)}$	quasi-dark or neutral-control benchmark

[CRITICAL NOTE]: naming a knot in this table is not a promotion to the dark sector. The table only fixes the minimum metadata needed to test the classifier without collapsing amphichirality, strong invertibility, and dynamical stability into one label.

[SPECULATIVE] Decision rules (symmetry-first, then stability-filtered):

$$\text{Dark}:\ \text{amphichiral candidate}\ \wedge\ \Delta_H(K)\ll 1\ \wedge\ N_u^{(1)}(K)=0, \tag{363}$$

$$\text{Quasi-dark}:\ \text{near-balanced symmetry/helicity}\ \wedge\ 0<N_u^{(1)}(K)\le 2, \tag{364}$$

$$\text{Visible}:\ \text{chiral-dominant and/or helicity-unbalanced sectors.} \tag{365}$$

[CANON WORKFLOW NOTE] Ambiguity tie-break: if classes overlap, choose the class with the strongest conjunction score in the order

$$\text{symmetry gate} \to \text{helicity balance gate} \to \text{stability gate}.$$

[SPECULATIVE] Baseline templates and falsifiers:

- **Visible baseline:** trefoil-like chiral template; falsifier = persistent amphichiral-plus-balanced signature under validated data updates.
- **Quasi-dark baseline:** Borromean-type weak-coupling template; falsifier = stable zero-instability assignment across independent scans.
- **Dark baseline:** figure-eight/amphichiral template; falsifier = robust nonzero chirality or high-instability assignment.

[CANON WORKFLOW NOTE]: amphichiral $\Rightarrow$ dark remains a classifier proposal tested against observables, not a closed theorem.

J.4 Stage-3 freeze: galactic law branch gate and closure route

[CANON WORKFLOW NOTE] Branch-gate decision: Branch A (legacy profile refine) is selected for current v0.8.x closure because it preserves tested limits while allowing benchmark traceability.

Working profile:

$$v_\theta(r) = \frac{C_{\rm core}}{\sqrt{1+(r_c/r)^2}} + C_{\rm tail}\left(1-e^{-r/r_c}\right). \tag{366}$$

[DERIVED] Required checks: small-r behavior, flat-tail behavior, parameter semantics, and pressure reconstruction via Eq. (360).

[DERIVED] Failure modes:

1. non-physical parameter combinations (wrong sign or unstable growth),
2. failure to satisfy tail-limit behavior in benchmark windows,
3. inability to reconstruct a consistent pressure profile from the same archived inputs.

J.5 Stage-4 freeze: probe falsification map

[CANON WORKFLOW NOTE] probe classes are fixed as astrophysical, laboratory pressure/swirl analog, HV-coil/thrust-inspired, optical/torsional coupling, and null dark-sector tests.

Probe class	Equation under test	Null-result meaning	Primary confounder	Expected sign/trend
Astrophysical rotation curves	Eq. (360) + profile closure	weakens pressure-law applicability window	baryonic-model bias	reconstructed dp/dr consistent with v_θ^2/r
Lab swirl analog	Euler radial balance scaling	weakens scale-transfer claim	viscosity/edge forcing	monotone pressure rise with imposed swirl
HV-coil/thrust-inspired	circulation-pressure bridge claims	kills quantized-bridge claim if no scaling	EM/thermal artifacts	response tracks circulation proxy
Optical/torsional coupling	swirl-clock bridge signatures	weakens coupling interpretation	detector drift/noise	phase/timing shift with controlled swirl proxy
Null dark tests	taxonomy conjunction rules	kills classifier branch if conjunction fails	mislabelled topology class	class score decreases under contradictory diagnostics

Table 7: Frozen equation-to-probe map for falsification-first workflow in the v0.8.x dark-sector program.

[CANON WORKFLOW NOTE]: probe outcomes can support or reject already-stated equations, but they do not replace derivation-level closure.

J.6 Promotion snapshot after execution

[DERIVED] post-freeze status in this package:

- Item 6 (topology toolkit): CANON-ready in scope and notation.
- Item 5 (benchmark protocol): CANON-CANDIDATE pending repository-wide protocol reuse.
- Item 1 (Euler pressure anchor): CANON-CANDIDATE with explicit assumptions, limits, and falsifier path.
- Item 2 (dark taxonomy): BRIDGE with measurable classifier gates and falsifiers.
- Item 3 (galactic swirl law): BRIDGE with branch gate closed and benchmark route fixed.
- Item 4 (probe map): CANON-BRIDGE with equation/null-result mapping frozen.

K Research Track: Geometric Impedance Bridge

K.1 Purpose and status

[RESEARCH TRACK / BRIDGE LEMMA / REPARAMETERIZATION]. This research-track appendix records the conservative geometric bridge between an SST topological mass ratio and the Schwarzschild source term. It does not derive Einstein dynamics and does not replace the gravitational sector of the main canon.

K.2 Electron-normalized source term

Using the electron Compton wavelength λ_c and the Planck length L_P, the electron Schwarzschild radius can be rewritten as

$$r_s(e) = \frac{2GM_e}{c^2} = 4\pi\,\frac{L_\mathrm{P}^2}{\lambda_c}. \tag{367}$$

For a topological state T with mass ratio

$$\widetilde{M}(T) = \frac{M(T)}{M_e}, \tag{368}$$

the dimensionless potential factor becomes

$$\Phi(T;r) = \frac{2GM(T)}{rc^2} = \frac{4\pi L_\mathrm{P}^2}{r\lambda_c}\,\widetilde{M}(T). \tag{369}$$

K.3 Use policy

Substituting a geometric SST mass branch into Eq. (369) yields a geometric impedance representation of the source term. The content is the factorization of an already-assumed mass ratio into length ratios and topological gates. It is not a proof that the Einstein equation follows from the swirl medium.

[RESEARCH-TRACK GUARD]. The bridge may be used for comparison, normalization, and falsifier design. It must not be used to claim a completed derivation of general relativity or a replacement of the retained Poisson/clock-field bridge.

L Candidate CANON-v0.8.x Modules from Swirl-Flux, Envelope, and Reconnection Analysis

L.1 Status Classification

The following modules are proposed as CANON-v0.8.x candidates.

1. The Schrödinger kinetic coefficient is retained as an effective envelope stiffness,

   $$\mathcal{S}_e = \frac{\hbar^2}{2M_e}.$$

   The replacement $-\mathbf{v}_{\circlearrowleft}^2\nabla^2/2$ is rejected on dimensional grounds.

2. The Meissner effect is interpreted as macroscopic swirl-flux phase locking in a coherent paired condensate.

3. R/T wave–string interaction is represented by conservation of radiative pseudomomentum plus vorticity impulse.

4. Reconnection is excluded in the ideal sector but allowed in measurement, decay, and dissipative sectors, where helicity is redistributed between topology, writhe, twist, radiation, and dissipation.

L.2 Envelope Stiffness and the Rejection of the $\mathbf{v}_{\mho}^2\nabla^2$ Kinetic Term

For an effective R/T envelope amplitude $\Psi(\mathbf{r}, t)$, the kinetic coefficient must have dimension $\mathrm{J\,m^2}$. The correct envelope stiffness is therefore

$$\mathcal{S}_e \equiv \frac{\hbar^2}{2M_e}, \qquad [\mathcal{S}_e] = \mathrm{J\,m^2}.$$

The effective low-energy envelope equation is

$$i\hbar\frac{\partial\Psi}{\partial t} = \left[-\mathcal{S}_e\nabla^2 + V_{\mathrm{SST}}(\mathbf{r}, t)\right]\Psi.$$

The operator $-\mathbf{v}_{\mho}^2\nabla^2/2$ is not a Schrödinger kinetic energy operator because

$$\left[\mathbf{v}_{\mho}^2\nabla^2\right] = \frac{\mathrm{m}^2}{\mathrm{s}^2}\frac{1}{\mathrm{m}^2} = \mathrm{s}^{-2},$$

which is not an energy.

For tunneling through a one-dimensional effective SST barrier,

$$T \sim \exp\left(-2\int_{x_1}^{x_2}\kappa_{\mathrm{SST}}(x)\,dx\right),$$

where

$$\kappa_{\mathrm{SST}}(x) = \sqrt{\frac{V_{\mathrm{SST}}(x) - E}{\mathcal{S}_e}} = \sqrt{\frac{2M_e\left[V_{\mathrm{SST}}(x) - E\right]}{\hbar^2}}.$$

This is dimensionally consistent because

$$\left[\frac{V - E}{\mathcal{S}_e}\right] = \mathrm{m}^{-2}, \qquad [\kappa_{\mathrm{SST}}] = \mathrm{m}^{-1}.$$

L.3 Numerical Bridge Anomaly

A numerically close but dimensionally non-identical bridge is

$$\mathcal{B}_e = \frac{F_{\mathrm{swirl}}^{\mathrm{max}}{r_c}^3}{5\lambda_c v_{\mho}^*}.$$

Using

$$\lambda_c = \frac{h}{M_e c},$$

one obtains

$$\frac{\hbar^2}{2M_e} = 6.10426431 \times 10^{-39}\,\mathrm{J\,m^2},$$

whereas

$$\mathcal{B}_e = 6.12394931 \times 10^{-39}\,\mathrm{J\,s}.$$

The numerical ratio is

$$\left.\frac{\mathcal{B}_e}{\mathcal{S}_e}\right|_{\text{SI numbers}} = 1.00322479.$$

This is classified as a Research Track numerical bridge, not a canonical identity, because

$$[\mathcal{B}_e] = \mathrm{J\,s} \neq \mathrm{J\,m^2} = [\mathcal{S}_e].$$

The dimensionally meaningful relation is instead

$$5\lambda_c v_{\mho}^* = 5\alpha\frac{h}{2M_e},$$

with both sides having dimension $\mathrm{m^2\,s^{-1}}$.

L.4 Swirl-Flux Locking and the Meissner Effect

[CRITICAL NOTE]: The values $2M_e$ and $q_{\text{pair}} = -2e$ are imported from the orthodox Cooper-pair description. Per Sec. 8.6 (Charge and Circulation), SST does not yet derive gauge charges from circulation alone, so this module is a [CALIBRATED] bridge using empirical pairing inputs, not a first-principles flux-exclusion derivation.

For a coherent paired electronic condensate of mass $2M_e$ and charge $q_{\text{pair}} = -2e$, phase single-valuedness gives

$$\oint_{\partial\Sigma} (2M_e\mathbf{v}_{\text{pair}} + q_{\text{pair}}\mathbf{A}) \cdot d\boldsymbol{\ell} = nh.$$

Since $q_{\text{pair}} = -2e$,

$$2M_e\Gamma_{\text{pair}} - 2e\Phi_B = nh.$$

Thus

$$\Phi_B = \frac{M_e}{e}\Gamma_{\text{pair}} - n\frac{h}{2e}.$$

In magnitude,

$$|\Phi_B| = \left| n\Phi_0 - \frac{M_e}{e}\Gamma_{\text{pair}} \right|, \qquad \Phi_0 = \frac{h}{2e}.$$

For a non-rotating coherent bulk,

$$\Gamma_{\text{pair}} \to 0,$$

and therefore

$$|\Phi_B| = n\Phi_0.$$

The Meissner effect is then interpreted in SST as exclusion of non-quantized magnetic flux from a phase-locked coherent R/T condensate.

L.5 R/T Pseudomomentum–Impulse Exchange

For a closed R/T interaction domain,

$$\frac{d}{dt}(\mathbf{P}_R + \mathbf{I}_T) = 0.$$

Here $\mathbf{P}_R$ denotes radiative R-phase pseudomomentum and $\mathbf{I}_T$ denotes T-phase vorticity impulse,

$$\mathbf{I}_T = \frac{\rho_f}{2}\int_\Omega \mathbf{x} \times \boldsymbol{\omega}\, d^3x.$$

The dimension is

$$[\mathbf{I}_T] = \mathrm{kg\,m\,s^{-1}},$$

so $\mathbf{I}_T$ is a true momentum-like quantity.

For a thin circular swirl string of circulation Γ and radius R,

$$\mathbf{I}_T = \rho_f \Gamma \pi R^2 \hat{\mathbf{n}}.$$

The core circulation scale is

$$\Gamma_0 = 2\pi r_c v_{\circlearrowleft}^* = 9.68361920 \times 10^{-9}\,\mathrm{m^2\,s^{-1}}.$$

L.6 Reconnection and Helicity Redistribution

In the ideal sector, topology is conserved:

$$\frac{D\mathcal{K}}{Dt} = 0,$$

under the assumptions

$$\nabla \cdot \mathbf{v} = 0, \qquad \nu = 0, \qquad \text{barotropic}, \qquad \text{no reconnection}.$$

In the dissipative, measurement, or decay sector, reconnection is allowed only as a non-ideal event:

$$\frac{D\mathcal{K}}{Dt} \neq 0.$$

This is not a free creation mechanism. A reconnection claim must state the trigger class, e.g. $\nu_{\text{eff}} > 0$, a boundary injection, a phase-slip singularity, a resolved-core collapse, or an explicit external forcing term. The helicity budget is written

$$\Delta H_{\text{topology}} + \Delta H_{\text{writhe}} + \Delta H_{\text{twist}} + \Delta H_{\text{radiative}} = -\Delta H_{\text{diss}}.$$

In the weak-dissipation limit,

$$\Delta H_{\text{topology}} + \Delta H_{\text{writhe}} + \Delta H_{\text{twist}} + \Delta H_{\text{radiative}} \approx 0.$$

For filamentary structures,

$$H_{\text{centerline}} = \sum_{i \neq j} \Gamma_i \Gamma_j \mathrm{Lk}_{ij} + \sum_i \Gamma_i^2 \left(\mathrm{Wr}_i + \mathrm{Tw}_i\right).$$

L.7 Spherical Pressure Envelopes

For inviscid incompressible SST flow,

$$\nabla p_{\text{SST}} = \rho_f (\mathbf{v} \cdot \nabla)\mathbf{v}.$$

For an azimuthal swirl profile $v_\theta(r)$,

$$\frac{dp}{dr} = \rho_f \frac{v_\theta^2(r)}{r}.$$

If

$$v_\theta(r) = \frac{\Gamma}{2\pi r},$$

then

$$\frac{dp}{dr} = \rho_f \frac{\Gamma^2}{4\pi^2 r^3},$$

and hence

$$p(r) = p_\infty - \frac{\rho_f \Gamma^2}{8\pi^2 r^2}.$$

Therefore

$$\Delta p(r) = p_\infty - p(r) = \frac{\rho_f \Gamma^2}{8\pi^2 r^2} = \frac{1}{2}\rho_f v_\theta^2(r).$$

At the canonical swirl speed,

$$\frac{1}{2}\rho_f {v_{\circlearrowleft}^{*}}^2 = 4.18774392 \times 10^5\,\text{Pa}.$$

At the core-density stress scale,

$$\rho_{\text{core}} {v_{\circlearrowleft}^{*}}^2 = 4.65848920 \times 10^{30}\,\text{Pa}.$$

The canonical maximum swirl force is recovered as

$$F_{\text{swirl}}^{\max} = \rho_{\text{core}} {v_{\circlearrowleft}^{*}}^2 \pi {r_c}^2 = 29.053507\,\text{N}.$$

L.8 Golden-Layer Scaling

The universal rule

$$E_n = \varphi^n E_0$$

is rejected as too strong. The weaker Research Track ansatz is

$$E_{K,k} = E_K^{(0)} \varphi^{-2k}, \qquad k \in \mathbb{Z}_{\geq 0}.$$

Here K labels a knot family and k labels a relaxation layer inside that family.

L.9 Non-Holonomic Gravity Claims

Podkletnov-type anti-gravity claims are not adopted as Canon. At most, the formal analogy

$$\text{non-holonomic twist} \longleftrightarrow \text{foliation twist/vorticity}$$

may be retained in a historical or speculative appendix. No experimental anti-gravity claim is inferred.

M Mode-Locked Swirl-Coil Excitation and $v_{\circlearrowleft}^{*} = f\Delta x$ Metrology

Status. [RESEARCH TRACK / CANON-CANDIDATE]. This appendix formulates a falsifiable experimental protocol for testing whether externally driven multi-phase coil geometries can inject coherent swirl modes into a bounded measurement region. The construction is not assumed as a new force law. It is a mode-selection and metrology rule for identifying when the canonical characteristic swirl speed

$$\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^{6}\,\mathrm{m\,s^{-1}} \tag{370}$$

is sampled by a laboratory-scale frequency–length product.

M.1 Motivation

The basic dimensional identity

$$\|\mathbf{v}_{\circlearrowleft}\| = f\,\Delta x \tag{371}$$

has the units of speed and is therefore not, by itself, a dynamical claim. Its SST use is metrological: a driven system is said to probe a swirl-resonant channel only when a temporal frequency f and a geometrically defined spatial period Δx satisfy Eq. (371) within experimental bandwidth.

For a circular or quasi-circular coil, the relevant spatial period is generally not an arbitrary external distance. It is the azimuthal wavelength selected by the coil radius and spatial mode number. Hence the physical coil radius enters the resonance condition directly.

M.2 Ring-mode closure condition

Let R_{coil} be the effective radius of the active winding, f_0 the electrical base frequency, $n \in \mathbb{N}$ the temporal harmonic, and $m \in \mathbb{Z} \setminus \{0\}$ the azimuthal spatial mode number around the coil. The spatial wavelength of mode m along the coil circumference is

$$\Delta x_m = \frac{2\pi R_{\mathrm{coil}}}{|m|}. \tag{372}$$

The (n)-th temporal harmonic has frequency

$$f_n = n f_0. \tag{373}$$

The dimensionless swirl-lock parameter is therefore

$$\boxed{\eta_{n,m} = \frac{f_n \Delta x_m}{\|\mathbf{v}_{\circlearrowleft}\|} = \frac{n f_0}{\|\mathbf{v}_{\circlearrowleft}\|}\frac{2\pi R_{\mathrm{coil}}}{|m|}}. \tag{374}$$

A ring mode is swirl-locked when

$$\boxed{\eta_{n,m} = 1.} \tag{375}$$

Equivalently, the resonant coil radius for a given (n, m, f_0) is

$$\boxed{R_{\mathrm{coil}}^{(n,m)} = \frac{|m|\,\|\mathbf{v}_{\circlearrowleft}\|}{2\pi n f_0}}. \tag{376}$$

Dimensional check.

$$[R_{\mathrm{coil}}] = \frac{\mathrm{m\,s^{-1}}}{\mathrm{s^{-1}}} = \mathrm{m}.$$

Thus Eq. (376) is dimensionally well-defined.

M.3 Numerical calibration examples

For the canonical value

$$\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^{6}\,\mathrm{m\,s^{-1}},$$

and a base drive frequency

$$f_0 = 150\,\mathrm{kHz},$$

the first three $m = 1$ radius locks are

$$R_{\mathrm{coil}}^{(1,1)} = \frac{1.09384563 \times 10^{6}}{2\pi(1)(150000)} = 1.1606\,\mathrm{m}, \tag{377}$$

$$R_{\mathrm{coil}}^{(2,1)} = 0.5803\,\mathrm{m}, \tag{378}$$

$$R_{\mathrm{coil}}^{(3,1)} = 0.3869\,\mathrm{m}. \tag{379}$$

For a smaller experimental coil with

$$R_{\mathrm{coil}} = 0.150\,\mathrm{m},$$

the $(n = 3,\ m = 1)$ lock frequency is

$$f_0 = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{2\pi n R_{\mathrm{coil}}} = \frac{1.09384563 \times 10^{6}}{2\pi(3)(0.150)} = 3.8687 \times 10^{5}\,\mathrm{Hz}. \tag{380}$$

Hence a $0.150\,\mathrm{m}$ coil driven at $150\,\mathrm{kHz}$ does not satisfy the $(n = 3, m = 1)$ ring-lock condition. Its lock parameter is

$$\eta_{3,1} = \frac{3(150000)\,2\pi(0.150)}{1.09384563 \times 10^{6}} = 0.3877. \tag{381}$$

This is sub-resonant for the $(3,1)$ ring mode, although it may still couple to different (n,m) channels.

M.4 Three-phase selection rule

Consider a three-phase coil drive with phase index $q = 0,1,2$. If the q-th phase is shifted by $2\pi q/3$, the discrete phase sum for temporal harmonic n and spatial azimuthal mode m is

$$T_{n,m}^{(3\phi)} = \frac{1}{3}\left|\sum_{q=0}^{2} \exp!\left[iq\frac{2\pi}{3}(m-n)\right]\right|. \tag{382}$$

The exact selection rule is

$$\boxed{T_{n,m}^{(3\phi)} = \begin{cases} 1, & m \equiv n \pmod 3, \\ 0, & m \not\equiv n \pmod 3. \end{cases}} \tag{383}$$

Thus a three-phase winding does not merely “add power”; it filters the allowed temporal–spatial mode pairs.

M.5 Double-stack phase-shift filter

Let the coil be built as two vertically separated stacks, with the second stack rotated azimuthally by

$$\delta = 30^\circ = \frac{\pi}{6}. \tag{384}$$

Let ψ denote the electrical inter-stack phase offset. The two-stack spatial filter is

$$L_m(\delta,\psi) = \left|1 + \exp[i(m\delta + \psi)]\right|. \tag{385}$$

For equal stack polarity and zero additional electrical offset ($\psi = 0$), the destructive condition is

$$m\delta = (2q+1)\pi, \qquad q \in \mathbb{Z}. \tag{386}$$

With $\delta = \pi/6$, this gives

$$\boxed{m = 6, 18, 30, \dots} \tag{387}$$

Therefore a 30° double-stack geometry exactly suppresses the sixth azimuthal spatial harmonic and its odd multiples in this idealized equal-amplitude model.

M.6 Golden-duty PWM spectrum

Let the duty cycle be the inverse golden-square fraction

$$d = \varphi^{-2} = \left(\frac{1+\sqrt{5}}{2}\right)^{-2} = 0.381966011\ldots. \tag{388}$$

For an ideal unipolar rectangular pulse train, the magnitude of the n-th Fourier coefficient, ignoring the DC normalization convention, is proportional to

$$P_n(d) = \left|\frac{\sin(\pi n d)}{\pi n}\right|. \tag{389}$$

For $d = \varphi^{-2}$, the first seven relative amplitudes normalized to P_1 are approximately

n	$P_n(d)$	$P_n(d)/P_1(d)$
1	0.296675	1.000
2	0.107508	0.362
3	0.046948	0.158
4	0.079273	0.267
5	0.017794	0.060
6	0.042102	0.142
7	0.038864	0.131

Thus golden-duty PWM strongly suppresses the fifth harmonic, but does not by itself null the sixth and seventh harmonics. In the proposed three-phase double-stack geometry, the suppression of the fifth, sixth, and seventh bands must be understood as a composite filter effect:

$$\boxed{\mathcal{C}_{n,m} = P_n(d)\, T^{(3\phi)}_{n,m}\, S_m(\mathrm{P40}, +11, -9)\, L_m(\delta, \psi)\, G_m(k_n R_{\mathrm{coil}})}. \tag{390}$$

Here:

- $P_n(d)$ is the PWM harmonic amplitude;
- $T^{(3\phi)}_{n,m}$ is the three-phase selection factor;
- $S_m(\mathrm{P40}, +11, -9)$ is the spatial winding factor of the SawShape path;
- $L_m(\delta, \psi)$ is the double-stack filter;
- $G_m(k_n R_{\mathrm{coil}})$ is the geometric near-field coupling factor.

The wave number of the n-th temporal harmonic is

$$k_n = \frac{2\pi n f_0}{\|\mathbf{v}_{\circlearrowleft}\|}. \tag{391}$$

M.7 SawShape winding factor for $\mathrm{P40}, +11, -9$

For a discrete winding on $N = 40$ angular sites, define the node angle

$$\theta_j = \frac{2\pi p_j}{40}, \qquad p_j \in \mathbb{Z}/40\mathbb{Z}. \tag{392}$$

A SawShape path with alternating skip sequence $(+11, -9)$ can be represented by

$$p_{j+1} = p_j + s_j \pmod{40}, \qquad s_j \in \{+11, -9\}. \tag{393}$$

The normalized spatial Fourier factor is then

$$\boxed{S_m(\mathrm{P40}, +11, -9) = \left|\frac{1}{N_s}\sum_{j=0}^{N_s-1} w_j \exp(-im\theta_j)\right|}. \tag{394}$$

Here w_j encodes segment weight, orientation, current direction, layer index, and local segment length. This factor is experimentally accessible: it can be inferred by measuring the spatial FFT of the near-field magnetic map over a circular sampling path.

M.8 Canonical interpretation

The proposed coil does not directly generate a canonical force. Instead, it prepares a structured, rotating, mode-filtered boundary condition for the local field. In SST language, the coil acts as a laboratory-scale swirl-mode injector.

The relevant energy-density comparison is

$$\rho_E = \frac{1}{2}\rho_f \|\mathbf{v}_{\circlearrowleft}\|^2. \tag{395}$$

Using

$$\rho_f = 7.0 \times 10^{-7}\,\mathrm{kg\,m^{-3}}, \qquad \|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^6\,\mathrm{m\,s^{-1}},$$

one obtains

$$\rho_E = \frac{1}{2}(7.0 \times 10^{-7})(1.09384563 \times 10^6)^2 = 4.187 \times 10^5\,\mathrm{J\,m^{-3}}. \tag{396}$$

The corresponding mass-equivalent density is

$$\rho_m = \frac{\rho_E}{c^2} = \frac{4.187 \times 10^5}{(2.99792458 \times 10^8)^2} = 4.659 \times 10^{-12}\,\mathrm{kg\,m^{-3}}. \tag{397}$$

These values should be understood as canonical scale references, not as a claim that a laboratory coil reaches the full canonical swirl speed in ordinary electromagnetic matter.

M.9 Experimental falsification protocol

A coil experiment is SST-relevant only if it passes the following null hierarchy.

1. **Electrical null:** repeat with identical RMS current and power but randomized phase ordering.
2. **Geometric null:** replace SawShape $(P40, +11, -9)$ by a symmetry-preserving circular winding with the same resistance and inductance.
3. **Duty-cycle null:** compare $d = 0.381966$ against $d = 0.500$, $d = 0.333$, and $d = 0.250$ at fixed RMS current.
4. **Stack null:** compare 30° double-stack shift against 0°, 15°, 45°, and 60°.
5. **Mode-lock scan:** sweep f_0 and test whether anomalous coherence peaks occur at
$$f_0^{(n,m)} = \frac{|m|\,\|\mathbf{v}_{\circlearrowleft}\|}{2\pi n R_{\text{coil}}}.$$
6. **Environmental null:** repeat in still air, shielded enclosure, and, where safely possible, reduced pressure to remove ion wind and acoustic coupling.
7. **Force-decomposition null:** decompose any measured force into electromagnetic, thermal, acoustic, EHD, and mechanical vibration contributions before assigning a residual.

Only a residual signal satisfying both

$$\eta_{n,m} \approx 1 \tag{398}$$

and

$$\mathcal{C}_{n,m} \neq 0 \tag{399}$$

is admissible as an SST research-track candidate.

M.10 Vorticity-force decomposition layer

Any measured force must first be written schematically as

$$\mathbf{F}_{\text{meas}} = \mathbf{F}_{\text{EM}} + \mathbf{F}_{\text{EHD}} + \mathbf{F}_{\text{thermal}} + \mathbf{F}_{\text{acoustic}} + \mathbf{F}_{\text{mechanical}} + \mathbf{F}_{\omega} + \mathbf{F}_{\text{res}}. \tag{400}$$

Here $\mathbf{F}_{\omega}$ denotes force reconstructed from ordinary vorticity transport and impulse balance. Only the remaining term $\mathbf{F}_{\text{res}}$, after conventional reconstruction and uncertainty propagation, can be tested against the SST mode-locking rule. The canonical research criterion is therefore

$$\boxed{\mathbf{F}_{\text{res}} = \mathbf{F}_{\text{res}}(\eta_{n,m}, \mathcal{C}_{n,m}) \quad \text{with a peak near} \quad \eta_{n,m} = 1.} \tag{401}$$

M.11 Recommended measured observables

The minimum observable set is:

1. drive current $I_A(t), I_B(t), I_C(t)$;
2. voltage $V_A(t), V_B(t), V_C(t)$;
3. real input power $P_{\text{in}}(t)$;
4. local magnetic field map $\mathbf{B}(\mathbf{x}, t)$;
5. pickup-coil FFT around the active circumference;
6. accelerometer spectrum on the coil frame;
7. air temperature and pressure field if operated in air;
8. force sensor output with synchronized phase reference.

A positive SST candidate requires a phase-stable spectral feature at a predicted (n, m) lock that disappears under at least one of the symmetry-breaking null tests.

M.12 Conclusion

The 3-phase SawShape (P40, +11, −9) double-stack coil with 30° inter-stack rotation and $d = \varphi^{-2}$ PWM is canonically useful as a mode-filtered swirl-excitation apparatus. Its SST value is not that it assumes a new force, but that it makes the frequency–length criterion

$$\|\mathbf{v}_{\circlearrowleft}\| = f\Delta x$$

experimentally precise.

The essential design rule is

$$\eta_{n,m} = \frac{nf_0}{\|\mathbf{v}_{\circlearrowleft}\|}\frac{2\pi R_{\text{coil}}}{|m|} \approx 1,$$

with observable coupling only when

$$\mathcal{C}_{n,m} = P_n(d)\, T^{(3\phi)}_{n,m}\, S_m(\text{P40}, +11, -9)\, L_m(\delta, \psi)\, G_m(k_n R_{\text{coil}})$$

is nonzero. This converts the coil concept into a falsifiable SST metrology protocol.

N Research Appendices: Topology, Stability, and Fluid Mechanics

N.1 Research Appendix: Non-Abelian Topological Charge for Protected Swirl Strings

Canon status. [SPECULATIVE] Research Track. This appendix does not promote a new particle assignment to canon. It records a canon-compatible extension of the SST topology sector: knot type and helicity may be insufficient to determine dynamical stability when reconnection and strand-crossing moves are allowed. A non-Abelian coloring invariant may supply an additional obstruction to decay.

N.1.1 Motivation

The current SST knot sector uses geometric and topological descriptors such as crossing number, ropelength, helicity, chirality, and hyperbolic-volume proxies. A minimal research-track screening form is

$$\frac{E_{\text{screen}}[K]}{E_0} = a_R\, \text{Rop}_{\text{rad}}(\gamma_K) + a_C\, c_{\min}(K) + a_H\, \widehat{\mathcal{H}}(K) + a_G\, I_G(K) + \cdots, \tag{402}$$

where Rop_{rad} is the orthodox radius-convention ropelength, $c_{\min}$ is a crossing/contact-complexity proxy, and $\widehat{\mathcal{H}}$ denotes nondimensional helicity or helicity-derived topological data. The term $I_G(K)$ may appear only after a non-Abelian sector label $Q_G(K)$ has been mapped to a real-valued index $I_G : Q_G(K) \to \mathbb{R}_{\geq 0}$. This is a scoring functional, not a derived Hamiltonian.

Recent work on non-Abelian topological defects shows that some linked structures cannot be removed by local reconnections and strand crossings because the strands carry non-commuting elements of an order-parameter fundamental group [55]. This suggests that SST should distinguish two layers:

1. the geometric knot type K of the swirl string centerline;
2. an internal topological coloring q valued in a discrete or continuous group G.

A concrete benchmark choice is the quaternion group

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}, \qquad ij = k, \qquad ji = -k, \tag{403}$$

so that $ij \neq ji$. The general symbol G below is retained because SST does not yet select a unique internal order-parameter group.

N.1.2 Group-colored swirl-string sector

Let a colored swirl-string configuration be denoted

$$\mathcal{K}_G = (K, q), \tag{404}$$

where

$$q : \pi_1\big(\mathbb{R}^3 \setminus K\big) \longrightarrow G \tag{405}$$

assigns a group element to each Wirtinger generator of the complement. For a diagram with arc generators a_i, the coloring must satisfy the usual crossing consistency relations. In SST language, $q(a_i)$ labels an internal director or phase-sector attached to a local swirl-string strand.

[ORTHODOX] Constraint. For a non-Abelian group G, strand exchange is obstructed when the corresponding labels do not commute:

$$[q(a_i), q(a_j)] = q(a_i)q(a_j)q(a_i)^{-1}q(a_j)^{-1} \neq e_G. \tag{406}$$

[SPECULATIVE] SST interpretation. Equation (406) is interpreted as a local reconnection barrier in the swirl medium. If two strands belong to non-commuting internal sectors, a direct crossing or reconnection requires an additional defect cord or phase bridge. In SST terms this is a candidate mechanism for topological protection beyond ordinary helicity conservation.

N.1.3 Protected-sector invariant

Define a protection marker

$$\mathcal{P}_G(\mathcal{K}_G) = \begin{cases} 1, & \text{if no allowed local reconnection/crossing sequence reduces } \mathcal{K}_G \text{ to the trivial sector,} \\ 0, & \text{otherwise.} \end{cases} \tag{407}$$

The marker $\mathcal{P}_G$ is not an energy term by itself. It is a sector label. Energetically, it enters as a barrier rule:

$$\Delta E_{\mathrm{rec}}(\mathcal{K}_G \to \mathcal{K}'_G) = \begin{cases} < \infty, & Q_G(\mathcal{K}_G) = Q_G(\mathcal{K}'_G), \\ \infty \text{ or macroscopically large}, & Q_G(\mathcal{K}_G) \neq Q_G(\mathcal{K}'_G), \end{cases} \tag{408}$$

where Q_G is a conserved coloring invariant under the permitted move set.

A useful research-track extension of Eq. (402) is therefore the scalarized screening form

$$\frac{E^{(G)}_{\mathrm{screen}}[\mathcal{K}_G]}{E_0} = a_R\, \mathrm{Rop}_{\mathrm{rad}}(\gamma_K) + a_C\, c_{\min}(K) + a_H\, \widehat{\mathcal{H}}(K) + a_G\, I_G(\mathcal{K}_G) + \cdots, \tag{409}$$

where

$$I_G : Q_G(\mathcal{K}_G) \longrightarrow \mathbb{R}_{\geq 0} \tag{410}$$

extracts a real-valued index from the conserved coloring sector. The barrier scale Δ_G belongs to reconnection dynamics through Eq. (408); it is not promoted to a fitted mass term. The notation E_{screen} is used deliberately: this is a scalarized screening/ordering functional, not a derived Hamiltonian and not a replacement for the thin-filament energy scale.

N.1.4 Dimensional check

The quantities $\mathrm{Rop}_{\mathrm{rad}}(\gamma_K)$, $c_{\min}(K)$, $\widehat{\mathcal{H}}(K)$, and I_G are dimensionless. Therefore the coefficients in Eq. (409) are dimensionless if the left-hand side is E_{screen}/E_0. A dimensional energy is recovered only after selecting an SST energy scale E_0, for example a calibrated tube energy, a core line-tension scale, or a controlled numerical relaxation scale. The group object Q_G itself is not added to the energy; only a scalar index extracted from it may enter a screening functional.

N.1.5 Minimal falsifier

If two SST candidate particles have the same $(C, L, \mathcal{H})$ but different observed stability, the model predicts that a missing internal sector label exists:

$$(C_1, L_1, \mathcal{H}_1) = (C_2, L_2, \mathcal{H}_2), \qquad \tau_1 \neq \tau_2 \quad \Rightarrow \quad Q_G(\mathcal{K}_1) \neq Q_G(\mathcal{K}_2). \tag{411}$$

Failure to find such a sector in simulation or experiment would demote this appendix from Research Track to rejected model branch.

N.2 Research Appendix: Reconnection Decay as a Stability Falsifier

Canon status. [ORTHODOX] Constraint + [SPECULATIVE] SST interpretation. This appendix records a negative constraint on SST. A single-component Gross–Pitaevskii-like medium with permitted reconnections does not automatically protect knotted filaments [56]. Therefore, SST particle stability requires additional structure beyond generic phase-defect topology.

N.2.1 Baseline statement

In a dissipative or reconnection-permitting medium, knotted filament topology can decay. The relevant orthodox lesson is not that SST is false, but that topology alone is insufficient unless the move set is restricted by the dynamics.

Let $K(t)$ denote the topology of a swirl-string centerline. A reconnection path is a sequence

$$K_0 \longrightarrow K_1 \longrightarrow \cdots \longrightarrow K_N, \tag{412}$$

where each arrow is a local reconnection or strand-crossing event. If K_N is an unlink or collection of unknotted loops, then K_0 is dynamically unprotected under that move set.

N.2.2 SST stability criterion

SST should not claim that every closed filament is stable. A canon-compatible stability criterion is

$$\tau_{\mathrm{life}}(K) \sim \tau_0 \exp\left(\frac{\Delta E_{\mathrm{rec}}(K)}{E_{\mathrm{drive}}}\right), \tag{413}$$

where $\Delta E_{\mathrm{rec}}(K)$ is the lowest reconnection barrier out of the topological sector, and E_{drive} is the effective environmental or internal driving energy available to trigger reconnection.

The stable-particle condition is therefore not merely

$$K \neq \bigcirc, \tag{414}$$

but rather

$$\Delta E_{\mathrm{rec}}(K) \gg E_{\mathrm{drive}} \quad \text{and/or} \quad Q_G(K) \neq Q_G(\bigcirc). \tag{415}$$

N.2.3 Relation to Chronos–Kelvin transport

For ideal SST evolution without reconnection or external source, the Chronos–Kelvin invariant remains

$$\frac{D}{Dt}\left(R^2\omega\right) = 0. \tag{416}$$

Reconnection is precisely a non-ideal event where this transport law is locally invalid or discontinuous. Thus a κ-type transition in the canon may be interpreted as the event layer at which topology can change.

[SPECULATIVE] Identification. A reconnection event is an SST Kairos event when it changes the topological sector:

$$\kappa: \quad K_i \to K_{i+1}, \qquad Q_G(K_i) \neq Q_G(K_{i+1}) \quad \text{or} \quad \mathcal{H}(K_i) \neq \mathcal{H}(K_{i+1}). \tag{417}$$

N.2.4 Dimensionless diagnostic

Define the stability ratio

$$\Xi_{\text{rec}}(K) = \frac{\Delta E_{\text{rec}}(K)}{E_{\text{tube}}(K)}. \tag{418}$$

For a slender swirl tube, the standard leading envelope energy scale is

$$E_{\text{tube}}(K) \simeq \frac{\rho_f \Gamma^2}{4\pi}\, \ell_K \ln\left(\frac{R}{r_c}\right), \tag{419}$$

where Γ is circulation, ℓ_K is centerline length, and R is the outer cutoff. The units check:

$$[\rho_f \Gamma^2 \ell_K] = \frac{\text{kg}}{\text{m}^3}\frac{\text{m}^4}{\text{s}^2}\text{m} = \text{kg}\,\text{m}^2\text{s}^{-2} = \text{J}. \tag{420}$$

A practical SST classification rule is

$$\Xi_{\text{rec}}(K) \lesssim 1 \Rightarrow \text{hydrodynamically fragile}, \qquad \Xi_{\text{rec}}(K) \gg 1 \Rightarrow \text{candidate stable T-phase state}. \tag{421}$$

N.2.5 Minimal simulation protocol

A candidate SST knot assignment should be tested under three levels of dynamics:

1. single-component phase-defect evolution;
2. constrained incompressible evolution without reconnection;
3. multi-director or non-Abelian colored evolution with reconnection rules.

Only states stable under the third protocol should be promoted as particle candidates.

N.3 Research Appendix: Fractional Torus-Phase Winding and Swirl-Clock Phase

Canon status. [SPECULATIVE] Research Track. This appendix records a bridge between coordinated rotational phase mechanics and SST R-phase torsional packets, using fractional-order optical torus knots as the mathematical template [57]. It does not assert that optical polarization knots are literal SST particles. It supplies a useful mathematical template for phase winding, internal rotation, and torus-knot closure.

N.3.1 Coordinated phase generator

Let θ be an external azimuthal coordinate and let ϕ be an internal director angle. A coordinated rotation acts as

$$\theta \mapsto \theta + \alpha, \qquad \phi \mapsto \phi + \gamma\alpha. \tag{422}$$

The generator has the schematic form

$$J_\gamma = L_{\text{orb}} + \gamma S_{\text{int}}, \tag{423}$$

where L_{orb} shifts external azimuthal phase and S_{int} shifts internal swirl-director phase.

For two commensurate internal frequencies $p\Omega$ and $q\Omega$ carrying azimuthal mode numbers m_1 and m_2, the coordinated-winding parameter is

$$\gamma = \frac{m_2 p - m_1 q}{p+q}. \tag{424}$$

The corresponding torus-winding pair is

$$(m, n) = (m_2 p - m_1 q,\ p+q)\,. \tag{425}$$

If $\gcd(m,n) = 1$, the internal lobe trajectory is a single (m,n) torus knot. If $d = \gcd(m,n) > 1$, the trajectory splits into d linked components.

N.3.2 SST phase-action translation

To avoid notation conflict with the Swirl-Clock factor $S^{\circlearrowleft}_{(t)}$, define a phase-action field $\mathscr{S}(\vec{x},t)$. In an R-phase torsional packet, write the reduced phase on a toroidal carrier as

$$\Theta(\theta,\phi,t) = n\phi - m\theta - \Omega t. \tag{426}$$

The closure condition is

$$\oint d\Theta = 2\pi N, \qquad N \in \mathbb{Z}. \tag{427}$$

The phase-action field may be represented locally as

$$\mathscr{S}(\vec{x},t) = \hbar_{\mathrm{SST}}\,\Theta(\vec{x},t), \tag{428}$$

so that the circulation quantization condition becomes

$$\oint \nabla\mathscr{S}\cdot d\vec{\ell} = 2\pi N\hbar_{\mathrm{SST}}. \tag{429}$$

N.3.3 Connection to Swirl-Clock modulation

The local Swirl-Clock factor remains the kinematic clock factor

$$S^{\circlearrowleft}_{(t)} = \sqrt{1-\frac{\|\mathbf{u}_{\circlearrowleft}(\mathbf{x},t)\|^2}{c^2}}. \tag{430}$$

The phase-action field controls wave-like periodicity, while $S^{\circlearrowleft}_{(t)}$ controls local clock rate. The research-track bridge is that a stable R-phase packet may require simultaneous closure of both:

$$\oint \nabla\mathscr{S}\cdot d\vec{\ell} = 2\pi N\hbar_{\mathrm{SST}}, \tag{431}$$

$$\frac{D}{Dt}\left(R^2\omega\right) = 0. \tag{432}$$

[SPECULATIVE] Interpretation. Equation (424) suggests a route to fractional internal winding without fractional circulation. The circulation quantum remains integer-valued, but the internal phase orientation can return only after multiple external turns. This may be useful for modeling R-phase torsional photons, spin-orbit-like packet structure, or generation-family phase layering.

N.3.4 Dimensional check

The variables m, n, p, q, N and γ are dimensionless. The phase Θ is dimensionless. Therefore $\mathscr{S} = \hbar_{\mathrm{SST}}\Theta$ has units of action, and Eq. (429) has units of action because $\nabla\mathscr{S}\cdot d\vec{\ell}$ has units $(\mathrm{J\,s\,m^{-1}})\mathrm{m} = \mathrm{J\,s}$.

N.4 Research Appendix: Topological Fluid Mechanics and the SST Energy Functional

Canon status. [ORTHODOX] Mathematical background + [SPECULATIVE] SST synthesis. This appendix records the topological-fluid mechanics background behind the SST knot-energy program [27, 58]. The orthodox part is that ideal incompressible flow preserves topology and helicity under the usual assumptions. The speculative part is the identification of selected relaxed knot classes with particle sectors.

N.4.1 Frozen-field topology

In an ideal incompressible medium, topology is transported by the flow map. For SST this supports the use of closed swirl strings as dynamically persistent structures under non-reconnecting evolution.

The primary invariants are

$$\Gamma = \oint_C \mathbf{u}_{\circlearrowleft}\cdot d\vec{\ell}, \tag{433}$$

$$\mathcal{H} = \int_V \mathbf{u}_{\circlearrowleft}\cdot\vec{\omega}\,d^3x, \tag{434}$$

$$\vec{\omega} = \nabla\times\mathbf{u}_{\circlearrowleft}. \tag{435}$$

For linked tubes with circulations Γ_i, the leading topological helicity decomposition is

$$\mathcal{H} \simeq \sum_i \Gamma_i^2 \, SL_i + 2 \sum_{i<j} Lk_{ij} \Gamma_i \Gamma_j, \tag{436}$$

where SL_i is self-linking and Lk_{ij} is Gauss linking number.

N.4.2 Energy relaxation under topological constraints

For a slender swirl tube, the leading hydrodynamic energy scale is

$$E_{\text{env}} \simeq \frac{\rho_f \Gamma^2}{4\pi} \, \ell_K \ln\left(\frac{R}{r_c}\right), \tag{437}$$

where ℓ_K is the tube centerline length. The core contribution can be represented by a line-tension term

$$E_{\text{core}} \simeq T_{\text{core}} \ell_K. \tag{438}$$

Here T_{core} has units $\mathrm{J\,m^{-1}} = \mathrm{N}$ and is a model-supplied line tension, not an additional canon primitive. A research calculation that uses it must state whether it comes from a resolved core profile, a numerical relaxation scale, or a calibration. Otherwise it remains symbolic and cannot be independently tuned alongside the scalarized screening coefficients. Near contacts and curvature corrections may be encoded by

$$E_{\text{geom}}[K] = T_{\text{core}} \ell_K + \frac{\rho_f \Gamma^2}{4\pi} \ell_K \ln\left(\frac{R}{r_c}\right) + \beta_\kappa \int_K \kappa_s^2 \, ds + \beta_{\text{nc}} \, C_{\text{near}}(K). \tag{439}$$

Here κ_s is centerline curvature and C_{near} is a near-contact or crossing-complexity proxy.

N.4.3 Reduction to the SST normal form

After nondimensionalization by a reference energy E_0, Eq. (439) reduces to the canon-style scoring form

$$\frac{E_{\text{screen}}[K]}{E_0} \rightsquigarrow a_R \, \mathrm{Rop}_{\text{rad}}(\gamma_K) + a_C \, c_{\min}(K) + a_H \, \widehat{\mathcal{H}}(K) + a_G \, I_G(K) + \cdots . \tag{440}$$

Here $I_G(K)$ is included only when a non-Abelian coloring sector $Q_G(K)$ has been specified and scalarized by a map $I_G : Q_G \to \mathbb{R}_{\geq 0}$. This justifies the SST normal form as a coarse-grained energy-ordering tool, not as a finished mass theorem. The unscalarized object $Q_G(K)$ is not an additive energy term.

N.4.4 Canon discipline

The functional in Eq. (440) should be used as follows:

1. **[ORTHODOX]** Use Γ, $\mathcal{H}$, Lk, writhe, twist, and ropelength as established geometric/topological descriptors.
2. **[DERIVED]** Within a chosen tube model, derive the units and leading energy scaling from ρ_f, Γ, ℓ_K, and r_c.
3. **[SPECULATIVE]** Only after this, map a relaxed topological class to an SST particle sector.

N.5 Research Appendix: Resolved-Tube Criticality and Contact-Stress Geometry

Canon status. [ORTHODOX GEOMETRIC BACKGROUND / RESEARCH-TRACK SST INTERPRETATION]. The orthodox input is the ropelength theory of finite-thickness embedded curves and the constrained-gradient formulation of polygonal knot tightening. The SST interpretation is that the active finite-thickness constraints of an ideal swirl-string may be read as a discrete contact-stress skeleton for the resolved tube. This appendix does not canonize a new mass formula; it canonizes a validation discipline for any future profile-resolved tube model.

N.5.1 Resolved tube radius as reach

For a resolved centerline γ_K, the physical tube radius is not the horn radius r_c. The resolved geometric radius is the reach/thickness of the centerline,

$$a_{\mathrm{core}}(K) \equiv \mathrm{Thi}_{\mathrm{rad}}(\gamma_K) = \mathrm{reach}(\gamma_K) = \mathrm{NIR}(\gamma_K), \tag{441}$$

with radius-convention ropelength

$$\mathrm{Rop}_{\mathrm{rad}}(K) = \frac{\mathrm{Len}(\gamma_K)}{a_{\mathrm{core}}(K)}. \tag{442}$$

For smooth centerlines the controlling geometric obstructions are curvature and doubly-critical self-distance,

$$a_{\mathrm{core}}(K) = \min\left(\inf_s \frac{1}{\kappa(s)},\ \frac{1}{2} d_{\mathrm{dcsd}}(\gamma_K)\right). \tag{443}$$

Equations (441)–(443) are **[ORTHODOX]** geometric definitions. They preserve the canon rule $a_{\mathrm{core}} \neq r_c$ unless a separate profile calculation proves otherwise.

N.5.2 Struts, kinks, and contact-stress balance

In a polygonal approximation V of the centerline, finite thickness is controlled by two active constraint classes:

$$\mathrm{MinRad}(v_i) \geq a_{\mathrm{core}}, \tag{444}$$

$$\frac{1}{2} d(p,q) \geq a_{\mathrm{core}}. \tag{445}$$

Vertices saturating Eq. (444) form the kink set $\mathrm{Kink}(V)$; pairs saturating Eq. (445) form the strut set $\mathrm{Strut}(V)$. The constrained-gradient criterion for a thickness-critical state has the schematic KKT form

$$-\nabla E[V] + \sum_{(p,q)\in \mathrm{Strut}(V)} \lambda_{pq} \nabla\left(\frac{d(p,q)}{2}\right) + \sum_{i\in \mathrm{Kink}(V)} \mu_i \nabla \mathrm{MinRad}(v_i) = 0, \qquad \lambda_{pq}, \mu_i \geq 0. \tag{446}$$

For the mathematical ropelength problem, $E[V] = \mathrm{Len}(V)$. For a profile-resolved SST tube one should instead test a functional of the form

$$E_{\mathrm{SST}}[\gamma] = T_{\mathrm{eff}}\,\mathrm{Len}(\gamma) + E_{\mathrm{bend}}[\gamma] + E_{\mathrm{twist}}[\gamma] + E_{\mathrm{swirl}}[\gamma] + E_{\mathrm{core}}[\gamma]. \tag{447}$$

The multipliers λ_{pq} and μ_i are then research-track candidates for a discrete swirl-stress/contact-force map, not yet canonical physical forces.

N.5.3 Contact-map refinement of the topological kernel

A scalar ropelength is not enough to determine all shape-sensitive SST observables. A resolved ideal tube carries a contact skeleton

$$\mathcal{C}_K = \big\{(p,q,\lambda_{pq})\big\}_{\mathrm{Strut}} \cup \big\{(i,\mu_i)\big\}_{\mathrm{Kink}}. \tag{448}$$

A future profile-resolved version of the dimensionless topology kernel may therefore have the schematic expansion

$$\Xi_K = \Xi_K^{(0)} + \epsilon_s \sum_{(p,q)} \lambda_{pq} + \epsilon_k \sum_i \mu_i + \epsilon_H \mathcal{H}_K + \cdots. \tag{449}$$

Equation (449) is a **[MATCHING ANSATZ]**. Its value is diagnostic: if the large calibrated normalization hidden in Ξ_K can be decomposed into reproducible contact, curvature, and helicity contributions, the mass kernel becomes mechanically inspectable rather than a single opaque coefficient.

N.5.4 Golden-layer factor as hyperbolic transfer parametrization

Status. [RESEARCH-TRACK / PARAMETRIC REWRITE]. This subsection records an exact algebraic reparametrization of the existing golden-layer factor in the SST kernel. It does not derive the factor from Euler dynamics, ropelength criticality, or knot stability.

The identity

$$\operatorname{asinh}\left(\frac{1}{2}\right) = \ln\left(\frac{1}{2} + \sqrt{1 + \frac{1}{4}}\right) = \ln\left(\frac{1+\sqrt{5}}{2}\right) = \ln\varphi \tag{450}$$

allows the layer term to be written as

$$\eta_\varphi \equiv \operatorname{asinh}\left(\frac{1}{2}\right) = \ln\varphi \simeq 0.4812118251, \tag{451}$$

$$\varphi^{-2k(K)} = \exp[-2k(K)\eta_\varphi]\,. \tag{452}$$

Equivalently,

$$\sinh\eta_\varphi = \frac{1}{2}, \qquad \cosh\eta_\varphi = \frac{\sqrt{5}}{2}, \qquad \tanh\eta_\varphi = \frac{1}{\sqrt{5}}. \tag{453}$$

The associated symmetric transfer matrix

$$B_\varphi = \begin{pmatrix} \sqrt{5}/2 & 1/2 \\ 1/2 & \sqrt{5}/2 \end{pmatrix}, \qquad \operatorname{spec}(B_\varphi) = \{\varphi, \varphi^{-1}\}, \tag{454}$$

therefore supplies a compact notation for one elementary layer-transfer step. Iteration gives attenuation eigenvalue

$$\lambda_-^{2k(K)} = \varphi^{-2k(K)}. \tag{455}$$

Canon discipline. This is not a hyperbolic-knot claim. In particular, the electron-sector assignment to the trefoil is not made hyperbolic by Eq. (452); the trefoil remains a torus knot. The useful content is only that the existing golden dressing factor can be written as a discrete exponential attenuation. To promote this from notation to a derived SST mechanism, a future calculation must identify $k(K)$ with a definite filament-transfer, impedance, phase-locking, or relaxation step and derive the elementary rapidity $\eta_\varphi = \operatorname{asinh}(1/2)$ from that model.

Dimensional check. The quantities φ, $k(K)$, η_φ, and all entries of B_φ are dimensionless. Therefore Eq. (452) can modify only a dimensionless kernel or impedance factor; it cannot by itself introduce a mass, length, energy, or force scale.

N.5.5 Geometric lower-bound gates for SSTcore

For any nontrivial knot in the radius-thickness convention, orthodox ropelength bounds give

$$\mathrm{Rop}_{\mathrm{rad}}(K_{\mathrm{nontrivial}}) \geq 4\pi + 2\pi\sqrt{2} \simeq 21.45213649. \tag{456}$$

Equivalently,

$$\mathrm{Rop}_{\mathrm{diam}}(K_{\mathrm{nontrivial}}) \geq 2\pi + \pi\sqrt{2} \simeq 10.72606825. \tag{457}$$

Using the canonical radius $r_c = 1.40897017 \times 10^{-15}$ m, the radius-convention bound corresponds to

$$L_{\mathrm{min}} = r_c\left(4\pi + 2\pi\sqrt{2}\right) \simeq 3.02254 \times 10^{-14}\,\mathrm{m}. \tag{458}$$

For the trefoil benchmark used elsewhere in the canon,

$$\mathrm{Rop}_{\mathrm{rad}}(3_1) \approx 2 \times 16.371637 = 32.743274, \tag{459}$$

$$L_{3_1} \approx 32.743274\, r_c \simeq 4.61343 \times 10^{-14}\,\mathrm{m}. \tag{460}$$

An SSTcore ideal-knot routine that reports a nontrivial knot below Eq. (456), or mixes the radius and diameter conventions without an explicit factor of two, fails the geometry gate.

N.5.6 Composite-link non-additivity warning

For composite links and connect-sum-like configurations, ideal ropelength is not generally additive. The ropelength literature reports a typical deficit of order

$$\Delta\mathcal{L}_{\text{comp}} \sim 4\pi - 4 \simeq 8.56637. \tag{461}$$

Therefore any SST baryon or multi-component particle branch using an additive length proxy must expose a correction term,

$$\mathcal{L}_{\text{comp}} = \sum_i \mathcal{L}_{K_i} - \Delta_{\text{link}} + \delta_{\text{stress}}, \tag{462}$$

rather than assuming $\mathcal{L}_{\text{comp}} = \sum_i \mathcal{L}_{K_i}$. This is a research-track guard, not a universal law for all SST composites.

N.5.7 Operational SSTcore diagnostics

A profile-resolved SSTcore tightening or imported ideal-knot pipeline should report at least

$$\mathcal{D}_K = (\text{Rop}_{\text{rad}}, a_{\text{core}}, \text{Strut}, \text{Kink}, \Lambda, \chi_{\text{contact}}), \tag{463}$$

where

$$\chi_{\text{contact}} = \frac{\|-\nabla E_{\text{SST}} + A\Lambda\|}{\|\nabla E_{\text{SST}}\|}, \qquad \Lambda \geq 0. \tag{464}$$

Here A is the rigidity/contact matrix built from strut and kink gradients. Small χ_{contact} means that the candidate state is not merely short; it is approximately contact-critical under the declared energy model.

Minimal falsifier for this appendix. If two independent high-resolution implementations, using the same thickness convention and topology, converge to incompatible contact skeletons $\mathcal{C}_K$ or incompatible χ_{contact} under the same declared E_{SST}, the proposed contact-kernel refinement Eq. (449) must be demoted from Research Track to numerical diagnostic only.

N.6 Minimal falsifier

If two distinct knots have the same SST particle assignment but relax to incompatible energy minima or different protected sectors under the same admissible dynamics, the assignment must be split or rejected:

$$K_1 \sim_{\text{SST}} K_2 \quad \text{but} \quad \min E[K_1] \neq \min E[K_2] \quad \Rightarrow \quad \text{taxonomy refinement required.} \tag{465}$$

O Swirl-String Response and Resonance Framework

O.1 Resonance Selection Principle

Status: [RESEARCH-TRACK; candidate Canon principle].

A swirl-string state K is assigned a causal response function $\chi_K(\omega)$. An external perturbation $F(t)$, with Fourier spectrum $\widetilde{F}(\omega)$, excites the state K with amplitude

$$\mathcal{A}_K[F] = \int_0^\infty \widetilde{F}(\omega)\, \chi_K(\omega)\, d\omega. \tag{466}$$

The excitation probability is

$$P_K[F] = |\mathcal{A}_K[F]|^2. \tag{467}$$

For isolated narrow resonances, the response function may be written as

$$\chi_K(\omega) = \sum_n \frac{B_{Kn}\gamma_{Kn}}{\omega - \omega_{Kn} + i\gamma_{Kn}}, \tag{468}$$

where ω_{Kn} are internal swirl-string eigenfrequencies, γ_{Kn} are linewidths, and B_{Kn} are coupling weights.

The natural SST frequency scale is

$$\Omega_{\mathrm{SST}} = \frac{v_{\circlearrowleft}^{*}}{r_c}, \tag{469}$$

so that internal modes may be parametrized by

$$\omega_{Kn} = \eta_{Kn}\Omega_{\mathrm{SST}}, \tag{470}$$

with dimensionless mode factors η_{Kn}.

This principle unifies delay-loop selection, Swirl-Clock phase-locking, mode-locking, and knot-state excitation as instances of one spectral overlap law.

O.2 Resonance Selection Principle for Swirl–String Excitation

Status and purpose

Status: [RESEARCH-TRACK]. This appendix promotes the frequency-overlap idea from a specific beam–swirl model to a general SST excitation principle. It is not a proof of nuclear activation or fusion enhancement. It is a controlled response ansatz compatible with the present canon language of swirl strings, mode selection, circulation, helicity, and Kelvin-compatible dynamics.

Claim

A swirl-string state K is excited most efficiently when the spectrum of an external perturbation overlaps the internal response spectrum of K. Define a normalized external spectral density

$$p_{\mathrm{ext}}(\omega) \geq 0, \qquad \int_0^\infty p_{\mathrm{ext}}(\omega)\, d\omega = 1, \tag{471}$$

where p_{ext} has units of seconds, because $d\omega$ has units s^{-1}. Let the dimensionless response function of the swirl-string state be

$$R_K(\omega) = \sum_n B_{Kn} \frac{\gamma_{Kn}^2}{(\omega - \omega_{Kn})^2 + \gamma_{Kn}^2}, \tag{472}$$

where:

- ω_{Kn} is the n-th internal mode frequency,
- γ_{Kn} is the corresponding linewidth,
- B_{Kn} is a dimensionless coupling weight.

The dimensionless excitation yield is then

$$\boxed{\mathcal{Y}_K = \int_0^\infty p_{\mathrm{ext}}(\omega)\, R_K(\omega)\, d\omega} \qquad \text{[RESEARCH-TRACK]}. \tag{473}$$

Dimensional check

Since

$$[p_{\mathrm{ext}}] = \mathrm{s}, \qquad [d\omega] = \mathrm{s}^{-1}, \qquad [R_K] = 1,$$

we obtain

$$[\mathcal{Y}_K] = \mathrm{s} \cdot 1 \cdot \mathrm{s}^{-1} = 1.$$

Thus $\mathcal{Y}_K$ is a dimensionless excitation efficiency.

Canonical scale

The natural SST angular-frequency scale is

$$\Omega_{\mathrm{SST}} = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c}. \tag{474}$$

Using the Canon values

$$\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^{6}\ \mathrm{m\,s^{-1}}, \qquad r_c = 1.40897017 \times 10^{-15}\ \mathrm{m},$$

gives

$$\Omega_{\mathrm{SST}} = 7.76344066 \times 10^{20}\ \mathrm{s^{-1}}. \tag{475}$$

The corresponding cycle period is

$$T_{\mathrm{SST}} = \frac{2\pi}{\Omega_{\mathrm{SST}}} = 8.09329985 \times 10^{-21}\ \mathrm{s}. \tag{476}$$

Interpretation

Equation (473) does not assert that all nuclear, atomic, or particle transitions are already derived in SST. It states a weaker and safer rule:

> External excitation is strongest when the perturbing spectrum overlaps an allowed internal swirl-string response spectrum.

This principle can later be specialized to:

- radiative transitions,
- qubit-like R/T phase transitions,
- knot-mode excitation,
- delayed mode selection,
- controlled response experiments in optical, plasma, or condensed-matter analogues.

Falsifiability

For a fixed target state K, SST predicts that changing the spectral center ω_0 of a narrow-band perturbation should change the activation yield according to $R_K(\omega_0)$, up to normalization and detector response. If no resonance-like dependence appears after controlling for beam power, fluence, and thermal effects, then Eq. (473) is falsified as a useful excitation law.

O.3 Time-Domain Kernel Form of Swirl–String Response

Status and purpose

Status: [RESEARCH-TRACK]. This appendix converts the frequency-domain selection rule into a time-domain response law. It is intended to connect SST mode selection to pulsed experiments, delay-loop systems, and Swirl-Clock phase locking.

Pulse-response formulation

Let $f(t)$ be a real external driving pulse. A Gaussian-windowed monochromatic pulse may be written as

$$f(t) = f_0 \exp\left(-\frac{t^2}{2\tau_p^2}\right) \cos(\omega_0 t + \phi_0), \tag{477}$$

where τ_p is the pulse duration and ω_0 is the carrier angular frequency.

Let $k_K(t)$ be the causal impulse-response kernel of the swirl-string state K, with

$$k_K(t < 0) = 0. \tag{478}$$

The induced response is

$$\boxed{s_K(t) = \int_{-\infty}^{t} f(t')\, k_K(t-t')\, dt'} \qquad \text{[ORTHODOX form; RESEARCH-TRACK SST interpretation].} \tag{479}$$

Frequency-domain equivalence

With the Fourier convention

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t) e^{i\omega t}\, dt,$$

the convolution theorem gives

$$\hat{s}_K(\omega) = \hat{f}(\omega)\, \chi_K(\omega), \tag{480}$$

where $\chi_K(\omega)$ is the complex response function of the swirl-string state. The measurable spectral response is proportional to

$$R_K(\omega) \propto |\chi_K(\omega)|^2. \tag{481}$$

Bandwidth–duration relation

A Gaussian pulse obeys the scaling

$$\Delta\omega\, \tau_p \sim 1. \tag{482}$$

Therefore:

- short pulses have broad spectra and can excite many nearby modes;
- long pulses have narrow spectra and selectively excite one mode;
- coherent excitation requires phase stability over at least one internal response time.

For the canonical SST scale,

$$T_{\mathrm{SST}} = 8.09329985 \times 10^{-21}\ \mathrm{s}.$$

Thus, a pulse with duration $\tau_p \gg T_{\mathrm{SST}}$ is narrow relative to the core-scale carrier, whereas $\tau_p \lesssim T_{\mathrm{SST}}$ is broadband at the canonical core scale.

Connection to Swirl Clock

If the internal phase of a swirl string is denoted by $\theta_K(t)$, then a resonant perturbation can be represented as a phase-locking term

$$\frac{d\theta_K}{dt} = \omega_K + \epsilon\, f(t) \sin(\omega_0 t - \theta_K), \tag{483}$$

where ϵ is a coupling parameter. Phase locking occurs when

$$|\omega_0 - \omega_K| \lesssim \epsilon |f_0|. \tag{484}$$

This gives a direct bridge between frequency-selective excitation and Swirl-Clock synchronization.

Falsifiability

For a given K, changing only the pulse duration τ_p while keeping total pulse energy fixed should alter the response through the bandwidth $\Delta\omega$. A null result after controlling for heating and total fluence would falsify this time-domain response model.

O.4 Swirl Impedance and Mode Linewidth

Status and critical correction

Status: [RESEARCH-TRACK]. The earlier expression

$$Z_{\text{eff}} \sim \frac{P_{\text{core}} r_c}{\|\mathbf{v}_{\circlearrowleft}\| \hbar}$$

should *not* be canonised as written. Its dimensions are

$$\left[\frac{P_{\text{core}} r_c}{\|\mathbf{v}_{\circlearrowleft}\| \hbar} \right] = \mathrm{m}^{-3},$$

so it is not an impedance and not dimensionless.

A Canon-compatible replacement is to define either:

1. a dimensionless rigidity ratio,
2. or an acoustic-like pressure-to-speed impedance.

Dimensionless swirl-rigidity ratio

Let V_K be the effective active volume of a swirl-string configuration and P_K the pressure scale associated with its deformation. Define

$$\boxed{\Pi_K(\omega) = \frac{P_K V_K}{\hbar \omega}} \qquad \text{[RESEARCH-TRACK]}. \tag{485}$$

This compares stored mechanical energy $P_K V_K$ to the quantum energy scale $\hbar\omega$. It is dimensionless:

$$[P_K V_K] = \mathrm{J}, \qquad [\hbar\omega] = \mathrm{J}.$$

Acoustic-like swirl impedance

An alternative intensive impedance is

$$\boxed{Z_{\text{swirl}} = \frac{P_K}{v_K}} \tag{486}$$

with units

$$[Z_{\text{swirl}}] = \frac{\mathrm{kg\,m^{-1}\,s^{-2}}}{\mathrm{m\,s^{-1}}} = \mathrm{kg\,m^{-2}\,s^{-1}},$$

the same units as acoustic impedance. Here v_K denotes the coarse-grained velocity amplitude of the same mode family K used to define P_K; it may be instantiated as a phase-transport speed, local carrier-speed amplitude, or resonant surface-speed amplitude, but the choice must be declared in any benchmark. It is not the canonical scalar speed $v_{\circlearrowleft}^*$ unless the benchmark explicitly specializes to the canonical carrier limit. This object measures resistance to velocity-driven pressure excitation.

Connection to linewidth and quality factor

A resonant mode with angular frequency ω_K and quality factor Q_K has linewidth

$$\gamma_K = \frac{\omega_K}{2Q_K}. \tag{487}$$

A minimal phenomenological relation is

$$Q_K = Q_0\, \Pi_K^{\eta}, \qquad \eta > 0, \tag{488}$$

so that larger stored-energy rigidity produces narrower response lines.

Numerical check using canonical scales

Using

$$F_{\text{swirl}}^{\max} = 29.053507 \text{ N}, \qquad r_c = 1.40897017 \times 10^{-15} \text{ m},$$

define the core pressure scale

$$P_{\max} = \frac{F_{\text{swirl}}^{\max}}{\pi {r_c}^2} = 4.65848920 \times 10^{30} \text{ Pa}. \tag{489}$$

[CRITICAL NOTE]: $V_{\text{horn}} = 2\pi^2 {r_c}^3$ is the r_c-scale envelope (horn-torus) volume, *not* the resolved tube cross-section, for which Sec. 2.2 stipulates $a_{\text{core}} \neq r_c$. The resulting $\Pi_K \approx \pi$ is hence a geometric normalization at the envelope scale, not a stiffness theorem for the physical tube; a profile-resolved a_{core} model is required for the latter. For a horn-torus active volume

$$V_{\text{horn}} = 2\pi^2 {r_c}^3 = 5.52122107 \times 10^{-44} \text{ m}^3, \tag{490}$$

the stored pressure energy is

$$P_{\max} V_{\text{horn}} = 2.57205487 \times 10^{-13} \text{ J}. \tag{491}$$

At the canonical Compton angular frequency

$$\omega_c = 7.76344066 \times 10^{20} \text{ s}^{-1},$$

we have

$$\hbar\omega_c = 8.18710572 \times 10^{-14} \text{ J} = 510998.946 \text{ eV}. \tag{492}$$

Therefore,

$$\Pi_{\text{horn}} = \frac{P_{\max} V_{\text{horn}}}{\hbar\omega_c} = 3.14159235 \approx \pi. \tag{493}$$

Interpretation

Equation (493) is not yet a derivation of a particle mass or coupling constant. It is a useful dimensional coincidence produced by canonical SST scales and the horn-torus active-volume choice. It should be labelled:

$$\boxed{\text{[CALIBRATED / RESEARCH-TRACK NUMERICAL STRUCTURE]}}$$

until derived from the SST action or from a topological minimization principle.

Falsifiability

If Π_K controls linewidth, then families of analog swirl-string excitations should satisfy

$$\gamma_K \propto \omega_K \Pi_K^{-\eta}.$$

A measured absence of correlation between active-volume rigidity and linewidth would falsify Eq. (488).

O.5 Layered Relaxation and Multi-Mode Decay

Status and warning

Status: [RESEARCH-TRACK]. This appendix preserves the useful part of the delayed-decay idea: a multi-layer swirl-string system may relax through multiple modes. It explicitly rejects the naive identification

$$\tau_i = \frac{r_i}{\|\mathbf{v}_\circlearrowleft\|}$$

with observed delayed-neutron group lifetimes. For nuclear-scale radii, this gives $\tau_i \sim 10^{-21}$ s, whereas delayed-neutron groups are macroscopic nuclear decay times. Therefore direct radial-transit mapping is dimensionally possible but physically wrong in scale.

General multi-mode relaxation

If a perturbed swirl-string state has several metastable channels, its observable relaxation can be expanded as

$$\boxed{A(t) = \sum_{i=1}^{N} A_i e^{-t/\tau_i} \cos(\omega_i t + \phi_i)} \qquad \text{[ORTHODOX form; RESEARCH-TRACK SST interpretation].} \tag{494}$$

The non-oscillatory limit is

$$A(t) = \sum_{i=1}^{N} A_i e^{-t/\tau_i}. \tag{495}$$

Pole interpretation

Equation (494) corresponds to a response function with poles

$$\omega = \omega_i - \frac{i}{\tau_i}. \tag{496}$$

Thus, each decay time τ_i is a lifetime of a response pole, not necessarily a physical radial crossing time.

Allowed SST interpretation

In SST, a layered or nested swirl-string configuration may contain:

- a core-scale response,
- a sheath-scale response,
- a linking/helicity response,
- a global Swirl-Clock phase response.

The decay constants τ_i should therefore be treated as *effective metastable relaxation times.* A safe parametrization is

$$\tau_i = \frac{Q_i}{\omega_i}, \tag{497}$$

where Q_i is the quality factor of the i-th mode.

Critical scale check

For the canonical circulation scale,

$$\tau_{\text{cross,c}} = \frac{r_c}{\|\mathbf{v}_{\circlearrowleft}\|} = 1.28808868 \times 10^{-21}\ \text{s}. \tag{498}$$

This is a single radial crossing time, distinct from the full circulation period $\tau_{\text{circ,c}} = 2\pi r_c / v_{\circlearrowleft}^{*}$ of Eq. (130) in the main canon. Therefore, an observed lifetime of 1 s would require

$$Q \sim \omega_i \tau_i \sim 7.76 \times 10^{20}. \tag{499}$$

Such a huge Q-factor is not impossible as a formal metastability parameter, but it cannot be interpreted as a simple geometric shell crossing.

Experimental analogy

Multi-exponential decay models are common in nuclear delayed-neutron kinetics and in linear response theory. SST may use this only as an analogy unless a topological barrier calculation derives the Q_i or τ_i values independently.

Falsifiability

The layered relaxation ansatz predicts that perturbing a structured target should produce reproducible decay constants τ_i independent of total pulse amplitude, provided the response remains linear. If the apparent τ_i vary strongly with drive amplitude, heating, or detector threshold, the model is likely observing instrumental or thermal relaxation rather than swirl-string metastability.

Summary of Canon Placement

The four additions should be placed as follows:

Appendix	Canon status	Recommended placement
Resonance Selection Principle	[RESEARCH-TRACK] strong candidate	CANON-v0.8.17 companion appendix; possible future main-canon principle after validation.
Time-Domain Kernel Response	[RESEARCH-TRACK] strong candidate	Near Swirl Clock, delay-mode selection, or quantum-response sections.
Swirl Impedance and Linewidth	[RESEARCH-TRACK] with corrected dimensions	Use as a response/linewidth ansatz; do not canonise the older dimensionally incorrect formula.
Layered Relaxation	[RESEARCH-TRACK] analogy only	Keep as metastable multi-mode response; reject direct delayed-neutron shell-crossing interpretation.

P Extended Vortex, Helicity, and Kairos Research Tracks

P.1 Research Track: Nested Vortex–Antivortex States as Entropy-Minimizing Swirl Structures

P.2 Motivation

Consider a coherent swirl structure consisting of an outer circulation region with vorticity $\omega_{\text{out}} > 0$ and an embedded inner core with opposite vorticity

$$\omega_{\text{in}} < 0.$$

Such configurations appear naturally in fluid dynamics, magnetic flux ropes, and vortex-ring interactions.

Within SST, opposite swirl chirality corresponds to opposite temporal orientation of the Swirl Clock:

$$S_t^{\circlearrowright} \quad \leftrightarrow \quad \text{matter},$$

$$S_t^{\circlearrowleft} \quad \leftrightarrow \quad \text{antimatter}.$$

P.3 Helicity Cancellation

The total helicity may be decomposed as

$$H_{\text{tot}} = H_{\text{out}} + H_{\text{in}}.$$

Because the embedded structure carries opposite chirality,

$$H_{\text{in}} < 0.$$

Hence

$$|H_{\text{tot}}| < |H_{\text{out}}|.$$

The configuration therefore possesses reduced net topological complexity.

P.4 Entropy Hypothesis

We introduce the conjecture

$$S_{\rm top} \propto |H|,$$

where $S_{\rm top}$ denotes topological entropy.

Consequently,

$$S_{\rm top}^{\rm nested} < S_{\rm top}^{\rm single},$$

suggesting that nested vortex–antivortex states may represent preferred low-energy attractors of the swirl medium.

P.5 Status

Research Track.

Not yet derived from the canonical SST Lagrangian. Proposed as a candidate mechanism for stability enhancement and matter–antimatter coexistence.

P.6 Research Track: Helicity Conversion and Torsional Radiation

P.7 Motivation

Experiments and simulations of knotted vortex structures indicate that reconnection events frequently preserve a substantial fraction of helicity.

Rather than disappearing, helicity is redistributed between knotting, linking, writhe, and twist.

P.8 Helicity Partition

Write total helicity as

$$H = H_{\rm knot} + H_{\rm twist} + H_{\rm link}.$$

During reconnection:

$$H_{\rm knot} \downarrow$$

while

$$H_{\rm twist} \uparrow .$$

Thus

$$H_{\rm tot} \approx \text{const.}$$

P.9 SST Interpretation

Within SST, mass-bearing T-phase structures correspond to localized knot helicity.

Radiative R-phase excitations correspond to propagating torsional modes.

We therefore conjecture

$$H_{\rm knot} \to H_{\rm twist} \to \gamma_R,$$

where γ_R denotes an R-phase torsional pulse.

P.10 Mass–Radiation Conversion Chain

The proposed conversion pathway is

$$M \leftrightarrow H_{\text{knot}} \leftrightarrow H_{\text{twist}} \leftrightarrow \gamma_R.$$

In this picture:

- stable particles store helicity as topology,
- reconnections release helicity as torsional twist,
- torsional twist propagates as photon-like R-phase radiation.

P.11 Connection to Photon Ontology

This hypothesis is consistent with the SST identification

$$\text{photon} = \text{torsional R-phase pulse},$$

and provides a possible mechanism for

- particle decay,
- annihilation,
- radiation emission,
- energy transport.

P.12 Falsifiable Prediction

If the hypothesis is correct, vortex reconnection experiments should show

$$\Delta H_{\text{knot}} \approx -\Delta H_{\text{twist}}$$

with emitted torsional wave packets carrying the missing helicity.

P.13 Status

Research Track.

Motivated by helicity-conservation studies in knotted vortex systems. Requires derivation from the SST radiation sector.

P.14 Research Track: Kairos Surfaces as Chirality Transition Layers

P.15 Definition

The Temporal Ontology introduces the Kairos event

$$\kappa,$$

representing an irreversible topological transition.
We extend this concept spatially.
Let

$$\omega(\mathbf{x}) = 0$$

define a surface separating opposite chiral domains:

$$\omega > 0 \quad \leftrightarrow \quad S_t^{\circlearrowleft},$$

$$\omega < 0 \quad \leftrightarrow \quad S_t^{\circlearrowright}.$$

We define the locus

$$\Sigma_\kappa = \{\mathbf{x} \mid \omega(\mathbf{x}) = 0\}$$

as a *Kairos Surface*.

P.16 Physical Interpretation

Across Σ_κ the swirl orientation reverses sign.

The surface therefore acts as a boundary between distinct temporal sectors.

$$S_t^{\circlearrowleft} \longleftrightarrow S_t^{\circlearrowright}.$$

Such regions may act as:

- annihilation interfaces,
- vortex reconnection sites,
- phase-transition boundaries,
- chirality-conversion layers.

P.17 Kairos Criterion

A local Kairos transition is conjectured whenever

$$\omega \to 0$$

and

$$\nabla\omega \neq 0.$$

The gradient supplies a preferred transition direction.

P.18 Status

Research Track.

Consistent with the Temporal Ontology but not yet derived from the SST field equations.

Q EM, Gravity, Diagnostics, Relativity, and Inertia Research Tracks

Q.1 Euler-Decomposed Swirl Gravity and Probe Transport

Q.2 Purpose and status

This appendix separates the canonical scalar pressure acceleration from orientation-dependent transverse transport effects. The pressure law is retained as the canon-level bridge, while Magnus-type terms are treated as probe-level corrections. The motivation is the orthodox transverse-force/Magnus result for moving vortices in acoustic-geometry and superfluid settings [24, 23].

[GUARDRAIL] The transverse term is not promoted here to the primary cause of universal free fall. In the present canon, universal fall and the Newtonian limit remain attached to the scalar pressure/clock sector. Magnus-type contributions are orientation-, circulation-, and chirality-sensitive probe transport corrections until a universal macroscopic coupling law is derived.

Q.3 Canonical pressure acceleration

$$\mathbf{a}_P = -\frac{1}{\rho_f}\nabla p_{\text{swirl}}. \tag{500}$$

Q.4 Euler decomposition

For an incompressible inviscid background flow,

$$\frac{\partial \mathbf{u}}{\partial t} + \boldsymbol{\omega} \times \mathbf{u} + \nabla\left(\frac{u^2}{2}\right) = -\frac{1}{\rho_f}\nabla p_{\text{swirl}}. \tag{501}$$

Q.5 Filament Magnus transport

For a circulation-carrying swirl-string probe K,

$$\mathbf{F}_{\mathrm{M}}^{(K)} = \int_K \rho_f \Gamma_K \mathbf{T}(s) \times [\mathbf{U}_K(s) - \mathbf{u}(\mathbf{X}(s))] \, ds. \tag{502}$$

Q.6 Curvature self-transport

In the local-induction approximation,

$$\mathbf{f}_\kappa \simeq -\frac{\rho_f \Gamma_K^2}{4\pi} \Lambda_K \kappa \mathbf{N}. \tag{503}$$

Q.7 Dimensional and density check

Equation (502) is a force on a closed filament. The corresponding local density

$$\mathbf{f}_{\mathrm{M}} = \rho_f \Gamma_K \, \mathbf{T} \times (\mathbf{U}_K - \mathbf{u}) \tag{504}$$

has units

$$[\rho_f \Gamma_K \Delta U] = \mathrm{kg\,m^{-3}\,m^2\,s^{-1}\,m\,s^{-1}} = \mathrm{N\,m^{-1}}. \tag{505}$$

With the canonical values

$$\Gamma_0 = 2\pi r_c v_{\circlearrowleft}^* = 9.6836192 \times 10^{-9} \ \mathrm{m^2\,s^{-1}}, \tag{506}$$

$$\rho_f \Gamma_0 = 6.7785334 \times 10^{-15} \frac{\mathrm{N\,m^{-1}}}{\mathrm{m\,s^{-1}}}, \tag{507}$$

$$\rho_f \Gamma_0 v_{\circlearrowleft}^* = 7.4146692 \times 10^{-9} \ \mathrm{N\,m^{-1}}. \tag{508}$$

Thus, with ρ_f, the Magnus layer is a weak background-transport term rather than a replacement for the saturated core force scale $F_{\mathrm{swirl}}^{\max} = 29.053507$ N. Replacing ρ_f by ρ_{core} would change the physical sector and is not allowed in this transport estimate.

Q.8 Axis/off-axis swirl interpretation

For an axisymmetric galactic swirl field

$$\mathbf{u} = u_\theta(r)\mathbf{e}_\theta, \qquad \boldsymbol{\omega} = \omega_z(r)\mathbf{e}_z, \tag{509}$$

an exactly axial probe suppresses the transverse Magnus correction by symmetry. Off-axis motion samples the azimuthal flow. For a flat-tail branch $u_\theta(r) \to v_0$,

$$\omega_z = \frac{1}{r}\frac{d}{dr}(r v_0) = \frac{v_0}{r}, \qquad \boldsymbol{\omega} \times \mathbf{u} = -\frac{v_0^2}{r}\mathbf{e}_r. \tag{510}$$

The inward sign is therefore already contained in the Euler decomposition of the background swirl. This observation should be used as an interpretation of probe drift and centripetal balance, not as an independent gravity postulate.

Q.9 Status

The scalar pressure term is canon-level. The Magnus and curvature terms are research-track transport corrections until a universal coupling law for macroscopic matter is derived. Their role is to describe path curvature, chiral drift, and frame-dragging-like probe effects on top of the scalar pressure/clock bridge.

Q.10 Research Track: Hybrid Density-Source Swirl-Clock Benchmark

Status. [RESEARCH TRACK / BENCHMARK CANDIDATE]. This subsection records a dimensionally corrected benchmark ansatz for coupling local Swirl-Clock dilation to density-source gravitation across microscopic and macroscopic regimes. It is not a canonical master equation. It may be promoted only if the transition function, source density, and rotational-length scale are independently derived from SST medium dynamics and pass the recovery tests listed below.

Hybrid clock ansatz. For a localized swirl-string configuration K, define the candidate clock factor

$$\boxed{S_{(t)\,\mathrm{hyb}}^{\circlearrowleft}(r;K) = \left[1 - \frac{2G_{\mathrm{hyb}}(r)M_{\mathrm{hyb}}(r;K)}{rc^2} - \frac{\|\mathbf{u}_{\mathrm{rel}} + \mathbf{v}_\theta(r;K)\|^2}{c^2} - \frac{\ell_\Omega^2(r;K)\Omega_{\mathrm{int}}^2(r;K)}{c^2}\right]^{1/2}} \tag{511}$$

where $\mathbf{u}_{\mathrm{rel}}$ is the relative velocity of the probe or localized configuration with respect to the background foliation, $\mathbf{v}_\theta$ is the resolved circular swirl velocity, and $\ell_\Omega\Omega_{\mathrm{int}}$ is the linear speed associated with an unresolved internal rotational mode.

Hybrid interpolation. The benchmark interpolation is written as

$$\mu(r) = \exp\left[-\left(\frac{r}{R_0}\right)^2\right], \tag{512}$$

$$G_{\mathrm{hyb}}(r) = \mu(r)G_{\mathrm{swirl}} + \big[1-\mu(r)\big]G_N, \tag{513}$$

$$M_{\mathrm{hyb}}(r;K) = \mu(r)M_{\mathrm{eff}}^{\mathrm{SST}}(r;K) + \big[1-\mu(r)\big]M_{\mathrm{bulk}}. \tag{514}$$

Here R_0 is a research-track transition length, not a canonical constant, and M_{bulk} denotes the independently measured macroscopic source mass used in the Newtonian recovery regime.

Density-source mass closure. The SST-compatible source is the mass-equivalent swirl-energy density, not a bare geometric density:

$$\boxed{M_{\mathrm{eff}}^{\mathrm{SST}}(r;K) = \int_{B_r} \rho_m^{\mathrm{eff}}(\mathbf{x};K)\,d^3x, \qquad \rho_m^{\mathrm{eff}} = \frac{\rho_E}{c^2} = \frac{\rho_f\,\|\mathbf{v}_\circlearrowleft\|^2}{2c^2}.} \tag{515}$$

This form keeps the source term aligned with the canonical density hierarchy $\rho_E = \frac{1}{2}\rho_f\|\mathbf{v}_\circlearrowleft\|^2$ and $\rho_m = \rho_E/c^2$.

Phenomenological spherical test profile. For numerical benchmarks one may use the non-canonical profile

$$\rho_m^{\mathrm{eff}}(r) = \rho_0 e^{-r/r_*}, \tag{516}$$

which gives the enclosed source mass

$$\boxed{M_{\mathrm{eff}}^{\exp}(r) = 8\pi\rho_0 r_*^3\left[1 - \left(1 + \frac{r}{r_*} + \frac{1}{2}\frac{r^2}{r_*^2}\right)e^{-r/r_*}\right].} \tag{517}$$

The small-radius limit is

$$M_{\mathrm{eff}}^{\exp}(r) = \frac{4\pi}{3}\rho_0 r^3 + O(r^4), \qquad r \ll r_*, \tag{518}$$

so the enclosed source remains regular at the origin. Omitting the quadratic term in Eq. (517) breaks this recovery limit and is therefore inadmissible.

Irrotational compression and boundary response. The exterior circular-flow branch used for bubble-boundary benchmarks is

$$\mathbf{v}_\theta(r;K) = \frac{\Gamma_K}{2\pi r}\,\hat{\boldsymbol{\theta}}, \qquad r \geq r_{\mathrm{core}}, \tag{519}$$

with pressure response inherited from Euler radial balance,

$$\frac{1}{\rho_f}\frac{dp_{\mathrm{swirl}}}{dr} = \frac{v_\theta^2}{r}. \tag{520}$$

Thus compression of the boundary at fixed circulation increases $v_\theta \propto r^{-1}$, raises the local kinetic energy density, and lowers the Swirl-Clock factor through Eq. (511). Energy addition may be represented by increasing Γ_K, by increasing the internal mode speed $\ell_\Omega\Omega_{\mathrm{int}}$, or by changing the boundary radius; these alternatives must be reported separately in benchmarks.

Dimensional guard for internal rotation. A bare factor Ω^2/c^2 is dimensionally inadmissible. The only admissible clock contribution from an angular rate is

$$\boxed{\frac{v_\Omega^2}{c^2} = \frac{\ell_\Omega^2(r;K)\Omega_{\rm int}^2(r;K)}{c^2},} \tag{521}$$

where ℓ_Ω is the length scale on which the internal angular mode is realized. If $\ell_\Omega = r_c$ and $\Omega_{\rm int} = \|\mathbf{v}_\circlearrowleft\|/r_c$, this term reduces to $\|\mathbf{v}_\circlearrowleft\|^2/c^2$ and must not be counted again as an independent correction.

Required benchmark reporting. Any numerical use of Eq. (511) must report the four dimensionless contributions separately:

$$\mathcal{B}_G = \frac{2G_{\rm hyb}M_{\rm hyb}}{rc^2}, \qquad \mathcal{B}_\theta = \frac{\|\mathbf{u}_{\rm rel} + \mathbf{v}_\theta\|^2}{c^2}, \qquad \mathcal{B}_\Omega = \frac{\ell_\Omega^2\Omega_{\rm int}^2}{c^2}, \qquad \mu(r). \tag{522}$$

At minimum the benchmark set must include electron-core, proton-core, Earth, Sun, and neutron-star limits. The candidate is rejected if it fails the weak field recovery $S^{\circlearrowleft}_{(t)\,\rm hyb} \to \sqrt{1 - 2G_N M_{\rm bulk}/(rc^2)}$ for $\mu(r) \to 0$, or if any contribution is dimensionally non-scalar.

Q.11 Research Track: Operator-Valued Swirl-Clock and Visibility Diagnostics

[RESEARCH TRACK / CONDITIONAL DIAGNOSTIC]. The core canon uses the Swirl-Clock as a scalar field. For interferometric or quantum-probe language, one may introduce an operator-valued clock readout only as a diagnostic layer. It is not a replacement for the scalar canonical law and it does not promote a full quantum-gravity operator algebra.

Let $\widehat{\mathcal{U}}_\circlearrowleft(x)$ be a positive self-adjoint operator representing the local squared swirl-speed diagnostic for a declared probe state, with spectrum restricted to $0 \le \widehat{\mathcal{U}}_\circlearrowleft < c^2$ on the domain of the experiment. Define

$$\boxed{\widehat{S}^{\circlearrowleft}_{(t)}(x) := \left(1 - \frac{\widehat{\mathcal{U}}_\circlearrowleft(x)}{c^2}\right)^{1/2}} \tag{523}$$

by functional calculus on that domain. In the weak-swirl limit,

$$\widehat{S}^{\circlearrowleft}_{(t)} = 1 - \frac{\widehat{\mathcal{U}}_\circlearrowleft}{2c^2} + O\left(\frac{\widehat{\mathcal{U}}_\circlearrowleft^2}{c^4}\right). \tag{524}$$

For a prepared probe state $|\psi\rangle$, the mean clock shift and the clock-noise diagnostic are

$$\left(\frac{\Delta\nu}{\nu}\right)_\psi := \langle\psi|\widehat{S}^{\circlearrowleft}_{(t)}|\psi\rangle - 1, \tag{525}$$

$$\sigma_S^2(\psi) := \langle\psi|\big(\widehat{S}^{\circlearrowleft}_{(t)}\big)^2|\psi\rangle - \langle\psi|\widehat{S}^{\circlearrowleft}_{(t)}|\psi\rangle^2. \tag{526}$$

Thus frequency shifts probe the expectation value of the clock operator, while visibility or coherence loss can probe its variance.

A minimal Gaussian dephasing diagnostic may be written as

$$\frac{\mathcal{V}(T)}{\mathcal{V}(0)} \simeq \exp\left[-\frac{1}{2}\,\mathrm{Var}_\psi\Big(\widehat{\Phi}_S(T)\Big)\right], \qquad \widehat{\Phi}_S(T) := \int_0^T \delta\widehat{\Omega}_S(N)\,dN, \tag{527}$$

where $\delta\widehat{\Omega}_S$ is the declared clock-induced phase-rate fluctuation for the experimental model. Equation (527) is a visibility ansatz, not a universal law.

[STATUS GUARD]. Equations (523)–(527) may be used only for research-track probe design. Promotion requires a declared Hilbert space, operator domain, preparation protocol, and a benchmark separating mean clock shifts from variance-driven decoherence.

Q.12 Research Track: Swirl–Electromagnetic Normalization by Vorticity

Q.13 Purpose and Status

This appendix records a research-track extension of Swirl–String Theory (SST) in which electromagnetic field variables are interpreted as SI-normalized coarse-grained measures of swirl kinematics. The result is not proposed as a replacement of Maxwell's equations at the level of observed electrodynamics. Instead, it is a constitutive bridge between the SST swirl-medium variables and the ordinary electromagnetic variables $\mathbf{E}$, $\mathbf{B}$, $\mathbf{A}$, $\mathbf{D}$, and $\mathbf{H}$.

The central observation is that a dimensionally correct identification of the vector potential is obtained by scaling the local swirl velocity by the electron mass-to-charge ratio:

$$\boxed{\mathbf{A}_{\mathrm{SI}} = \frac{m_e}{e}\,\mathbf{u}_{\circlearrowleft}(\mathbf{x},t)}. \tag{528}$$

Since $[\mathbf{A}] = \mathrm{kg\,m\,C^{-1}\,s^{-1}}$ and $[(m_e/e)\mathbf{u}_{\circlearrowleft}] = \mathrm{kg\,C^{-1}} \cdot \mathrm{m\,s^{-1}}$, this equation has the correct SI units. This use of $\mathbf{u}_{\circlearrowleft}(\mathbf{x},t)$ is intentional: the EM bridge is a local-field bridge, not a curl of the calibrated scalar speed $v_{\circlearrowleft}^{*}$.

[CRITICAL NOTE] Eq. (528) is a constitutive normalization, not a proof that the full nonlinear Euler vorticity dynamics are isomorphic to vacuum Maxwell theory. The bridge is admissible only in a coarse-grained, transverse, weak-response regime where advective and vortex-stretching terms are either projected out, averaged away, or shown to be subleading by an explicit perturbation estimate.

The associated magnetic field is then

$$\boxed{\mathbf{B}_{\mathrm{SST}} = \nabla \times \mathbf{A}_{\mathrm{SI}} = \frac{m_e}{e}\,\boldsymbol{\omega}}, \qquad \boldsymbol{\omega} = \nabla \times \mathbf{u}_{\circlearrowleft}(\mathbf{x},t). \tag{529}$$

This is the cleanest form of the statement that magnetic flux arises from structured vorticity, not merely from charge flow. The charge-current description remains the effective macroscopic description; the SST interpretation places the microscopic origin of the magnetic field in quantized circulation and vorticity.

Q.14 Why Direct Vorticity Source Terms Are Rejected

Earlier trial equations attempted to write vorticity as an additional source term directly inside Maxwell's equations, for example terms of the schematic form

$$\nabla \cdot \mathbf{E} \sim \frac{\rho}{\varepsilon_0} + \frac{F_{\max}\omega}{\|\mathbf{v}_{\circlearrowleft}\| {r_c}^2}, \qquad \nabla \times \mathbf{B} \sim \mu_0 \mathbf{J} + \frac{\|\mathbf{v}_{\circlearrowleft}\|\hbar}{q{r_c}^2}\boldsymbol{\Omega}. \tag{530}$$

These forms are not retained as canonical equations because the added terms do not generally have the SI dimensions required by Maxwell's source equations. In particular, $\nabla \cdot \mathbf{E}$ has units $\mathrm{V\,m^{-2}}$, while the proposed vorticity-force expression does not reduce to this unit without an additional calibrated constitutive coefficient. Likewise, the Ampere–Maxwell equation requires units of $\mathrm{T\,m^{-1}}$, not a force-per-charge expression multiplied by an unfixed vorticity vector.

Therefore, the research-track correction is:

> Vorticity should not be inserted as an uncalibrated extra Maxwell source. It should enter through a constitutive identification of the potential and field variables.

The retained bridge is thus Eq. (528), not Eq. (530).

Q.15 Electric Field from Time-Dependent Swirl

With the vector potential bridge established, the electric field follows from the usual potential relation:

$$\boxed{\mathbf{E}_{\mathrm{SST}} = -\frac{\partial \mathbf{A}_{\mathrm{SI}}}{\partial t} - \nabla\Phi_{\mathrm{SST}} = -\frac{m_e}{e}\frac{\partial \mathbf{u}_{\circlearrowleft}(\mathbf{x},t)}{\partial t} - \nabla\Phi_{\mathrm{SST}}}. \tag{531}$$

Here Φ_{SST} is the SI-normalized scalar potential associated with compressive, pressure-like, or boundary-induced swirl response. In strictly incompressible regions, Φ_{SST} should be interpreted as an effective potential imposed by boundary constraints, topology, and coarse-grained pressure balance rather than by local volume compression.

Equation (531) gives the SST analogue of Faraday induction: time-varying swirl produces an electric field. Taking the curl gives

$$\nabla \times \mathbf{E}_{\mathrm{SST}} = -\frac{\partial}{\partial t}(\nabla \times \mathbf{A}_{\mathrm{SI}}) = -\frac{\partial \mathbf{B}_{\mathrm{SST}}}{\partial t}, \tag{532}$$

since $\nabla \times \nabla \Phi_{\mathrm{SST}} = 0$. Thus the ordinary Faraday law is preserved exactly.

Q.16 Maxwell Equations as Effective Medium Equations

The correct research-track formulation is to retain Maxwell's equations in their macroscopic form while adding swirl-induced polarization and magnetization terms:

$$\nabla \cdot \mathbf{D}_{\mathrm{SST}} = \rho_{\mathrm{free}}, \tag{533}$$

$$\nabla \cdot \mathbf{B}_{\mathrm{SST}} = 0, \tag{534}$$

$$\nabla \times \mathbf{E}_{\mathrm{SST}} = -\frac{\partial \mathbf{B}_{\mathrm{SST}}}{\partial t}, \tag{535}$$

$$\nabla \times \mathbf{H}_{\mathrm{SST}} = \mathbf{J}_{\mathrm{free}} + \frac{\partial \mathbf{D}_{\mathrm{SST}}}{\partial t}. \tag{536}$$

The constitutive relations are

$$\mathbf{D}_{\mathrm{SST}} = \varepsilon_0 \mathbf{E}_{\mathrm{SST}} + \mathbf{P}_{\mathrm{swirl}}, \tag{537}$$

$$\mathbf{H}_{\mathrm{SST}} = \frac{1}{\mu_0}\mathbf{B}_{\mathrm{SST}} - \mathbf{M}_{\mathrm{swirl}}. \tag{538}$$

Here $\mathbf{P}_{\mathrm{swirl}}$ and $\mathbf{M}_{\mathrm{swirl}}$ encode the coarse-grained response of bound swirl structures, knot cores, circulating R-phase modes, and boundary-induced polarization.

This formulation has an important advantage: it does not violate the observed Maxwell structure. It instead interprets charge and current as effective descriptions of deeper circulation and vorticity organization.

Q.17 Canonical Circulation-Scale Calibration and the QED Critical Field

At the SST canonical circulation scale the characteristic angular frequency scale is

$$\Omega_{\mathrm{core}} = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c}. \tag{539}$$

Using the canonical SST values

$$\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^{6}\ \mathrm{m\,s^{-1}}, \qquad r_c = 1.40897017 \times 10^{-15}\ \mathrm{m}, \tag{540}$$

one obtains

$$\Omega_{\mathrm{core}} = 7.76344066 \times 10^{20}\ \mathrm{s^{-1}}. \tag{541}$$

The corresponding SST magnetic field scale is

$$B_{\mathrm{core}}^{\mathrm{SST}} = \frac{m_e}{e}\Omega_{\mathrm{core}}. \tag{542}$$

Numerically,

$$\frac{m_e}{e} = 5.68563010 \times 10^{-12}\ \mathrm{kg\,C^{-1}}, \tag{543}$$

and therefore

$$\boxed{B_{\mathrm{core}}^{\mathrm{SST}} = 4.41400519 \times 10^{9}\ \mathrm{T}}. \tag{544}$$

This is not an arbitrary scale. The standard QED critical magnetic field is

$$B_{\mathrm{QED}} = \frac{m_e^2 c^2}{e\hbar} = 4.41400522 \times 10^{9}\ \mathrm{T}. \tag{545}$$

Thus

$$\frac{B_{\mathrm{core}}^{\mathrm{SST}}}{B_{\mathrm{QED}}} = 0.999999993. \tag{546}$$

The equality follows analytically from the SST Compton–horn-radius relation

$$\lambda_c = \frac{2\pi c r_c}{\|\mathbf{v}_{\circlearrowleft}\|}, \tag{547}$$

together with the standard identity

$$\lambda_c = \frac{2\pi\hbar}{m_e c}. \tag{548}$$

Equating these two expressions gives

$$\frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c} = \frac{m_e c^2}{\hbar}. \tag{549}$$

Substitution into Eq. (542) yields

$$B_{\mathrm{core}}^{\mathrm{SST}} = \frac{m_e}{e}\frac{m_e c^2}{\hbar} = \frac{m_e^2 c^2}{e\hbar} = B_{\mathrm{QED}}. \tag{550}$$

This is the strongest numerical result of this appendix: the SI-normalized vorticity bridge naturally reproduces the Schwinger/QED magnetic field scale at the SST core.

The associated electric field scale is

$$E_{\mathrm{core}}^{\mathrm{SST}} = cB_{\mathrm{core}}^{\mathrm{SST}} = 1.32328547 \times 10^{18}\ \mathrm{V\,m^{-1}}, \tag{551}$$

which matches the usual Schwinger electric field

$$E_{\mathrm{Schwinger}} = \frac{m_e^2 c^3}{e\hbar}. \tag{552}$$

Q.18 Corrected Maximum Force Relation

The same calibration fixes the correct factor in the SST maximum-force relation. The expression

$$\frac{\|\mathbf{v}_{\circlearrowleft}\|\hbar}{{r_c}^2} \tag{553}$$

has units of force, but it is twice the canonical force scale. The consistent relation is

$$\boxed{F_{\mathrm{max}}^{\mathrm{swirl}} = \frac{\|\mathbf{v}_{\circlearrowleft}\|\hbar}{2{r_c}^2}}. \tag{554}$$

Using the canonical constants gives

$$F_{\mathrm{max}}^{\mathrm{swirl}} = 29.0535098\ \mathrm{N}, \tag{555}$$

in agreement with the canonical value

$$F_{\mathrm{max}}^{\mathrm{swirl}} \simeq 29.053507\ \mathrm{N}. \tag{556}$$

The electron rest energy relation is then

$$\boxed{m_e c^2 = 2F_{\mathrm{max}}^{\mathrm{swirl}} r_c}. \tag{557}$$

Equivalently,

$$m_e = \frac{2F_{\mathrm{max}}^{\mathrm{swirl}} r_c}{c^2}. \tag{558}$$

Numerically,

$$\frac{2F_{\mathrm{max}}^{\mathrm{swirl}} r_c}{c^2} = 9.10938 \times 10^{-31}\ \mathrm{kg}, \tag{559}$$

which matches the electron mass within the stated calibration precision.

Q.19 Photon and R-Phase Normal Modes

A second research-track result is that photon frequency should not be introduced through an ad hoc energy-density law of the form

$$u(r,\omega,T) \propto \frac{F_{\max}\omega^3}{\|\mathbf{v}_{\circlearrowleft}\| r^2}\frac{1}{\exp(\hbar\omega/k_BT)-1}. \tag{560}$$

This expression is not retained as a fundamental energy density because its bare prefactor does not have the units of energy per volume without additional geometric or constitutive factors.

Instead, the research-track photon sector is formulated as a normal-mode condition on a closed or quasi-closed R-phase path:

$$\boxed{k_n = \frac{2\pi n}{L_{\text{eff}}(K)}, \qquad \omega_n = c_\gamma k_n = \frac{2\pi n c_\gamma}{L_{\text{eff}}(K)}}. \tag{561}$$

Here $L_{\text{eff}}(K)$ is the effective optical/swirl path length of the supporting configuration K, and c_γ is the propagation speed of the R-phase excitation. In the vacuum limit one expects $c_\gamma = c$. For a circular ring of radius R,

$$L_{\text{eff}} = 2\pi R, \tag{562}$$

so that

$$\boxed{\omega_n = \frac{n c_\gamma}{R}}. \tag{563}$$

This gives a clean formulation of the statement:

> Photon frequency depends on the supporting swirl-ring or path structure.

For knotted, linked, or boundary-deformed paths, $L_{\text{eff}}(K)$ may include writhe, torsion, self-linking, and medium-response corrections:

$$L_{\text{eff}}(K) = L(K) + \Delta L_{\text{writhe}} + \Delta L_{\text{torsion}} + \Delta L_{\text{boundary}} + \Delta L_{\text{medium}}. \tag{564}$$

This places the photon sector in the same mathematical family as the SST delay-loop and mode-selection principles.

Q.20 Kelvin Impulse Bridge

For a thin circular swirl ring with circulation Γ, major radius R, and core radius a, the classical ring scalings are

$$E_{\text{ring}} = \frac{1}{2}\rho_f \Gamma^2 R\left[\ln\left(\frac{8R}{a}\right) - \alpha_{\text{ring}}\right], \tag{565}$$

$$P_{\text{ring}} = \rho_f \Gamma \pi R^2, \tag{566}$$

$$V_{\text{ring}} = \frac{\Gamma}{4\pi R}\left[\ln\left(\frac{8R}{a}\right) - \beta_{\text{ring}}\right]. \tag{567}$$

These relations provide a bridge between Kelvin's impulse theory and an SST momentum assignment. A research-track identification may be written as

$$\boxed{\hbar k \leftrightarrow \eta_I P_{\text{swirl}}}, \qquad P_{\text{swirl}} = \rho_f \Gamma \pi R^2. \tag{568}$$

The coefficient η_I is not a free fit parameter in the final theory. It must be determined from the same SI-normalization that fixes Eqs. (528)–(550), or from an independently specified R-phase normalization.

This suggests a useful research direction:

> Planck momentum may correspond to normalized Kelvin impulse of a circulating R-phase excitation.

Q.21 Topology, Helicity, and Reconnection

The ideal SST sector assumes conservation of circulation, linkage, and helicity in the absence of reconnection. This corresponds to the Helmholtz–Kelvin regime:

$$\frac{D\Gamma}{Dt} = 0, \qquad \frac{D\mathcal{H}}{Dt} = 0, \qquad \mathcal{H} = \int \mathbf{u}_{\circlearrowleft} \cdot \boldsymbol{\omega}\, d^3x. \tag{569}$$

In this ideal sector, knot class is preserved.

However, laboratory and numerical studies of knotted and linked fluid structures show that reconnection can transfer helicity between linking, twist, and writhe. Therefore, SST should distinguish two regimes:

$$\text{Ideal sector:} \quad K(t) = K(0), \qquad \Gamma = \text{constant}, \qquad \mathcal{H} = \text{constant}, \tag{570}$$

$$\text{Transition sector:} \quad K_i \to K_j, \qquad \Delta\mathcal{H} = \Delta\mathcal{L} + \Delta\mathcal{T} + \Delta\mathcal{W}, \tag{571}$$

where $\mathcal{L}$, $\mathcal{T}$, and $\mathcal{W}$ denote linking, twist, and writhe contributions respectively.

This is relevant for the SST interpretation of measurement, decay, and mode conversion. A stable particle-like excitation belongs to the ideal sector. A decay or transition corresponds to a controlled failure of ideal topological conservation, usually through reconnection or boundary interaction.

Q.22 Pressure Envelopes and Spherical Equilibrium

The electromagnetic bridge above should be kept distinct from the pressure-envelope sector of SST. The pressure sector is governed by the Bernoulli-type swirl balance

$$p + \frac{1}{2}\rho_f \|\mathbf{v}_{\circlearrowleft}\|^2 = p_0, \tag{572}$$

or, for a perturbation,

$$\Delta p = -\frac{1}{2}\rho_f \Delta\left(\|\mathbf{v}_{\circlearrowleft}\|^2\right). \tag{573}$$

This pressure defect can generate attraction-like behavior in the gravitational/swirl-clock sector. It should not be confused with the magnetic vorticity bridge

$$\mathbf{B}_{\mathrm{SST}} = \frac{m_e}{e}\boldsymbol{\omega}. \tag{574}$$

Thus the research-track separation is:

$$\text{EM-like sector:} \quad \mathbf{A}_{\mathrm{SI}} = \frac{m_e}{e}\mathbf{u}_{\circlearrowleft}(\mathbf{x}, t), \qquad \mathbf{B}_{\mathrm{SST}} = \frac{m_e}{e}\boldsymbol{\omega}, \tag{575}$$

$$\text{pressure/gravity sector:} \quad \Delta p = -\frac{1}{2}\rho_f \Delta(\|\mathbf{v}_{\circlearrowleft}\|^2), \qquad d\tau_{\mathrm{loc}}/dt_\infty = \sqrt{1 - \|\mathbf{v}_{\circlearrowleft}\|^2/c^2}. \tag{576}$$

Q.23 Negative-Temperature and Entropy Analogy

A further research-track idea is that bounded swirl ensembles may possess entropy-ordering behavior analogous to negative-temperature vortex states. This is not required for the electromagnetic normalization, but it may be useful in the classification of high-energy knot ensembles.

A possible entropy functional is

$$S_{\mathrm{swirl}} = k_B \ln \Omega_{\mathrm{topo}}(E, \Gamma, \mathcal{H}), \tag{577}$$

where Ω_{topo} counts accessible configurations at fixed energy, circulation, and helicity. A negative-temperature-like regime would satisfy

$$\frac{\partial S_{\mathrm{swirl}}}{\partial E} < 0. \tag{578}$$

Such a regime would describe increasing order with increasing energy, for example through clustering, alignment, or coherent large-scale swirl organization. This idea should remain in the research track until it is connected to a concrete SST Hamiltonian and a well-defined finite phase space.

Q.24 Research Claims Retained

The following claims are retained for research-track development:

1. Magnetic field strength can be SI-normalized as vorticity:

$$\mathbf{B}_{\mathrm{SST}} = \frac{m_e}{e}\boldsymbol{\omega}.$$

2. The vector potential is the SI-normalized swirl velocity:

$$\mathbf{A}_{\mathrm{SI}} = \frac{m_e}{e}\mathbf{u}_{\circlearrowleft}(\mathbf{x}, t).$$

3. The core-scale SST magnetic field equals the QED critical field:

$$B_{\mathrm{core}}^{\mathrm{SST}} = B_{\mathrm{QED}} = \frac{m_e^2 c^2}{e\hbar}.$$

4. The associated electric field equals the Schwinger scale:

$$E_{\mathrm{core}}^{\mathrm{SST}} = cB_{\mathrm{core}}^{\mathrm{SST}} = \frac{m_e^2 c^3}{e\hbar}.$$

5. Photon frequency should be modeled by normal-mode closure:

$$\omega_n = \frac{2\pi n c_\gamma}{L_{\mathrm{eff}}(K)}.$$

6. Kelvin impulse may provide the bridge to photon momentum:

$$\hbar k \leftrightarrow \eta_I \rho_f \Gamma \pi R^2.$$

7. Reconnection belongs to a separate transition sector and should not be mixed with ideal knot conservation.

Q.25 Claims Not Retained as Canonical

The following statements are not retained as canonical equations:

1. Direct insertion of uncalibrated vorticity terms into Maxwell source equations.
2. The statement that charge is unnecessary in all electromagnetic phenomena. In this appendix charge remains the effective macroscopic source variable.
3. The claim that Podkletnov-type gravitational shielding is established evidence for SST. Such material may be mentioned only as speculative historical motivation, not as support for the canon.
4. Photon energy-density laws containing $F_{\max}\omega^3/(\|\mathbf{v}_{\circlearrowleft}\| r^2)$ without a complete dimensional and geometric derivation.

Q.26 Minimal Experimental Tests

The vorticity-normalization bridge suggests several falsifiable directions.

1. Structured plasmonic or polaritonic media. If electromagnetic fields are coupled to structured circulation-like degrees of freedom, then anisotropic or hyperbolic media should show geometry-dependent emission and propagation consistent with an effective path length $L_{\mathrm{eff}}(K)$.

2. Superfluid or quantum-fluid circulation. Quantized circulation structures should generate measurable effective magnetization only when the system contains charged, spinful, or electromagnetically coupled degrees of freedom. A neutral inviscid fluid alone should not emit ordinary electromagnetic fields unless an additional coupling channel exists.

3. Ring-resonator delay tests. The photon/R-phase normal-mode law predicts that frequency selection should scale as

$$\omega_n \propto \frac{1}{L_{\rm eff}}.$$

Boundary deformation, writhe-like path changes, or torsional loading should shift the selected resonances.

4. Core-scale extrapolation. The equality

$$B_{\rm core}^{\rm SST} = B_{\rm QED}$$

is not directly reachable in ordinary laboratory magnetic fields, but it can be tested indirectly by checking whether the same normalization reproduces Compton, Schwinger, and electron-rest-energy scales without additional tuning.

Q.27 Numerical Verification Table

Using the canonical SST constants

$$\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^6 \ \mathrm{m\,s^{-1}}, \qquad r_c = 1.40897017 \times 10^{-15} \ \mathrm{m},$$

one obtains:

Quantity	Expression	Numerical value
Core angular scale	$\Omega_{\rm core} = \|\mathbf{v}_{\circlearrowleft}\|/r_c$	$7.76344066 \times 10^{20}\ \mathrm{s^{-1}}$
Mass-charge ratio	m_e/e	$5.68563010 \times 10^{-12}\ \mathrm{kg\,C^{-1}}$
SST core magnetic field	$(m_e/e)\Omega_{\rm core}$	$4.41400519 \times 10^9\ \mathrm{T}$
QED critical magnetic field	$m_e^2c^2/(e\hbar)$	$4.41400522 \times 10^9\ \mathrm{T}$
Ratio	$B_{\rm core}^{\rm SST}/B_{\rm QED}$	0.999999993
SST electric field scale	$cB_{\rm core}^{\rm SST}$	$1.32328547 \times 10^{18}\ \mathrm{V\,m^{-1}}$
Circulation quantum	$2\pi r_c\|\mathbf{v}_{\circlearrowleft}\|$	$9.68361920 \times 10^{-9}\ \mathrm{m^2\,s^{-1}}$
Corrected force ceiling	$\|\mathbf{v}_{\circlearrowleft}\|\hbar/(2{r_c}^2)$	$29.0535098\ \mathrm{N}$
Electron mass from force	$2F_{\rm max}^{\rm swirl}r_c/c^2$	$9.10938 \times 10^{-31}\ \mathrm{kg}$

Q.28 Conclusion

The strongest result of this appendix is the SI-normalized vorticity bridge

$$\mathbf{A}_{\rm SI} = \frac{m_e}{e}\mathbf{u}_{\circlearrowleft}(\mathbf{x},t), \qquad \mathbf{B}_{\rm SST} = \frac{m_e}{e}\boldsymbol{\omega}.$$

It preserves Maxwell's equations as effective field equations, avoids dimensionally invalid source terms, and reproduces the QED critical field at the SST core scale. For this reason it is suitable for inclusion in the research-track sections of CANON-v0.8.x.

The proposed canonical status is:

Result	Status
$\mathbf{A}_{\rm SI} = (m_e/e)\mathbf{u}_{\circlearrowleft}(\mathbf{x},t)$	Research Track, strong candidate
$\mathbf{B}_{\rm SST} = (m_e/e)\boldsymbol{\omega}$	Research Track, strong candidate
$B_{\rm core}^{\rm SST} = B_{\rm QED}$	Numerically verified calibration result
Direct vorticity source terms in Maxwell equations	Rejected / not canonical
Photon frequency from $L_{\rm eff}(K)$	Research Track, compatible with delay-loop sector
Kelvin impulse to $\hbar k$ bridge	Research Track, unresolved normalization
Negative-temperature swirl entropy	Speculative Research Track

Q.29 Research Track: Flame-Lock, Photonless-Caustic, and Elastic-Shell Diagnostics

Status. [RESEARCH TRACK / DIAGNOSTIC ONLY]. This section records a three-channel diagnostic motivated by anomalous-field witness reports. The reports are not used as evidence. They serve only as a catalogue of separable experimental signatures compatible with the canonical sector-projection law of Section 10.15.

Q.30 Mode I: flame-lock / plume stabilization

A flame or thermal plume may be stabilized without significant optical ray deflection if the dominant SST response is local impedance or damping rather than photon-index curvature. A minimal diagnostic equation is

$$\frac{\partial \mathbf{u}_{\text{plume}}}{\partial t} + (\mathbf{u}_{\text{plume}} \cdot \nabla)\mathbf{u}_{\text{plume}} = -\frac{1}{\rho_{\text{plume}}}\nabla p + \nu \nabla^2 \mathbf{u}_{\text{plume}} - \Gamma_{\circlearrowleft} \mathbf{u}_{\text{plume}}. \tag{579}$$

The diagnostic signature is

$$P_{\text{plume}}(f) \downarrow, \qquad \Delta\phi_\gamma \approx 0, \qquad \nabla_\perp n_\gamma \approx 0. \tag{580}$$

This channel is a boundary/impedance response, not a standalone proof of a gravitational bulk force.

Primary orthodox confounders: ion wind, acoustic forcing, thermal convection suppression, electric-field stabilization of charged flame species, camera exposure artifacts, and ordinary airflow.

Q.31 Mode II: photonless sphere / optical caustic

A dark or photon-depleted region is interpreted, if real, as an optical caustic, shadow, or phase-gradient effect, not as a literal astrophysical black hole. The minimal ray equation is

$$\frac{d}{ds}\left(n_\gamma \frac{d\mathbf{x}}{ds}\right) = \nabla n_\gamma, \qquad n_\gamma = {S^{\circlearrowleft}_{(t)}}^{-1}. \tag{581}$$

The diagnostic signature is

$$\nabla_\perp n_\gamma \neq 0, \qquad \Delta\phi_\gamma = \frac{2\pi}{\lambda}\int (n_\gamma - 1)\, ds \neq 0, \qquad \mathbf{f}_{\text{matter}} \approx 0. \tag{582}$$

Macroscopic photon depletion or black-ball-like appearance over laboratory path lengths requires a resonant enhancement factor:

$$n_\gamma - 1 = Q_\gamma \frac{\|\mathbf{u}_{\circlearrowleft}\|^2}{2c^2}, \qquad Q_\gamma \gg 1. \tag{583}$$

The gain Q_γ is apparatus-specific and is not a canonical constant.

Primary orthodox confounders: thermal lensing, plasma absorption, smoke or aerosol scattering, detector saturation, schlieren artifacts, camera auto-exposure, and ordinary optical shadowing.

Q.32 Mode III: elastic shell / boundary-stress response

A localized repulsive shell is represented as a boundary-stress potential,

$$U_{\text{shell}}(r) = U_0 \exp\left[-\frac{(r - R_s)^2}{2\sigma_s^2}\right], \tag{584}$$

with radial force

$$F_r(r) = -\frac{dU_{\text{shell}}}{dr} = \frac{U_0 (r - R_s)}{\sigma_s^2} \exp\left[-\frac{(r - R_s)^2}{2\sigma_s^2}\right]. \tag{585}$$

The diagnostic signature is

$$\nabla \cdot \boldsymbol{\sigma}^{\text{swirl}} \neq 0, \qquad \Delta\phi_\gamma \approx 0 \quad \text{or independently measurable.} \tag{586}$$

This mode is a stress/boundary response. It must not be rebranded as gravity shielding unless the bulk acceleration sector is independently measured.

Primary orthodox confounders: electrostatic repulsion, magnetic force, acoustic radiation pressure, airflow, vibration, thermal expansion, hidden mechanical coupling, and contact charging.

Q.33 Required measurement vector

A valid test must record the full sectoral observable set

$$\boxed{\left\{F(t), \Delta\phi_\gamma(t), \frac{\Delta f}{f}(t), P_{\mathrm{plume}}(f,t), \mathbf{E}(t), \mathbf{B}(t), T(t), p_{\mathrm{air}}(t)\right\}}. \tag{587}$$

Any reported force must first be decomposed as

$$\mathbf{F}_{\mathrm{meas}} = \mathbf{F}_{\mathrm{EM}} + \mathbf{F}_{\mathrm{EHD}} + \mathbf{F}_{\mathrm{thermal}} + \mathbf{F}_{\mathrm{acoustic}} + \mathbf{F}_{\mathrm{mechanical}} + \mathbf{F}_{\omega} + \mathbf{F}_{\mathrm{res}}. \tag{588}$$

Only $\mathbf{F}_{\mathrm{res}}$, after uncertainty propagation and null tests, is admissible as an SST research-track candidate.

Q.34 Null hierarchy

The minimum null hierarchy is:

1. reproduce the measurement with all high-voltage and high-current channels disabled;
2. randomize phase order at fixed RMS power;
3. replace the structured geometry with a symmetry-preserving dummy load of matched impedance;
4. repeat in still air, shielded enclosure, and reduced pressure where safe;
5. compare optical, force, and plume observables under the same drive waveform.

A positive SST candidate requires a correlated sectoral signature that survives these nulls and scales with the predicted mode-selection or stress-closure parameter.

Q.35 Research Track: No-Monopole Audit of the Transport-Stress Gravity Candidate

Status. [RESEARCH TRACK / DERIVED CONSTRAINT]. This section records a constraint on the SST-73 transport-stress gravity candidate. It does not modify the canon-level weak-field closure $\nabla^2\Phi_{\circlearrowleft} = 4\pi G_{\mathrm{swirl}}\rho_m$. It shows that a double-divergence stress source belongs to the local pressure/stress channel unless an additional non-divergence density closure or boundary source is specified.

Q.36 Pressure equivalence lemma

For an incompressible inviscid medium with constant ρ_f,

$$\partial_i u_i = 0, \tag{589}$$

and no external body force, the Euler equation is

$$\rho_f\left(\partial_t u_i + u_j\partial_j u_i\right) = -\partial_i p. \tag{590}$$

Taking the divergence and using $\partial_i u_i = 0$ gives

$$\partial_i\partial_j\left(\rho_f u_i u_j\right) = -\nabla^2 p. \tag{591}$$

Therefore the SST-73 transport-stress candidate

$$\nabla^2\Phi_{\mathrm{tr}} = \lambda\,\partial_i\partial_j\left(\rho_f u_i u_j\right) \tag{592}$$

is equivalent to

$$\nabla^2\left(\Phi_{\mathrm{tr}} + \lambda p\right) = 0. \tag{593}$$

For decaying exterior boundary conditions,

$$\boxed{\Phi_{\mathrm{tr}} = -\lambda\left(p - p_\infty\right)} \tag{594}$$

up to an irrelevant harmonic constant. Candidate C is therefore not an independent long-range gravity mechanism in the smooth incompressible bulk; it is the Euler pressure channel written in Poisson form.

Q.37 No-monopole theorem for smooth compact stress sources

Let

$$S(\mathbf{x}) = \partial_i \partial_j \Sigma_{ij}, \qquad \Sigma_{ij} = \rho_f u_i u_j, \tag{595}$$

with Σ_{ij} smooth and compactly supported, or decaying sufficiently fast at infinity. The monopole moment is

$$Q_0 = \int_{\mathbb{R}^3} S(\mathbf{x})\, dV = \oint_{\partial\mathbb{R}^3} \partial_j \Sigma_{ij}\, dA_i = 0. \tag{596}$$

The dipole moment also vanishes after integration by parts:

$$Q_k = \int_{\mathbb{R}^3} x_k S(\mathbf{x})\, dV = 0. \tag{597}$$

Hence the first possible nonzero exterior multipole is quadrupolar:

$$\Phi_{\mathrm{tr}}(r) = O(r^{-3}), \qquad \|\nabla \Phi_{\mathrm{tr}}\| = O(r^{-4}). \tag{598}$$

This cannot supply a Newtonian $1/r^2$ acceleration by itself.

Q.38 Boundary and singular-core exception

The no-monopole result assumes a smooth regularized field on all of $\mathbb{R}^3$. If vortex tubes are excised, singular cores are retained, or material boundaries/shells are present, the surface term

$$\oint_{\partial V} \partial_j \Sigma_{ij}\, dA_i \tag{599}$$

need not vanish. Such terms are boundary-stress or shell-response physics $\mathcal{R}_\Pi$, not bulk Poisson gravity. A nonzero Newtonian monopole therefore requires either a genuine scalar density source, such as $\rho_m = \rho_E/c^2$, or an explicitly modeled boundary source.

Q.39 Equivalence-principle consequence of the density source

If the gravitational source density is the same effective mass density that defines inertial mass,

$$\rho_m^{\mathrm{eff}} = \frac{\rho_E}{c^2}, \qquad \rho_E = \frac{1}{2}\rho_f \|\mathbf{u}_{\circlearrowleft}\|^2, \tag{600}$$

then

$$M_{\mathrm{grav}} = \int \rho_m^{\mathrm{eff}}\, dV = \frac{1}{c^2}\int \rho_E\, dV = M_{\mathrm{inert}}. \tag{601}$$

For full topological SST states this statement must be applied to the same coarse-grained mass functional used for inertial mass, including topology, geometric gate, and clock-impedance factors. The numerical value of G_{swirl} remains **[CALIBRATED]** until derived independently of orthodox G or t_p.

Q.40 Minimal numerical tests

A verification script for this sector must include:

1. a smooth compact divergence-free test field for which $\int S\, dV \approx 0$ and $\int x_k S\, dV \approx 0$;
2. a far-field multipole fit verifying $\Phi_{\mathrm{tr}} \sim r^{-3}$, not r^{-1};
3. a pressure-equivalence test verifying $\Phi_{\mathrm{tr}} + \lambda p$ is harmonic/constant under the stated boundary conditions;
4. a separate density-source Poisson test verifying $\Phi \sim -G_{\mathrm{swirl}} M/r$ and $\|\mathbf{g}\| \sim r^{-2}$ for compact positive ρ_m^{eff}.

Q.41 Research Track: Atomic Gravity Closure and Condensate Coherence

Status. [CANON-CANDIDATE / RESEARCH BRIDGE]. This section records the electron-envelope and condensate-coherence clue for future gravity-sector tests. It does not claim gravitational shielding.

Q.42 Electron envelope as atomic closure layer

In a neutral atom, the electron envelope cancels the far electromagnetic charge sector but does not cancel positive swirl-energy density. Therefore the electron is not the dominant mass reservoir, but it is the outer phase and linking boundary through which the neutral atom closes its electromagnetic projection:

$$Q_{\text{atom}} = 0, \qquad \rho_m = \frac{\rho_E}{c^2} > 0. \tag{602}$$

The canon-safe interpretation is

> *The nucleus supplies most mass-energy, while the electron envelope supplies the neutral phase boundary and long-range matching layer.*

Q.43 Condensate coherence extension

For coherent electronic or atomic condensates, a macroscopic order parameter

$$\Psi_{\text{cond}}(\mathbf{x}) = \sqrt{n_s(\mathbf{x})}\, e^{i\theta(\mathbf{x})} \tag{603}$$

may coherently reweight boundary-stress and optical/clock projections:

$$\boldsymbol{\sigma}^{\text{swirl}} \to \boldsymbol{\sigma}^{\text{swirl}}_{\text{coh}}[n_s, \theta, \nabla\theta], \tag{604}$$

$$n_\gamma \to n_{\gamma,\text{coh}}. \tag{605}$$

The proposed clue from Meissner/BEC physics is therefore not mass cancellation, but collective phase closure: many microscopic carriers can impose one macroscopic boundary condition.

[CRITICAL NOTE]. This does not imply gravitational shielding or mass cancellation. The testable claim is only a possible phase-dependent change in boundary-stress, optical phase, or clock response across a coherence transition such as T_c.

Q.44 Condensate test signature

A minimally admissible superconducting or BEC test compares the same apparatus above and below the coherence transition:

$$\Delta\mathcal{O}_{\text{coh}} = \mathcal{O}(T < T_c) - \mathcal{O}(T > T_c), \qquad \mathcal{O} \in \{F, \Delta\phi_\gamma, \Delta f/f, \Delta\Phi\}\,. \tag{606}$$

A positive result must track coherence and vanish under matched non-coherent thermal, electromagnetic, and mechanical controls.

Q.45 Relativity Emergence Ladder in SST

Q.46 Purpose and status

This section records the precise epistemic status of special-relativistic and general-relativistic structures inside Swirl–String Theory (SST). The goal is not to overclaim that general relativity has been derived from incompressible Euler dynamics. The goal is to separate:

1. kinematic consequences of a common propagation cone;
2. clock and metric bridge structures;
3. weak-field gravitational closures;
4. the still-open problem of deriving the Einstein field equations.

Status summary. SST presently supports a conditional reconstruction of relativistic kinematics for its excitation sector, a conditional weak equivalence principle, and a conditional Newtonian limit. The full Einstein–Hilbert dynamics remain an open research target.

Q.47 Primitive structure

The minimal SST substrate is taken to be an incompressible, inviscid medium on a preferred foliation,

$$\mathcal{M}_{\mathrm{SST}} = \mathbb{R}^3 \times \mathbb{R}, \qquad \nabla \cdot \mathbf{v} = 0, \tag{607}$$

with structured vorticity, quantized circulation, a canonical transverse propagation speed c, and a local Swirl–Clock factor

$$S_{(t)}(x) = \sqrt{1 - \frac{u(x)^2}{c^2}}. \tag{608}$$

[ORTHODOX] The functional form of Eq. (608) is the standard Lorentz time-dilation factor.

[SST INTERPRETATION] The velocity $u(x)$ is interpreted as a local medium-kinematic state rather than as an abstract spacetime postulate.

[CRITICAL NOTE] If c is inserted as a primitive signal speed, then the existence of a limiting cone is not itself derived. What may be derived is the operational Lorentz kinematics of clocks and rods whose internal dynamics are mediated only by that cone.

Q.48 Light-clock derivation of the Lorentz factor

Consider a clock whose tick is defined by a transverse signal moving at speed c in the substrate frame. If the clock moves with speed u relative to the substrate, then during one half-cycle

$$c^2 \Delta t^2 = u^2 \Delta t^2 + c^2 \Delta \tau^2. \tag{609}$$

Therefore

$$\Delta \tau = \Delta t \sqrt{1 - \frac{u^2}{c^2}}. \tag{610}$$

[DERIVED, conditional] Equation (610) derives Lorentz time dilation for clocks whose internal communication is entirely mediated by substrate excitations with speed c [35].

[OPEN CANON GAP] Universal Lorentz kinematics for all matter species requires the stronger monometricity condition below.

Q.49 Monometricity theorem target

Let A label low-energy SST excitation species. The principal symbol of the linearized equations for species A defines an effective characteristic surface

$$P_A(k) = 0. \tag{611}$$

Relativistic universality requires that, at low energy,

$$P_A(k) = 0 \quad \Longleftrightarrow \quad g^{\mu\nu}_{\mathrm{eff}} k_\mu k_\nu = 0 \qquad \text{for all } A, \tag{612}$$

up to irrelevant conformal factors:

$$g^{\mu\nu}_{\mathrm{eff},A} = \Omega_A^2(x)\, g^{\mu\nu}_{\mathrm{eff}}. \tag{613}$$

[OPEN CANON GAP: MONOMETRICITY] SST currently contains at least two distinguished speeds,

$$c, \qquad \mathbf{v}_{\circlearrowleft} = \frac{\alpha c}{2}, \tag{614}$$

so exact low-energy Lorentz universality is not automatic. The required theorem is that matter knots, gauge/torsion pulses, and clock excitations share the same effective cone in the infrared. In condensed-matter language this is the Fermi-point universality problem [32].

Falsifier. Failure of Eq. (612) predicts species-dependent Lorentz violation and preferred-frame effects.

Q.50 Analogue-metric bridge

For a hyperbolic perturbation sector with common characteristic speed c, the linearized excitation equation can be written schematically as

$$\frac{1}{\sqrt{-g_{\mathrm{eff}}}}\partial_\mu\left(\sqrt{-g_{\mathrm{eff}}}\,g_{\mathrm{eff}}^{\mu\nu}\partial_\nu\phi\right)=0. \tag{615}$$

[ORTHODOX] Equations of the form (615) are standard in analogue-gravity systems [10, 37].

[SST STATUS: CONDITIONAL BRIDGE] In SST this bridge applies only to the explicit hyperbolic excitation sector. It does not follow from incompressibility alone, because incompressible Euler pressure is a constraint field rather than an ordinary compressional sound mode. The correct conditional statement is: if the torsion/transverse excitation sector has a hyperbolic principal operator with speed c, then an effective Lorentz cone follows.

[CRITICAL NOTE] Analogue gravity reproduces curved-spacetime kinematics for perturbations. It does not, by itself, derive the Einstein field equations.

Q.51 Weak equivalence principle

There are two independent conditional routes to the weak equivalence principle.

Energy-source route. If both inertial mass and gravitational source strength are determined by the same localized energy functional,

$$m_i(T)c^2=E_T,\qquad m_g(T)c^2=E_T, \tag{616}$$

then

$$m_i(T)=m_g(T). \tag{617}$$

[DERIVED, conditional] This route is valid only if the gravitational clock/pressure field couples universally to the total localized SST energy functional.

Metric-coupling route. If matter is minimally coupled to a Lorentzian metric and $\nabla_\mu T^{\mu\nu}=0$, then the point-particle limit follows geodesic motion,

$$u^\mu\nabla_\mu u^\nu=0. \tag{618}$$

[ORTHODOX CONDITIONAL THEOREM] This is the standard GR result. In SST it is not yet a primitive-substrate derivation, because the Lorentzian metric and minimal coupling are assumed at the effective-field-theory level.

Q.52 Newtonian limit

A Newtonian limit requires a scalar potential satisfying

$$\nabla^2\Phi=4\pi G_{\mathrm{eff}}\rho_m. \tag{619}$$

SST can motivate $\rho_m=\rho_E/c^2$ from the local energy accounting, but Eq. (619) remains a closure law unless derived from the coarse-grained vortex ensemble.

[CONDITIONAL] If Eq. (619) is admitted as the long-wavelength closure, the inverse-square force and Newtonian limit follow.

[OPEN CANON GAP] Derive Eq. (619) directly from the statistical mechanics or coarse-grained pressure response of many swirl strings.

Q.53 Research Track: Compton–Horn Balance Angle

[RESEARCH-TRACK GEOMETRIC DIAGNOSTIC]. This subsection records a dimensionless visualization of the calibrated Compton–horn closure used in the core canon. It does not introduce a new gravity law and it must not be used as an independent derivation of Newton's constant.

Define ϕ_G by

$$G = \frac{v_{\circlearrowleft}^{*} c^3 t_p^2}{M_e r_c} \tan^2 \phi_G. \tag{620}$$

Equivalently,

$$\tan^2 \phi_G = \frac{G M_e r_c}{v_{\circlearrowleft}^{*} c^3 t_p^2}. \tag{621}$$

Using the v0.8.18 calibration chain

$$r_c = \frac{v_{\circlearrowleft}^{*}}{\omega_c}, \qquad \omega_c = \frac{M_e c^2}{\hbar}, \qquad t_p^2 = \frac{\hbar G}{c^5}, \tag{622}$$

one obtains

$$\tan^2 \phi_G = \frac{M_e c^2}{\hbar \omega_c} = 1, \qquad \phi_G = \frac{\pi}{4}. \tag{623}$$

Thus the 45° value means that the calibrated electron benchmark weights radial horn confinement and tangential swirl transport equally in this particular dimensionless projection. It does *not* mean that gravity acts at a spatial angle of 45°, and it does *not* add a new coupling constant.

[CRITICAL NOTE] Because $t_p^2 = \hbar G/c^5$ already contains G, Eq. (623) is a closure diagnostic, not a prediction of G from non-gravitational inputs. A future promotion would require deriving the effective long-range coupling or the Planck-time scale from SST medium dynamics without importing orthodox G.

Q.53.1 Rejected microscopic Kelvin-angle ansatz

The dimensional ansatz

$$\tan \phi = \sqrt{\frac{I^{3/2}}{N \mu K^{1/2} \sqrt{\pi}}} \tag{624}$$

is not canon-ready. In SI units $[I] = \mathrm{kg\,m^2}$, while the present canon uses $[K] = \mathrm{kg\,m^{-3}\,s}$ for the density–rotation bridge. Unless μ and K are replaced by explicitly normalized quantities, Eq. (624) is not dimensionless, whereas $\tan\phi$ must be dimensionless.

A permissible research-track replacement must use only dimensionless knot functionals:

$$\tan^2 \phi_K = \frac{\mathcal{I}_K^{3/2}}{N_K \mathcal{M}_K \mathcal{K}_K^{1/2} \sqrt{\pi}}, \tag{625}$$

with, for example,

$$\mathcal{I}_K = \frac{I_K}{M_K {r_c}^2}, \qquad \mathcal{K}_K = \frac{K_K}{K_0}, \qquad \mathcal{M}_K = \frac{\mu_K}{\mu_0^*}. \tag{626}$$

For the electron/trefoil benchmark the admissible normalization condition is

$$\tan^2 \phi_{3_1} = 1, \tag{627}$$

but this is a benchmark closure condition, not yet a microscopic theorem.

Q.54 Einstein dynamics: three candidate routes

Route I: thermodynamic gravity. Jacobson's route derives the Einstein equation as an equation of state from $\delta Q = T\,dS$ applied to local Rindler horizons, assuming an entropy-area law and the Unruh temperature [38]. In SST this requires:

$$S_{\text{defect}} \propto A, \qquad T_{\text{SST}} = \frac{\hbar a}{2\pi c k_B}. \tag{628}$$

[RESEARCH TRACK] This is the most promising route because it connects directly to horizon, boundary, and holographic SST sectors.

Source-paper integration for Route I. The thermodynamic-gravity route should be linked explicitly to three source papers. SST-63 supplies the constraint-holographic boundary reconstruction layer. SST-23 supplies the accelerated torsion / Unruh-response candidate. SST-56 supplies the superfluid topology diagnostics—Hall angle, writhe barrier, and lifetime hierarchy—needed to test whether a line-tangle or fabric sector has stable protected response. Thus the route is not a single paper claim but a three-asset programme:

$$\boxed{\text{SST-63} + \text{SST-23} + \text{SST-56} \longrightarrow (S = \eta A,\ T_{\text{SST}},\ \delta Q = T\,dS)}. \tag{629}$$

The next open lemma is the Unruh-response derivation in the hyperbolic torsion sector. The area law may be attacked through a line-piercing entropy model, but its coefficient remains calibrated unless the vacuum line density is derived independently.

Route II: Sakharov induced gravity. With ultraviolet cutoff $\ell_{\text{UV}} \sim r_c$, a schematic induced gravity estimate gives [39]

$$G_{\text{ind}} \sim \frac{r_c^2 c^3}{\hbar N}. \tag{630}$$

Matching Newton's constant requires

$$N_{\text{req}} \sim \frac{r_c^2 c^3}{\hbar G} = \left(\frac{r_c}{L_p}\right)^2 = 7.60 \times 10^{39}. \tag{631}$$

[CRITICAL NOTE] Unless N_{req} is independently derived from SST mode counting, this route only restates the hierarchy r_c/L_p.

Route III: infrared EFT universality. If coarse-grained SST organizes into a diffeomorphism-invariant metric EFT, then the leading two-derivative action has the form

$$S_{\text{IR}} = \frac{c^3}{16\pi G_{\text{eff}}} \int d^4x \sqrt{-g}\,(R - 2\Lambda) + S_{\text{matter}} + S_{\text{clock}}. \tag{632}$$

[RESEARCH TRACK] This route explains why Einstein–Hilbert appears as the leading IR term, but it does not explain why a preferred-foliation incompressible medium becomes diffeomorphism invariant in the infrared.

[CRITICAL NOTE] The cosmological-constant problem is severe [44]:

$$\frac{\rho_f}{\rho_\Lambda} \approx 1.2 \times 10^{20}, \qquad \frac{\rho_{\text{core}}}{\rho_\Lambda} \approx 6.6 \times 10^{44}. \tag{633}$$

Q.55 Tensor-speed naturalness

In the Einstein–Æther comparison layer,

$$c_T^2 = \frac{c^2}{1 - c_{13}}, \qquad c_{13} = c_1 + c_3. \tag{634}$$

If SST tensor modes are identified with transverse substrate waves, and if the canonical transverse wave speed is c, then

$$c_T = c \quad \Rightarrow \quad c_{13} = 0. \tag{635}$$

[DERIVED, conditional] The luminal spin-2 speed is then not a tuning but an internal consistency condition of the mode identification [41, 42]. This mirrors the main-canon proposition of Section 10.12.

[LIMIT] This does not prove the Einstein field equations. It only fixes the spin-2 propagation speed inside the effective comparison theory.

Q.56 Final status statement

The accurate SST statement is:

> SST conditionally reconstructs relativistic kinematics for its low-energy excitation sector, provided monometricity holds. It provides conditional routes to the weak equivalence principle and to the Newtonian limit. It naturally explains luminal tensor-mode speed if gravitational tensor modes are transverse substrate waves. The full Einstein field equations remain an open research target, with thermodynamic, induced-gravity, and infrared-EFT routes under active investigation.

Q.57 Observational falsifier ladder

The relativity-emergence programme has three immediate observational gates:

1. **Tensor-speed gate.** Multimessenger propagation requires $c_T = c$, equivalently $c_{13} = 0$ in the Einstein–Æther comparison layer. Quantitatively, GW170817/GRB170817A gives

$$-3 \times 10^{-15} < \frac{c_T}{c} - 1 < 7 \times 10^{-16}, \qquad |c_{13}| \lesssim 10^{-15} \tag{636}$$

   in the Einstein–Æther comparison convention [50, 51]. This is the spin-2 version of the transverse-mode identification.

2. **Preferred-frame gate.** Any residual preferred-foliation coupling must satisfy the post-Newtonian preferred-frame constraints of the comparison theory. In practice this means that the effective α_1 and α_2 parameters cannot be freely chosen; they must be derived from the same substrate couplings that set the clock sector. Strong-field pulsar timing gives representative bounds

$$\hat{\alpha}_1 = (-0.4^{+3.7}_{-3.1}) \times 10^{-5} \quad (95\% \text{ CL; PSR } J1738 + 0333), \tag{637}$$

$$|\hat{\alpha}_2| < 1.6 \times 10^{-9} \quad (95\% \text{ CL; PSR } B1937 + 21, J1744 - 1134). \tag{638}$$

   Binary-pulsar damping and PSR J0337+1715 further reduce the post-GW170817 viable Einstein–Æther parameter region by about one order of magnitude [52, 53, 54].

3. **Species-cone gate.** Matter knots, gauge/torsion pulses, and clock excitations must share the same infrared characteristic cone. Failure of this gate predicts species-dependent Lorentz violation and directly falsifies exact monometricity.

[CRITICAL NOTE] In the orthodox Einstein–Æther comparison theory, matter is usually coupled only to the metric, not directly to the Æther field. SST cannot inherit this protection automatically, because matter is modeled as a substrate excitation. The species-cone gate is therefore stricter for SST than for a purely metric matter-coupled Einstein–Æther EFT.

[STATUS] This is not an additional derivation. It is the observational test hierarchy attached to the monometricity and tensor-speed theorem targets.

Q.58 Research Track: Core–Torsion Impedance Matching for Inertia Closure

Q.59 Purpose and status

[RESEARCH-TRACK]. This section records the canon-safe form of the inertia-closure programme. It separates the internal NLSE/Gross–Pitaevskii core layer from the transverse torsion/shear radiation layer. The separation is mandatory because the canonical swirl speed $v_{\circlearrowleft}^{*}$ is much smaller than the observed causal speed c:

$$\frac{v_{\circlearrowleft}^{*}}{c} = 3.64867628 \times 10^{-3}. \tag{639}$$

A derivation that produces $u^2/v_{\circlearrowleft}^{*2}$ or u^2/c_s^2 therefore does not yet derive the Lorentz-compatible factor u^2/c^2.

Q.60 NLSE/Gross–Pitaevskii core diagnostic

For a defocusing NLSE/Gross–Pitaevskii core with Madelung phase velocity,

$$\mathbf{u}_{\rm NLS} = \frac{\hbar_*}{m_*}\nabla S, \tag{640}$$

the circulation quantum and healing length may be written

$$\Gamma_q = \frac{h_*}{m_*}N_\Gamma, \qquad \xi = \frac{\hbar_*}{\sqrt{2}\,m_* c_s}. \tag{641}$$

For the single-winding convention $N_\Gamma = 1$, imposing

$$\Gamma_q = \Gamma_0 = 2\pi r_c v_{\circlearrowleft}^*, \qquad \xi = r_c \tag{642}$$

gives

$$c_s = \frac{\Gamma_0}{2\sqrt{2}\pi r_c} = \frac{v_{\circlearrowleft}^*}{\sqrt{2}} = 7.73465663 \times 10^5 \ \mathrm{m\,s^{-1}}. \tag{643}$$

[DERIVED-CONDITIONAL]. Equation (643) is a useful internal diagnostic of the core layer. It is not a derivation of the Lorentz signal speed.

Q.61 Transverse torsion/shear causal layer

A minimal transverse torsion displacement field $\mathbf{A}$, with $\nabla\cdot\mathbf{A} = 0$, may be assigned the quadratic bridge Lagrangian

$$\mathcal{L}_{\rm torsion} = \frac{1}{2}\rho_f\|\partial_t\mathbf{A}\|^2 - \frac{1}{2}K_T\|\nabla\times\mathbf{A}\|^2. \tag{644}$$

The transverse propagation speed is

$$c_T^2 = \frac{K_T}{\rho_f}. \tag{645}$$

The canon-compatible light-speed identification belongs only to this transverse layer:

$$c_T = c. \tag{646}$$

Its intensive impedance is

$$Z_T = \rho_f c_T = \sqrt{\rho_f K_T}, \tag{647}$$

which has the units of acoustic impedance, $\mathrm{kg\,m^{-2}\,s^{-1}}$.

Q.62 Core–Torsion Impedance Matching Lemma

Let K be a localized closed swirl-string configuration with rest-swirl energy $E_0[K]$. Define the torsion-dressed inertial tensor by the quadratic stored-energy response

$$\Delta E_{\rm torsion}[K;\mathbf{u}] = \frac{1}{2}\mathbf{u}^{\mathsf{T}}\mathsf{M}_{\rm torsion}[K]\mathbf{u} + O(\|\mathbf{u}\|^4). \tag{648}$$

The required Lorentz-compatible closure is

$$\boxed{\mathsf{M}_{\rm torsion}[K] \overset{?}{=} \frac{2E_0[K]}{c_T^2}\,\mathsf{I}} \tag{649}$$

or equivalently

$$\Delta E_{\rm torsion}[K;u] \overset{?}{=} E_0[K]\frac{u^2}{c_T^2} + O(u^4). \tag{650}$$

If $c_T = c$, Eq. (649) supplies the missing bridge from core inertia to the Lorentz-compatible clock factor.

Define the diagnostic residual

$$\chi_K^{(T)} = \frac{c_T^2}{2E_0[K]} \lambda_{\mathrm{iso}}(\mathsf{M}_{\mathrm{torsion}}[K]) . \tag{651}$$

The bridge closes only if

$$\chi_K^{(T)} = 1 \tag{652}$$

within numerical and topological tolerance.

[OPEN LEMMA]. The equality $\chi_K^{(T)} = 1$ is the falsifiable theorem target. If numerical torsion-dressing returns a stable non-unity value, the Lorentz clock factor remains a constitutive bridge rather than a derived theorem from the core model.

Q.63 Required numerical tests

A canon-safe numerical programme must report:

1. the NLSE boost coefficient $M_{\mathrm{core}}[K]$ and its comparison to $2E_0[K]/c_s^2$, not to $2E_0[K]/c^2$;
2. the torsion-dressed tensor $\mathsf{M}_{\mathrm{torsion}}[K]$ and the residual $\chi_K^{(T)}$;
3. a stiffness sweep K_T verifying whether $\chi_K^{(T)} = 1$ occurs naturally at $c_T = c$ or only after retuning;
4. the circulation-normalization ratio $\mathcal{R}_\Gamma = (h_*/m_*)/\Gamma_0$, which must equal unity under the canonical single-winding convention;
5. the trefoil-locked source test comparing $T_{2,3}$ against generic helical sources.

[CANON INTERPRETATION]. The canon may retain the two-speed discipline and the open impedance lemma. It should not promote the Lorentz clock factor to hard canon through the NLSE core alone.

Q.63.1 Research Track: EM-to-Swirl Force-Density Correspondence

Status labels. The correspondence proposed in this section is [**SPECULATIVE**] and [**PENDING DERIVATION**]. It is not a canonical replacement of Maxwell–Lorentz electrodynamics. It is a Rosetta-level research bridge between electromagnetic force density and SST substrate stress.

Structural correspondence. The electromagnetic Lorentz force density is

$$\mathbf{f}_{\mathrm{EM}} = \rho_e \mathbf{E} + \mathbf{J} \times \mathbf{B}.$$

The magnetic part has the same force-density dimension as the SST swirl-force density,

$$[\mathbf{J} \times \mathbf{B}] = \mathrm{N\,m^{-3}}, \qquad [\rho_f \mathbf{v} \times \boldsymbol{\omega}] = \mathrm{N\,m^{-3}}.$$

Hence the most conservative Rosetta correspondence is

$$\boxed{\mathbf{J} \times \mathbf{B} \;\longleftrightarrow\; \lambda_{\mathrm{EM}\to\circlearrowleft}\, \rho_f \mathbf{v} \times \boldsymbol{\omega}.}$$

Since both sides already carry units of force density,

$$\boxed{[\lambda_{\mathrm{EM}\to\circlearrowleft}] = 1.}$$

The coefficient $\lambda_{\mathrm{EM}\to\circlearrowleft}$ is therefore dimensionless. It is not fixed by dimensional analysis and remains [**SPECULATIVE / PENDING DERIVATION**] until a field-level dictionary or normalization condition is specified.

Relation to the canonical flux-impulse channel. The canonical SST electromagnetic bridge proceeds through phase, flux, and flux-piercing variables, including the effective Faraday/flux-impulse channel and the flux quantum

$$\Phi_0 = \frac{h}{2e}.$$

The bulk correspondence

$$\mathbf{J} \times \mathbf{B} \;\longleftrightarrow\; \lambda_{\mathrm{EM}\to\circlearrowleft} \rho_f \mathbf{v} \times \boldsymbol{\omega}$$

is therefore not an independent canonical EM bridge. It is a candidate continuum-limit or bulk-stress projection of the already canonical phase/flux channel. A canonical upgrade requires showing how the flux observable, phase winding, and current density reduce to a definite $\lambda_{\mathrm{EM}\to\circlearrowleft}$ in the continuum limit.

Possible field-level dictionary. A possible but non-canonical dictionary is the classical hydrodynamic analogy

$$\mathbf{A} \;\longleftrightarrow\; \alpha_A \mathbf{v}, \qquad \mathbf{B} = \nabla \times \mathbf{A} \;\longleftrightarrow\; \alpha_A \boldsymbol{\omega},$$

and

$$\mu_0 \mathbf{J} = \nabla \times \mathbf{B} \;\longleftrightarrow\; \alpha_A \nabla \times \boldsymbol{\omega}.$$

The constants and normalization implied by α_A are not fixed here. This dictionary is retained only as a research-track route toward deriving $\lambda_{\mathrm{EM}\to\circlearrowleft}$.

Stress closure and SST-44 tensor. The canonical swirl-stress candidate is the symmetric Cauchy-type stress used in the SST-44 appendix,

$$\sigma_{ij}^{(44)} = \rho_f v_i v_j + \rho_f {r_c}^2 \left(\omega_i \omega_j - \frac{1}{2} \delta_{ij} |\boldsymbol{\omega}|^2 \right) - \delta_{ij} \frac{1}{2} \rho_f |\mathbf{v}|^2.$$

Because this tensor already contains the isotropic dynamic-pressure term

$$-\delta_{ij} \frac{1}{2} \rho_f |\mathbf{v}|^2,$$

one must not also add an independent $-\nabla p_\circlearrowleft$ term with

$$p_\circlearrowleft = \frac{1}{2} \rho_f |\mathbf{v}|^2.$$

Otherwise the Bernoulli contribution is counted twice.

Two admissible decompositions. The full-stress convention is

$$\boxed{\mathbf{f}_{\mathrm{SST}} = \nabla \cdot \boldsymbol{\sigma}^{(44)} + \mathbf{f}_{\mathrm{link}}.}$$

The split-stress convention is

$$\boxed{\mathbf{f}_{\mathrm{SST}} = -\nabla p_\circlearrowleft + \nabla \cdot \boldsymbol{\sigma}_\circlearrowleft^{\mathrm{dev}} + \mathbf{f}_{\mathrm{link}},}$$

where

$$\boldsymbol{\sigma}_\circlearrowleft^{\mathrm{dev}} \equiv \rho_f \mathbf{v}\mathbf{v} + \rho_f {r_c}^2 \left(\boldsymbol{\omega}\boldsymbol{\omega} - \frac{1}{2} \mathbf{I} |\boldsymbol{\omega}|^2 \right)$$

contains the advective and torsional anisotropic parts but excludes the isotropic dynamic-pressure term. These two conventions must not be mixed.

Topological link force. The link-force term is not yet canonical. The intended structure is a topological potential functional

$$U_{\mathrm{link}} = U_{\mathrm{link}}[Lk(C_a, C_b), \Gamma_a, \Gamma_b, H],$$

with force density obtained schematically by variation,

$$\boxed{\mathbf{f}_{\mathrm{link}} = -\frac{\delta U_{\mathrm{link}}}{\delta \mathbf{x}}}$$

or, for a collective coordinate q,

$$\boxed{F_{\text{link},q} = -\frac{\partial U_{\text{link}}}{\partial q}.}$$

The linking number is

$$Lk(C_a, C_b) = \frac{1}{4\pi} \oint_{C_a} \oint_{C_b} \frac{(d\mathbf{x}_a \times d\mathbf{x}_b) \cdot (\mathbf{x}_a - \mathbf{x}_b)}{|\mathbf{x}_a - \mathbf{x}_b|^3}.$$

Until U_{link} is specified and normalized, $\mathbf{f}_{\text{link}}$ remains **[PENDING DERIVATION]**.

Research-track conclusion. The candidate bulk EM-to-swirl mapping is therefore

$$\boxed{\mathbf{f}_{\text{EM}} = \rho_e \mathbf{E} + \mathbf{J} \times \mathbf{B} \quad \rightsquigarrow \quad \mathbf{f}_{\text{SST}} = \nabla \cdot \boldsymbol{\sigma}^{(44)} + \mathbf{f}_{\text{link}}}$$

under the full-stress convention, or equivalently

$$\boxed{\mathbf{f}_{\text{SST}} = -\nabla p_{\circlearrowleft} + \nabla \cdot \boldsymbol{\sigma}^{\text{dev}}_{\circlearrowleft} + \mathbf{f}_{\text{link}}}$$

under the split-stress convention. The correspondence becomes canonizable only after $\lambda_{\text{EM}\to\circlearrowleft}$, the field-level dictionary, and U_{link} are derived or experimentally fixed.

Q.64 Foucault Pendulum as a Macroscopic Rotation-to-Clock Inference Probe

[RESEARCH TRACK / ORTHODOX INPUT / SST INFERENCE PROTOCOL]. A Foucault pendulum is not a direct clock-comparison experiment, and its ordinary swing period is not itself a direct measurement of relativistic or SST time dilation. Its useful role in SST is narrower: it provides a mechanical readout of the local vertical projection of Earth's rotation vector. That measured precession may then be used as an input to the already-canonical Swirl-Clock law. This subsection is therefore a Rosetta-level inference protocol, not a new axiom and not a replacement for precision clock transport.

Orthodox precession input. For a pendulum at geodetic latitude ϕ, the standard Foucault precession rate is

$$\Omega_F(\phi) = \Omega_\oplus \sin\phi, \tag{653}$$

where $\Omega_\oplus$ is the sidereal angular velocity of Earth [59, 60]. Equation (653) is an orthodox rotating-frame result; in SST it is used only as an operational inference channel for the coarse-grained terrestrial rotation rate.

Solving Eq. (653) gives

$$\Omega_\oplus = \frac{\Omega_F}{\sin\phi}. \tag{654}$$

The tangential speed of a surface clock relative to the non-rotating foliation frame is

$$v_{\text{rot}}(\phi, h) = \Omega_\oplus R_\perp(\phi, h), \tag{655}$$

where $R_\perp(\phi, h)$ is the distance from the clock to Earth's rotation axis. In the spherical-Earth approximation used for the numerical benchmark below,

$$R_\perp(\phi, 0) \simeq R_\oplus \cos\phi, \qquad v_{\text{rot}}(\phi) \simeq R_\oplus \Omega_F \cot\phi. \tag{656}$$

This step is the inference bridge: the pendulum measures Ω_F, and Eq. (656) converts that measurement into the rotational speed entering the local clock law.

Swirl-Clock mapping. Using the canonical SST clock factor

$$S^{\circlearrowleft}_{(t)}(x,t) = \sqrt{1 - \frac{\|\mathbf{u}_{\circlearrowleft}(x,t)\|^2}{c^2}}, \tag{657}$$

with the local coarse-grained rotational speed substituted as the kinematic input, the Earth-rotation contribution inferred from a measured Foucault rate is

$$\boxed{S^{\circlearrowleft}_{(t)\,\mathrm{rot}}(\phi) = \sqrt{1 - \frac{R_\oplus^2 \Omega_F^2 \cot^2\phi}{c^2}}} \tag{658}$$

and, for $R_\oplus \Omega_F \cot\phi \ll c$,

$$\boxed{1 - S^{\circlearrowleft}_{(t)\,\mathrm{rot}} \simeq \frac{R_\oplus^2 \Omega_F^2 \cot^2\phi}{2c^2}}. \tag{659}$$

Equivalently, using the SST calibration identity $\alpha = 2v^*_{\circlearrowleft}/c$,

$$\boxed{1 - S^{\circlearrowleft}_{(t)\,\mathrm{rot}} \simeq \frac{\alpha^2 R_\oplus^2 \Omega_F^2 \cot^2\phi}{8 v^{*\,2}_{\circlearrowleft}}}. \tag{660}$$

This expression uses the measured Foucault precession only as a macroscopic rotation readout. It does not claim that the pendulum bob internally samples the microscopic core swirl speed $v^*_{\circlearrowleft}$.

Height correction. A height displacement h above the local geoid adds the standard weak-field redshift term used in precision clock and GPS modelling [61],

$$\Delta S^{\circlearrowleft}_{(t)\,h} \simeq +\frac{gh}{c^2} = +\frac{\alpha^2 gh}{4 v^{*\,2}_{\circlearrowleft}}. \tag{661}$$

This is an orthodox weak-field benchmark term imported into the inference protocol; this subsection does not derive the gravitational redshift from Foucault dynamics.

Thus the combined first-order inference formula relative to the non-rotating foliation time N is

$$\boxed{\frac{d\tau_{\mathrm{loc}}}{dN} \simeq 1 - \frac{\alpha^2 R_\oplus^2 \Omega_F^2 \cot^2\phi}{8 v^{*\,2}_{\circlearrowleft}} + \frac{\alpha^2 gh}{4 v^{*\,2}_{\circlearrowleft}}}. \tag{662}$$

Equation (662) is an inference protocol, not a new axiom.

Energy-density form. The same rotational contribution may be written as a coarse-grained swirl-energy density,

$$\rho_E^{(\oplus)}(\phi) = \frac{1}{2}\rho_f v_{\mathrm{rot}}^2(\phi), \qquad 1 - S^{\circlearrowleft}_{(t)\,\mathrm{rot}} \simeq \frac{\rho_E^{(\oplus)}}{\rho_f c^2}. \tag{663}$$

This is the safest SST interpretation: the pendulum reads a macroscopic rotation state, and the clock offset follows from the energy-density version of the Swirl-Clock law.

Projected swirl-vorticity frequency. The preceding Foucault construction is the terrestrial special case of a more general SST diagnostic. For any SST velocity field $\mathbf{u}_{\circlearrowleft}(x,t)$ and any declared unit projection direction $\hat{\mathbf{a}}(x,t)$, define the projected swirl-vorticity frequency

$$\boxed{f_{\circlearrowleft}(x,t;\hat{\mathbf{a}}) := \hat{\mathbf{a}}(x,t)\cdot(\nabla\times\mathbf{u}_{\circlearrowleft}(x,t))} \tag{664}$$

with units s^{-1}. This quantity is the SST analogue of the orthodox Coriolis parameter

$$f_{\mathrm{Cor}} = 2\boldsymbol{\Omega}\cdot\hat{\mathbf{n}} = \hat{\mathbf{n}}\cdot(\nabla\times\mathbf{u}), \tag{665}$$

not a new force law. It is a projected-vorticity readout. In a terrestrial laboratory, taking $\mathbf{u}_\circlearrowleft$ to be the coarse-grained laboratory congruence velocity and $\hat{\mathbf{a}} = \hat{\mathbf{n}}$ gives $f_{\mathrm{Cor}} = 2\Omega_F$. Thus Foucault precession measures half the vertical projected vorticity of the macroscopic rotating frame.

For an atomic or knot-scale SST state K, Eq. (664) becomes a local diagnostic

$$f_K(x;\hat{\mathbf{a}}) := \hat{\mathbf{a}}(x)\cdot\boldsymbol{\omega}_K(x), \qquad \boldsymbol{\omega}_K := \nabla\times\mathbf{u}_K. \tag{666}$$

The projection direction must be declared. Natural choices include the local core tangent $\hat{\mathbf{t}}(s)$, a Frenet normal $\hat{\mathbf{n}}_F(s)$, a radial direction $\hat{\mathbf{r}}$, or an external quantization axis. The parallel strand diagnostic is

$$f_\parallel(s) := \hat{\mathbf{t}}(s)\cdot\boldsymbol{\omega}_K(s), \tag{667}$$

while a radial pressure-envelope diagnostic is

$$f_r(r,\theta,\varphi) := \hat{\mathbf{r}}\cdot\boldsymbol{\omega}_K(r,\theta,\varphi). \tag{668}$$

For a closed knot or approximately spherical envelope, the signed surface average of Eq. (668) may vanish by symmetry even when the local vorticity is dynamically important. The research-track scalar for spherical pressure support is therefore the RMS projected swirl-vorticity frequency

$$\boxed{f_{\mathrm{rms}}^{(K)}(r) := \left[\frac{1}{4\pi}\int_{S^2}\left(\hat{\mathbf{r}}\cdot\boldsymbol{\omega}_K(r,\theta,\varphi)\right)^2 d\Omega\right]^{1/2}} \tag{669}$$

which is non-negative and remains meaningful under sign cancellations. It may be used as a rotational-frequency diagnostic for spherically averaged pressure support, while the pressure balance itself remains governed by the Euler–vorticity channel

$$\nabla\left(p+\frac{1}{2}\rho_f\|\mathbf{u}_\circlearrowleft\|^2\right) = \rho_f\,\mathbf{u}_\circlearrowleft\times(\nabla\times\mathbf{u}_\circlearrowleft) \tag{670}$$

within the usual inviscid, incompressible SST approximation.

A local solid-body approximation gives $\|\mathbf{u}_\circlearrowleft\| \simeq \frac{1}{2}|f_\circlearrowleft|r_\perp$. Substitution into the canonical Swirl-Clock factor gives the diagnostic estimate

$$\boxed{S_{(t)}^{\circlearrowleft} \simeq \sqrt{1-\frac{f_\circlearrowleft^2 r_\perp^2}{4c^2}}} \tag{671}$$

valid only where the declared projection direction and local solid-body approximation are justified.

[STATUS]. Equations (664)–(665) are research-track definitions/orthodox Rosetta bridges. Equations (666)–(671) are conditional diagnostics for atomic, knot-core, or pressure-envelope modelling. They do not identify the projected frequency with ρ_f, ρ_{core}, or $v_\circlearrowleft^*$, and they do not add a direct vorticity source term to Maxwell or gravity equations.

Groningen benchmark. Using

$$\phi = 53.219^\circ, \qquad \Omega_\oplus = 7.2921150\times10^{-5}\ \mathrm{s}^{-1}, \qquad R_\oplus = 6.378137\times10^{6}\ \mathrm{m}, \tag{672}$$

one obtains

$$\Omega_F = 5.84047\times10^{-5}\ \mathrm{s}^{-1}, \qquad T_F = \frac{2\pi}{\Omega_F} = 29.883\ \mathrm{h}, \tag{673}$$

$$v_{\mathrm{rot}} = 278.48\ \mathrm{m\,s}^{-1}, \qquad 1-S_{(t)\,\mathrm{rot}}^{\circlearrowleft} = 4.31446\times10^{-13}. \tag{674}$$

The corresponding accumulated rotational offset per day is

$$\Delta\tau_{\mathrm{rot}} = 86400\left(4.31446\times10^{-13}\right)\ \mathrm{s} = 37.28\ \mathrm{ns/day}, \tag{675}$$

with the clock slower relative to the non-rotating foliation frame. For $h = 100\,\mathrm{m}$,

$$\frac{gh}{c^2} = 1.09114\times10^{-14}, \qquad \Delta\tau_h = 0.943\ \mathrm{ns/day}, \tag{676}$$

so the net first-order Groningen offset at 100 m above the geoid is approximately

$$\Delta\tau_{\rm net} \simeq -37.28 \text{ ns/day} + 0.94 \text{ ns/day} = -36.34 \text{ ns/day}. \tag{677}$$

The macroscopic rotational energy density at Groningen is

$$\rho_E^{(\oplus)} = \frac{1}{2}(7.0\times10^{-7})(278.48)^2 = 2.714\times10^{-2}\ \mathrm{J\,m^{-3}}. \tag{678}$$

Compared with the canonical core-scale reference

$$\rho_E^* = \frac{1}{2}\rho_f v_{\circlearrowleft}^{*\,2} = 4.18774\times10^5\ \mathrm{J\,m^{-3}}, \tag{679}$$

the ratio is

$$\frac{\rho_E^{(\oplus)}}{\rho_E^*} = 6.48\times10^{-8}. \tag{680}$$

This explains why the effect is measurable by precision clock comparison but remains weak relative to microscopic core-swirl scales.

Guardrails. The following claims are explicitly not canonized here.

1. The pendulum swing period $T_{\rm pend} = 2\pi\sqrt{L/g_{\rm eff}}$ is not a direct time-dilation measurement. It mainly probes the local effective gravity gradient, while clock rate probes potential and kinematic factors.
2. A laboratory-size term $v_{\rm lab} = \Omega_F R_{\rm eff}$ must not be promoted to canon. Foucault precession is not automatically a solid-body swirl around an arbitrary lab radius.
3. Engineered-swirl claims require a constitutive probe model

$$\delta\Omega_F = \mathcal{C}_{\rm probe}\big[\delta\mathbf{u}_{\circlearrowleft}, \delta p_{\circlearrowleft}, \delta\boldsymbol{\sigma}^{\rm swirl}\big] \tag{681}$$

before any local rotor, coil, or superfluid-ring perturbation can be interpreted as a clock-rate perturbation.

4. No density law of the form $\rho_f = K\Omega$, nor any identification of projected vorticity with ρ_f, $\rho_{\rm core}$, or $v_{\circlearrowleft}^*$, is asserted in this patch. Such a relation would require a separate micro-to-macro averaging theorem and independent calibration.

Research-track conclusion. A Foucault pendulum is therefore a macroscopic phase-precession probe of the same coarse-grained rotational state that enters the SST Swirl-Clock inference chain. It supports a clean falsifiable protocol: measure Ω_F, infer $v_{\rm rot}$, compute the expected clock offset from Eq. (662), and compare against precision clock transport or co-located clock-comparison data.

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